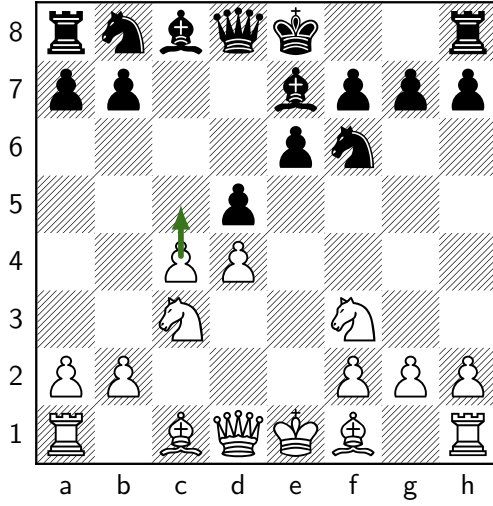


Melanie-BOTS (CCRL engines) — Caro-Kann, Panov

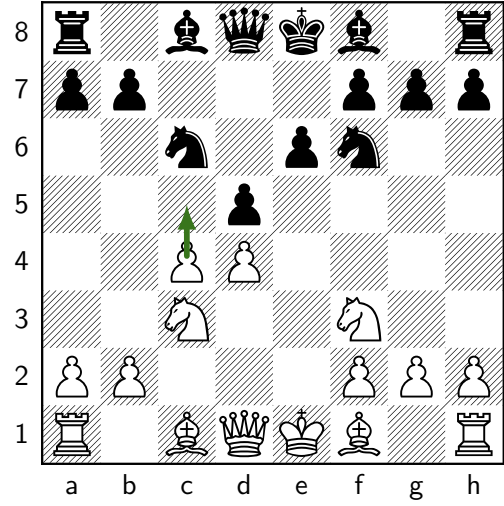
Melanie (Deep_Vision)

Abstract

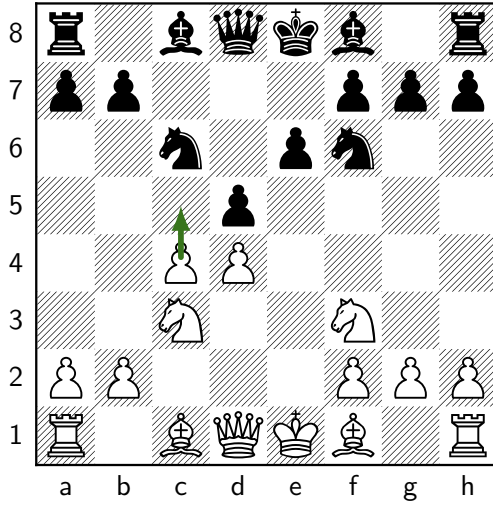
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



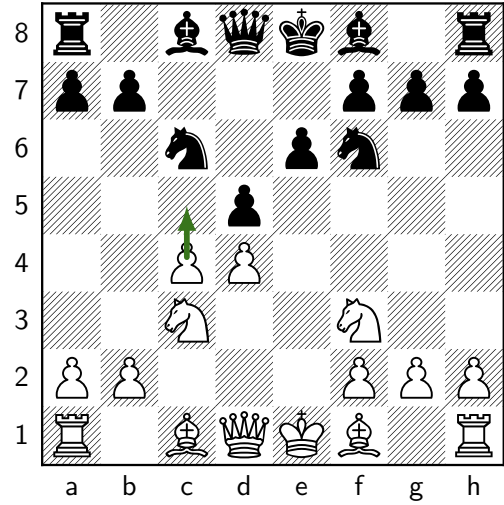
(a)



(b)

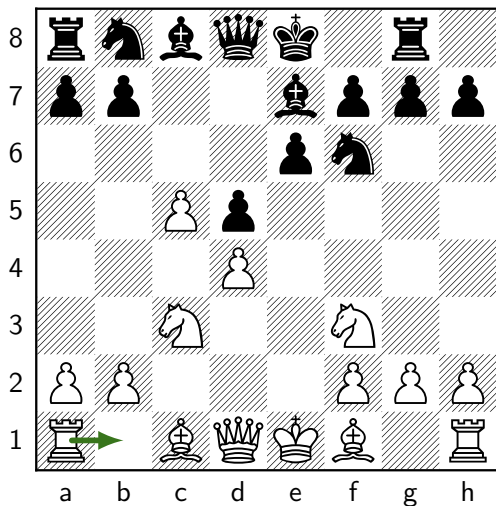


(c)

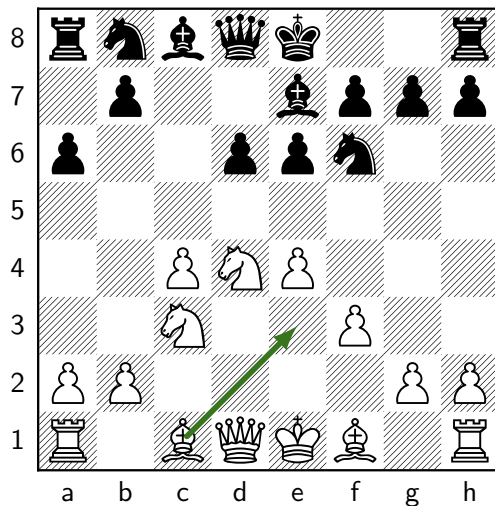


(d)

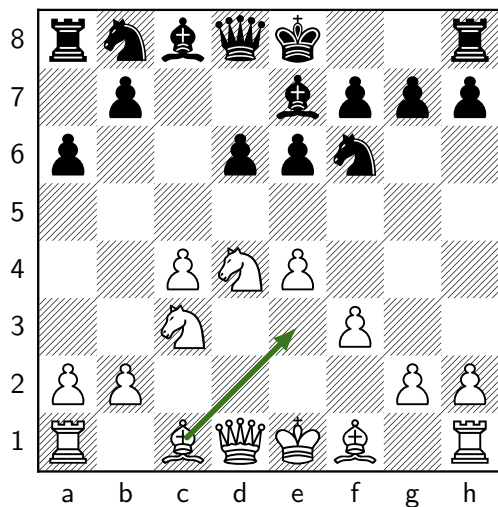
Figure 1: **Move 7, White: c5.** (a) Our position; Melanie plays c5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3405` (ply 12) [structural distance $d=29$, QAP obj=109]: the master played c5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Bf1$, our $be7$ as their $Pb2$; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3406` (ply 12) [structural distance $d=29$, QAP obj=109]: the master played c5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Bf1$, our $be7$ as their $Pb2$; transferred move c4c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#180` (ply 12) [structural distance $d=29$, QAP obj=109]: the master played c5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Bf1$, our $be7$ as their $Pb2$; transferred move c4c5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to c5 (summed QAP objective 789, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

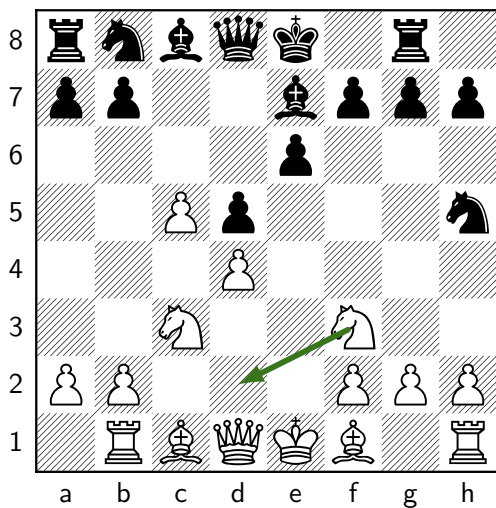


(b)

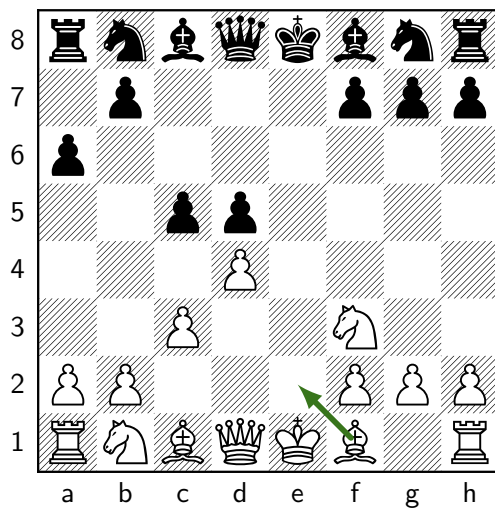


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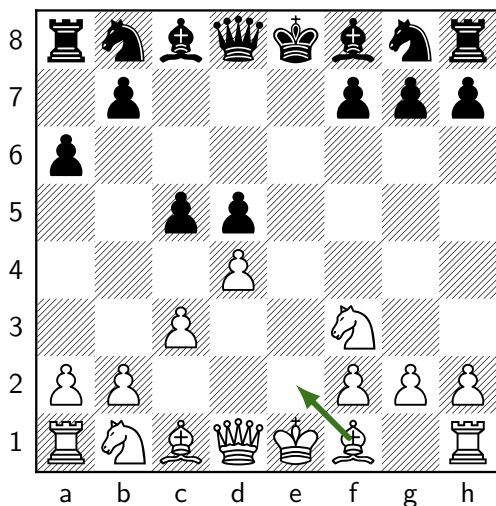
Figure 2: **Move 8, White: Rb1.** (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2684` (ply 14) [structural distance $d=35$, QAP obj=41]: the master played Be3; under μ our *Ra1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pd4* as their *Ra1*, our *Nf3* as their *Qd1*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2685` (ply 14) [structural distance $d=35$, QAP obj=41]: the master played Be3; under μ our *Ra1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pd4* as their *Ra1*, our *Nf3* as their *Qd1*; transferred move a1b1 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Rb1 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



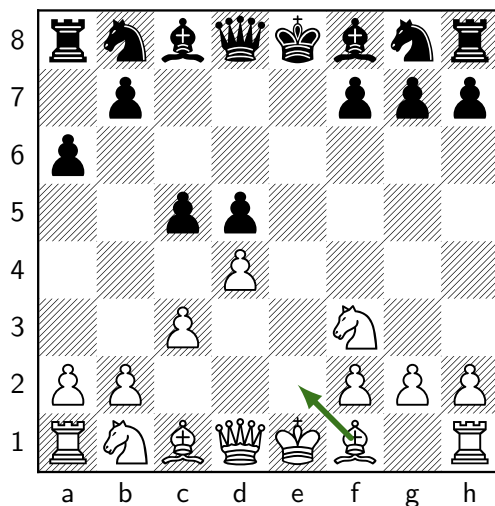
(a)



(b)

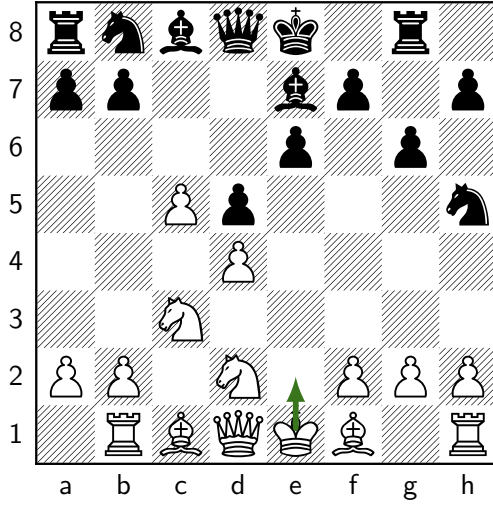


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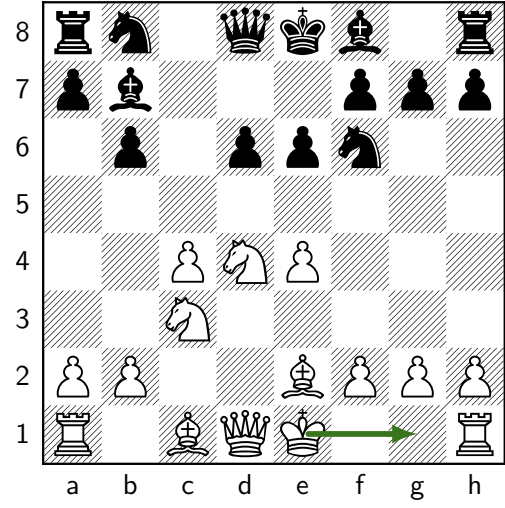


(d)

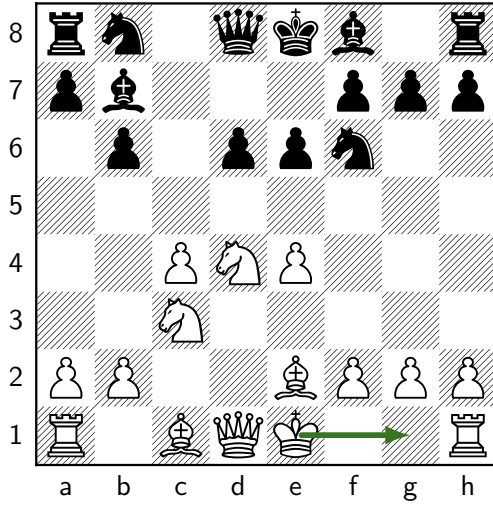
Figure 3: **Move 9, White: Nd2.** (a) Our position; Melanie plays Nd2 (f3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3308` (ply 10) [structural distance $d=41$, QAP obj=16]: the master played **Be2**; under μ our *Nf3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *pd5* as their *Ra1*, our *Ph2* as their *Nb1*; transferred move f3d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3309` (ply 10) [structural distance $d=41$, QAP obj=16]: the master played **Be2**; under μ our *Nf3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *pd5* as their *Ra1*, our *Ph2* as their *Nb1*; transferred move f3d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1337` (ply 10) [structural distance $d=41$, QAP obj=16]: the master played **Be2**; under μ our *Nf3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *pd5* as their *Ra1*, our *Ph2* as their *Nb1*; transferred move f3d2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd2 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



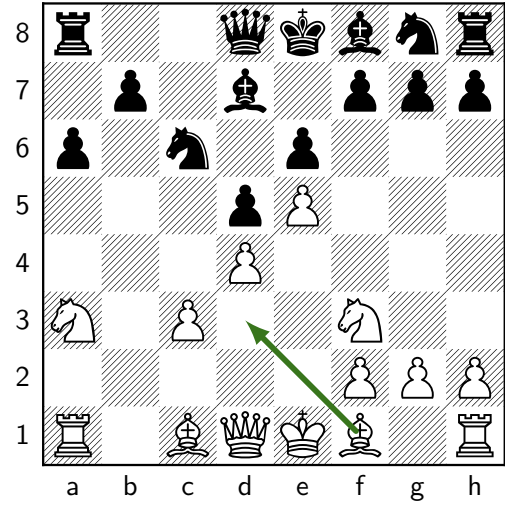
(a)



(b)

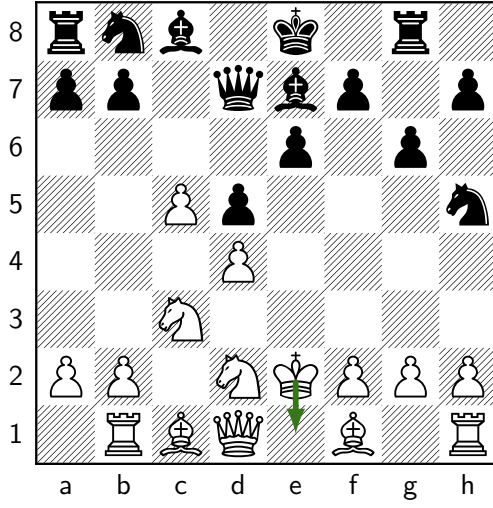


(c)

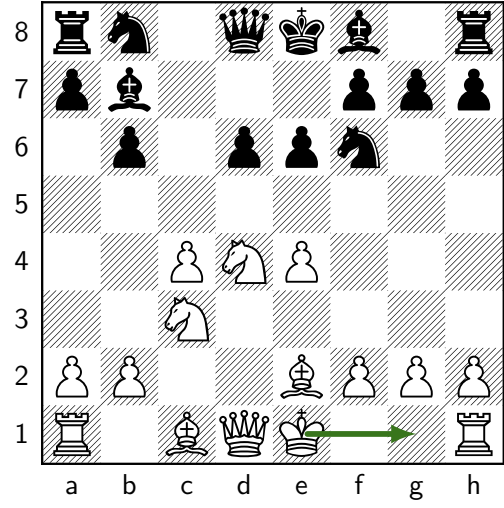


(d)

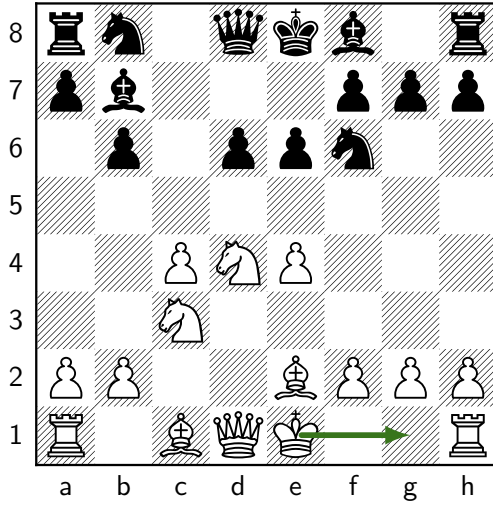
Figure 4: **Move 10, White: Ke2.** (a) Our position; Melanie plays Ke2 (**e1e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#2988` (ply 14) [structural distance $d=71$, QAP obj=37]: the master played **0-0**; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg2* as their *Ra1*, our *Rb1* as their *Pa2*; transferred move **e1e2** (**endorses this move**). (c) Retrieved from `cctl_top_p0003#2989` (ply 14) [structural distance $d=71$, QAP obj=37]: the master played **0-0**; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg2* as their *Ra1*, our *Rb1* as their *Pa2*; transferred move **e1e2** (**endorses this move**). (d) Retrieved from `cctl_top_p0002#1900` (ply 16) [structural distance $d=71$, QAP obj=48]: the master played **Bd3**; under μ our *Nc3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *ph7* as their *Ra1*, our *Nd2* as their *Bc1*; transferred move **c3e4**. Synthesis: 2 retrieved analogs transferred to Ke2 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



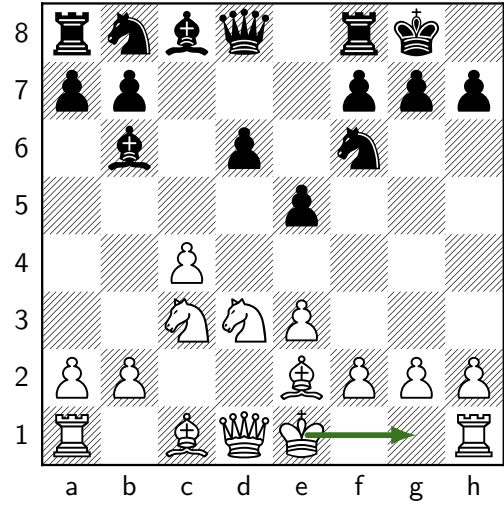
(a)



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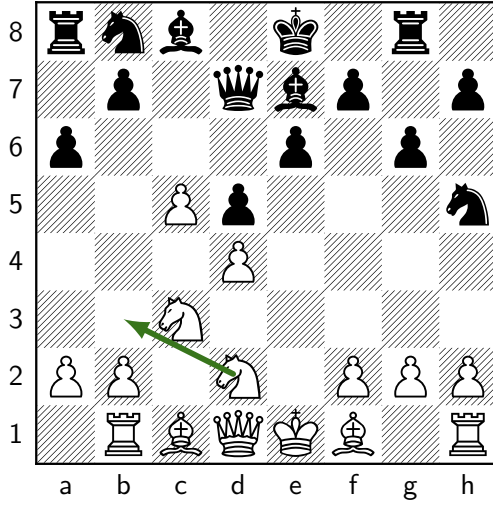


(c)

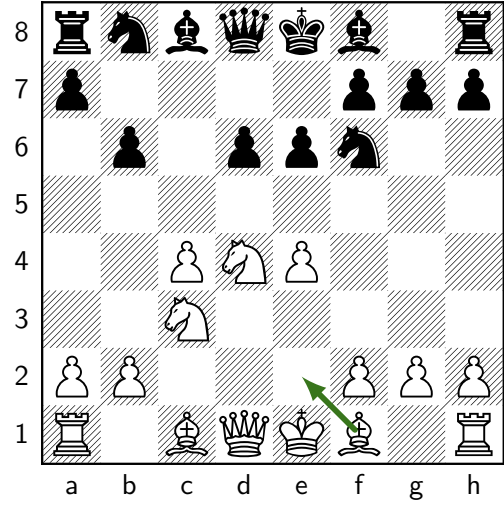


(d)

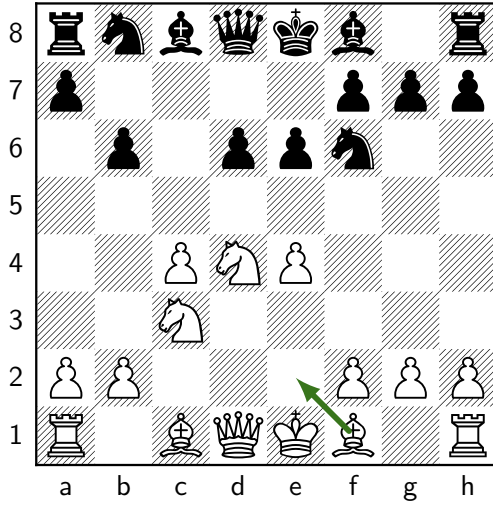
Figure 5: **Move 11, White: Ke1.** (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2988` (ply 14) [structural distance $d=79$, QAP obj=51]: the master played 0-0; under μ our Ke2 plays the role of that game's Ke1; cross-type roles: our pe6 as their Ra1, our ra8 as their Qd1; transferred move e2e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2989` (ply 14) [structural distance $d=79$, QAP obj=51]: the master played 0-0; under μ our Ke2 plays the role of that game's Ke1; cross-type roles: our pe6 as their Ra1, our ra8 as their Qd1; transferred move e2e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3586` (ply 18) [structural distance $d=71$, QAP obj=37]: the master played 0-0; under μ our Ke2 plays the role of that game's Ke1; cross-type roles: our Pd4 as their Ra1, our Rb1 as their Bc1; transferred move e2e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke1 (summed QAP objective 139, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



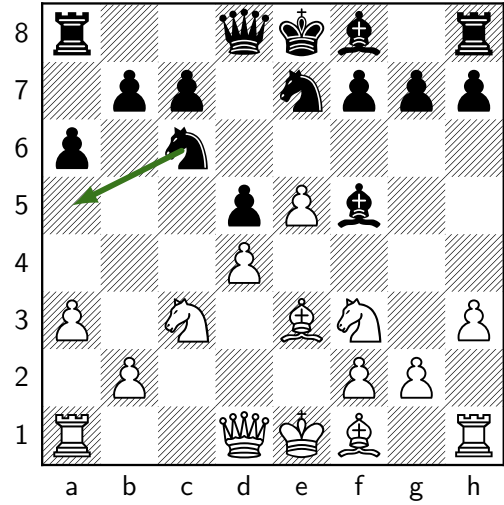
(a)



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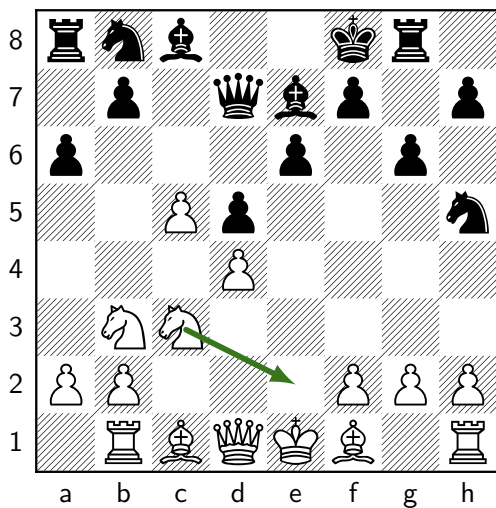


(c)

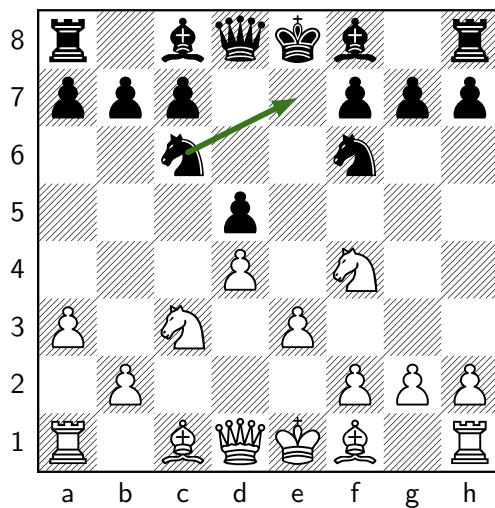


(d)

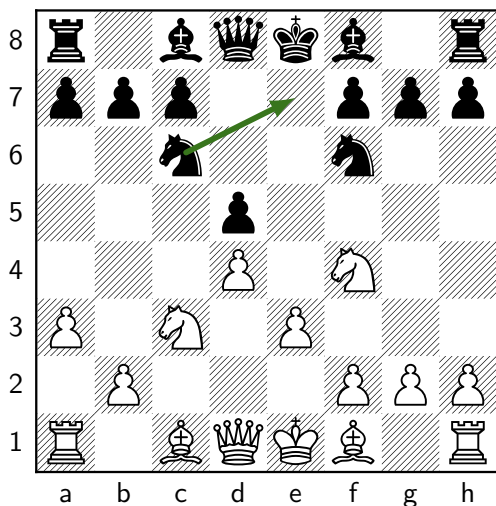
Figure 6: **Move 12, White: Nb3.** (a) Our position; Melanie plays Nb3 (d2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2988` (ply 12) [structural distance $d=84$, QAP obj=29]: the master played Be2; under μ our Nd2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our nb8 as their Ra1, our Nc3 as their Bc1; transferred move d2b3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2989` (ply 12) [structural distance $d=84$, QAP obj=29]: the master played Be2; under μ our Nd2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our nb8 as their Ra1, our Nc3 as their Bc1; transferred move d2b3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#377` (ply 19) [structural distance $d=84$, QAP obj=52]: the master played Na5; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our pg6 as their Ra8, our Nd2 as their Qd8; transferred move c3a4. Synthesis: 2 retrieved analogs transferred to Nb3 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



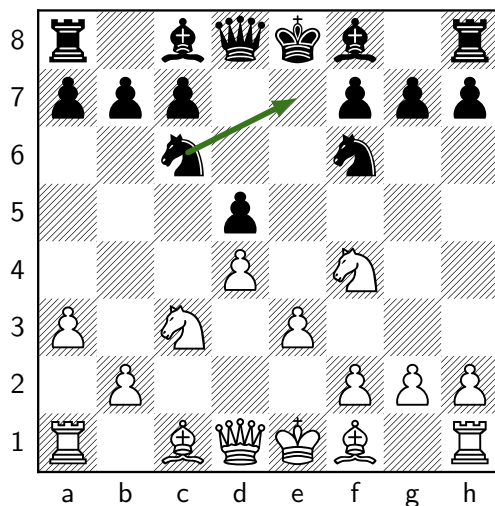
(a)



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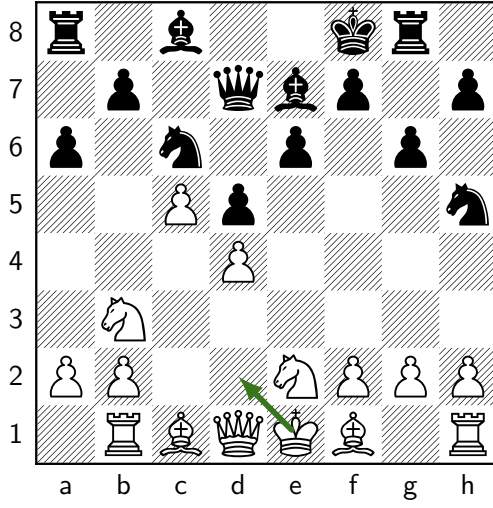


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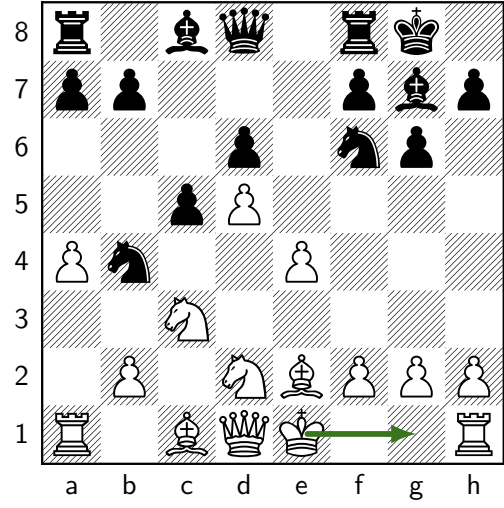


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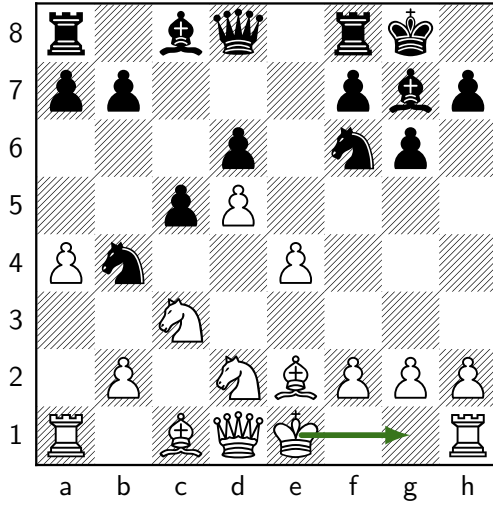
Figure 7: **Move 13, White: Ne2**. (a) Our position; Melanie plays Ne2 (c3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3807` (ply 15) [structural distance $d=50$, QAP obj=46]: the master played Ne7; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Pd4 as their Bc8, our qd7 as their Pg7; transferred move c3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3808` (ply 15) [structural distance $d=50$, QAP obj=46]: the master played Ne7; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Pd4 as their Bc8, our qd7 as their Pg7; transferred move c3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#437` (ply 15) [structural distance $d=50$, QAP obj=46]: the master played Ne7; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Pd4 as their Bc8, our qd7 as their Pg7; transferred move c3e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ne2 (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



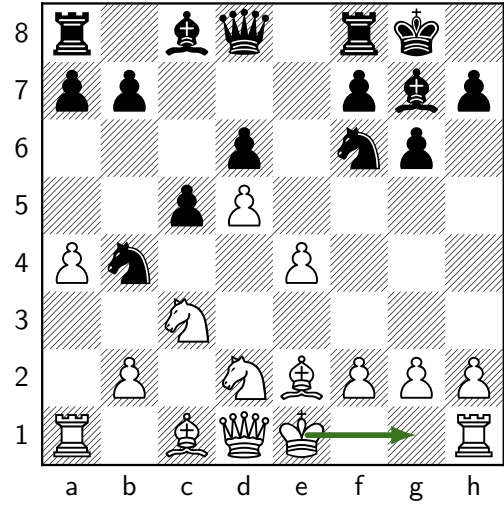
(a)



(b)

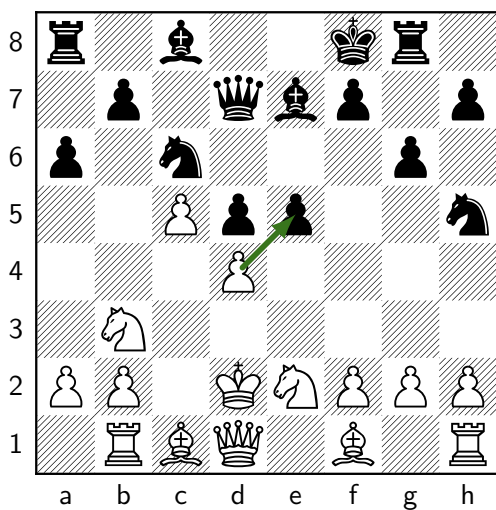


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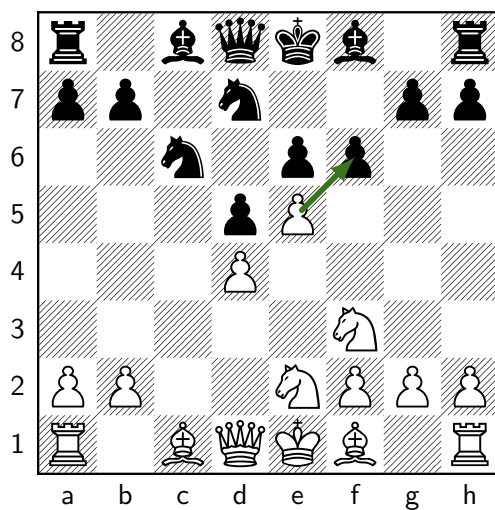


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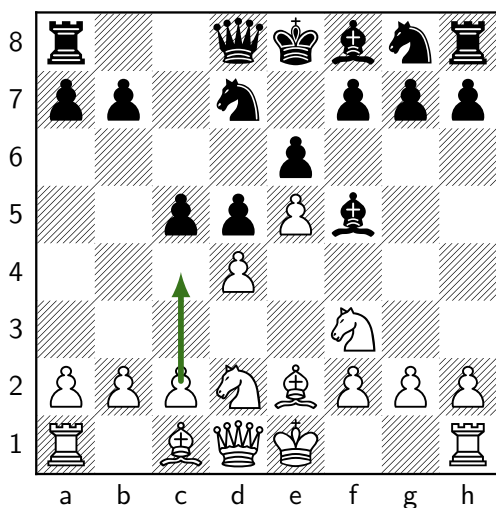
Figure 8: **Move 14, White: Kd2.** (a) Our position; Melanie plays Kd2 (e1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1801` (ply 20) [structural distance $d=77$, QAP obj=60]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ra1*, our *Pd4* as their *Be2*; transferred move e1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1221` (ply 20) [structural distance $d=77$, QAP obj=60]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ra1*, our *Pd4* as their *Be2*; transferred move e1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2843` (ply 20) [structural distance $d=77$, QAP obj=60]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ra1*, our *Pd4* as their *Be2*; transferred move e1d2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kd2 (summed QAP objective 543, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



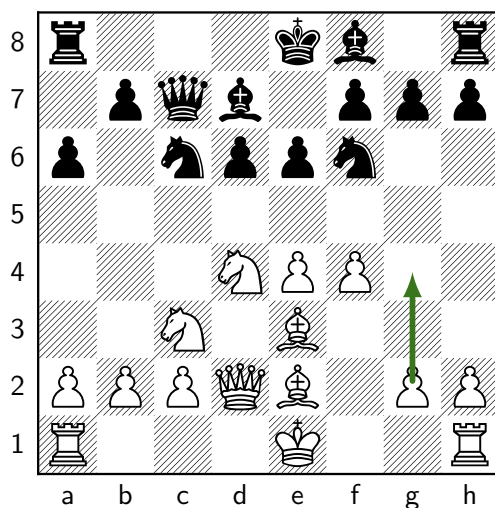
(a)



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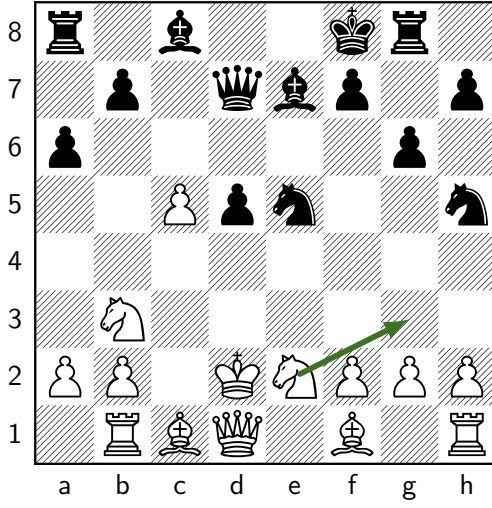


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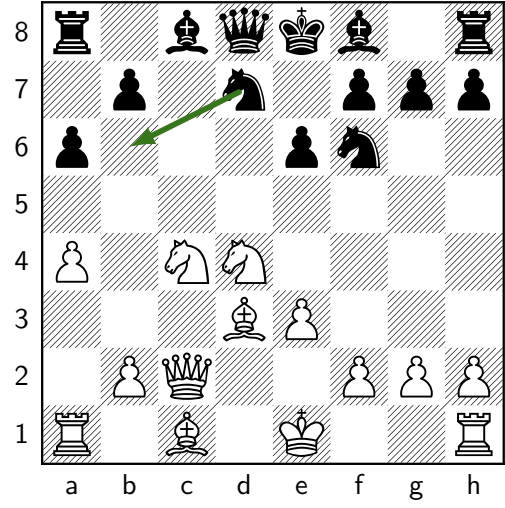


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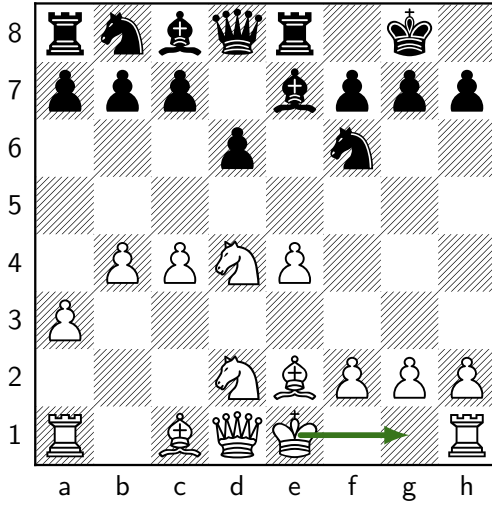
Figure 9: **Move 15, White: dxe5.** (a) Our position; Melanie plays dxe5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#410` (ply 16) [structural distance $d=80$, QAP obj=93]: the master played **exf6**; under μ our *Pd4* plays the role of that game’s *Pe5*; cross-type roles: our *nh5* as their *Bc1*, our *Ne2* as their *Rh1*; transferred move **d4e5 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#320` (ply 12) [structural distance $d=84$, QAP obj=54]: the master played **c4**; under μ our *Pd4* plays the role of that game’s *Pc2*; cross-type roles: our *qd7* as their *Ra1*, our *Nb3* as their *Qd1*; transferred move **d4e5 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#65` (ply 18) [structural distance $d=86$, QAP obj=43]: the master played **g4**; under μ our *Ne2* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *Nb3* as their *Ra1*, our *Bc1* as their *Rh1*; transferred move **e2g1**. Synthesis: 2 retrieved analogs transferred to dxe5 (summed QAP objective 148, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



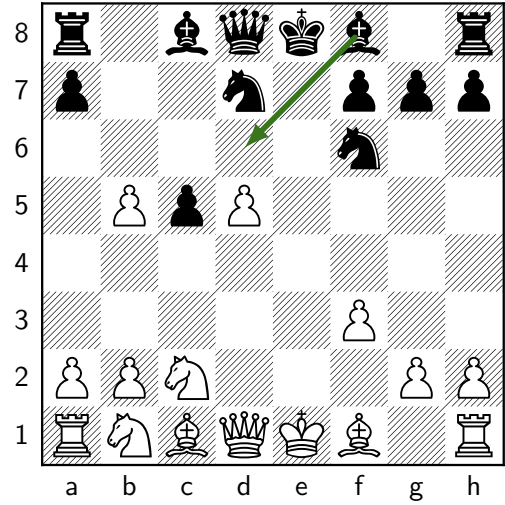
(a)



(b)

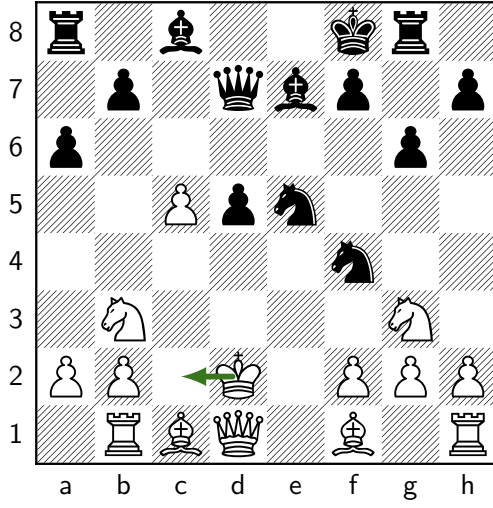


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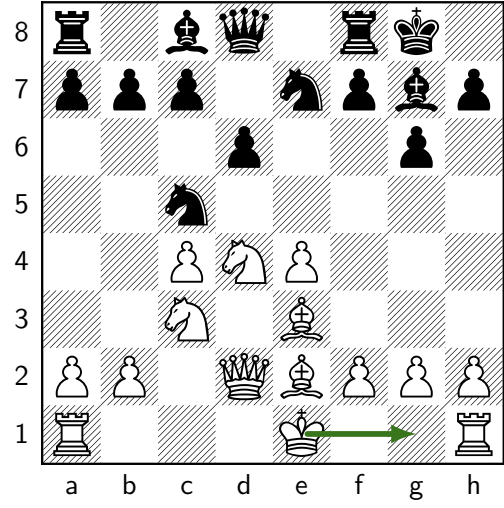


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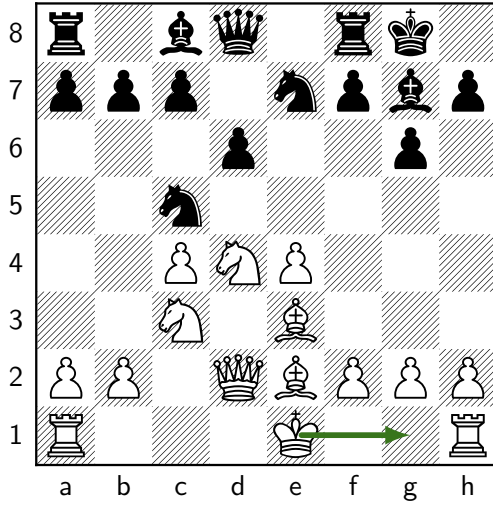
Figure 10: **Move 16, White: Ng3.** (a) Our position; Melanie plays Ng3 (e2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3909` (ply 19) [structural distance $d=95$, QAP obj=24]: the master played Nb6; under μ our Ne2 plays the role of that game’s Nd2; cross-type roles: our Bc1 as their Ra8, our ra8 as their Bc8; transferred move e2g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3988` (ply 18) [structural distance $d=87$, QAP obj=25]: the master played 0-0; under μ our Kd2 plays the role of that game’s Ke1; cross-type roles: our Bc1 as their Rh1, our ra8 as their Nd2; transferred move d2c2. (d) Retrieved from `ccrl_top_p0003#1434` (ply 17) [structural distance $d=90$, QAP obj=24]: the master played Bd6; under μ our Pf2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra8, our Ne2 as their Bc8; transferred move f2f4. Synthesis: 1 retrieved analog transferred to Ng3 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



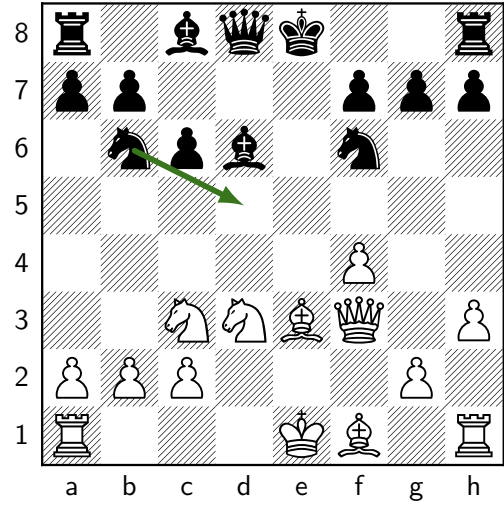
(a)



(b)

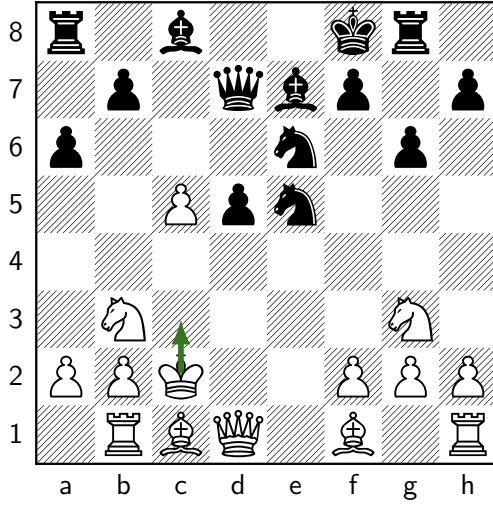


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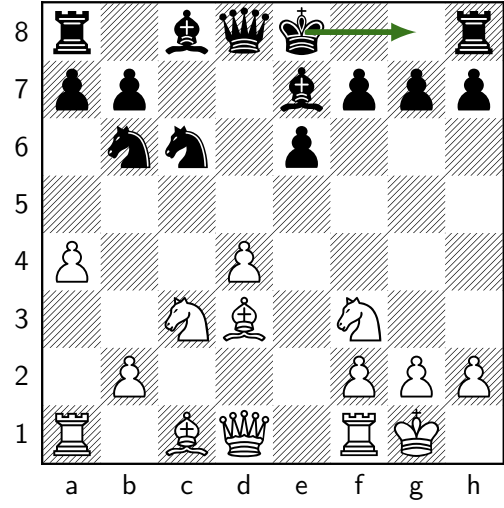


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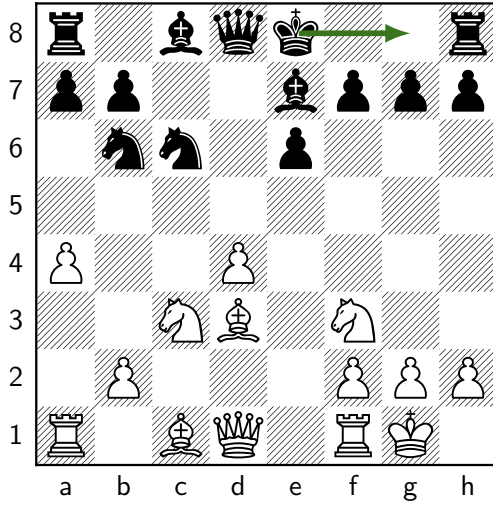
Figure 11: **Move 17, White: Kc2.** (a) Our position; Melanie plays Kc2 (d2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#840` (ply 18) [structural distance $d=103$, QAP obj=22]: the master played 0-0; under μ our *Kd2* plays the role of that game's *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Ph2* as their *Rh1*; transferred move d2c2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#841` (ply 18) [structural distance $d=103$, QAP obj=22]: the master played 0-0; under μ our *Kd2* plays the role of that game's *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Ph2* as their *Rh1*; transferred move d2c2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3319` (ply 19) [structural distance $d=89$, QAP obj=24]: the master played Nbd5; under μ our *Pc5* plays the role of that game's *Nb3* (a **cross-type** correspondence); cross-type roles: our *Ng3* as their *Ra8*, our *qd7* as their *Bc8*; transferred move c5c6. Synthesis: 2 retrieved analogs transferred to Kc2 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



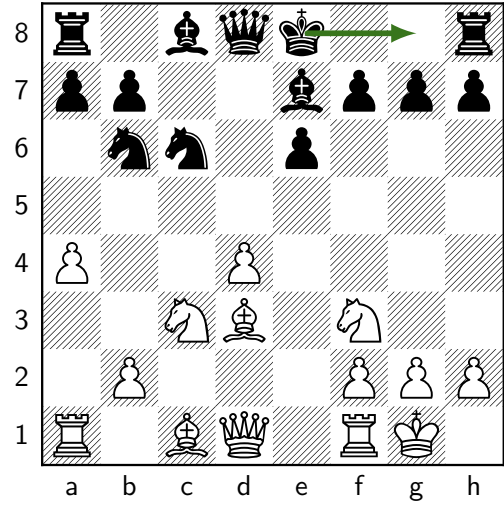
(a)



(b)

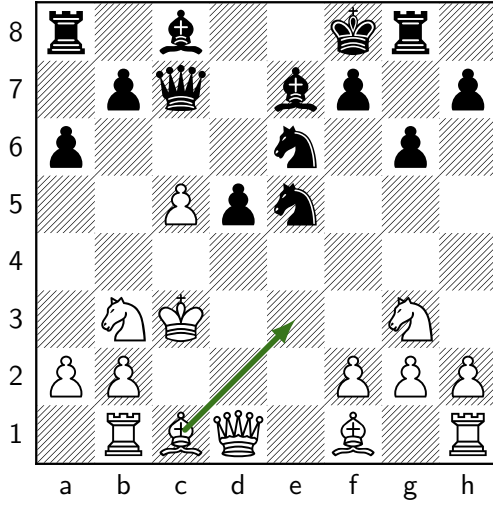


(c)

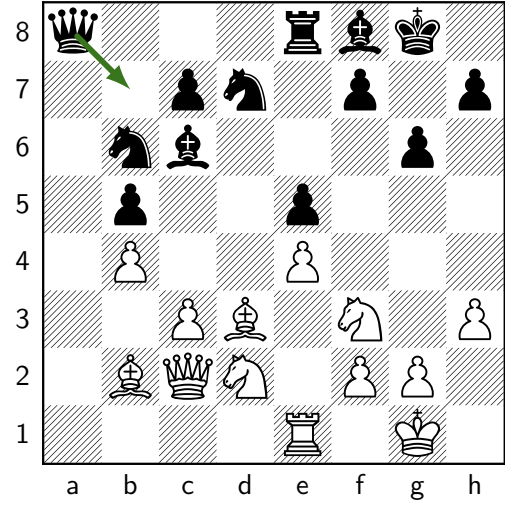


(d)

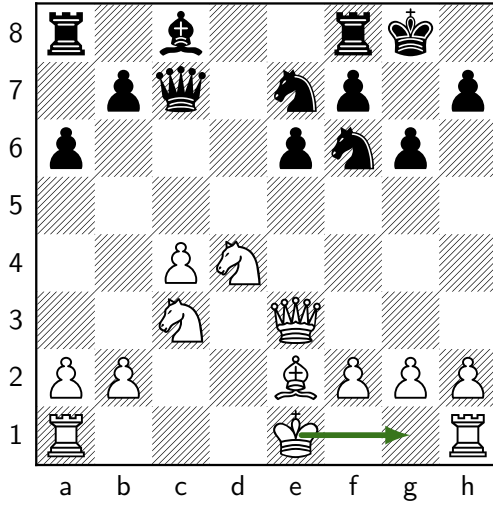
Figure 12: **Move 18, White: Kc3.** (a) Our position; Melanie plays Kc3 (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2131` (ply 19) [structural distance $d=100$, QAP obj=74]: the master played 0-0; under μ our Kc2 plays the role of that game's Ke1; cross-type roles: our Nb3 as their Ra8, our Bc1 as their Pa7; transferred move c2c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2132` (ply 19) [structural distance $d=100$, QAP obj=74]: the master played 0-0; under μ our Kc2 plays the role of that game's Ke1; cross-type roles: our Nb3 as their Ra8, our Bc1 as their Pa7; transferred move c2c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#611` (ply 19) [structural distance $d=100$, QAP obj=74]: the master played 0-0; under μ our Kc2 plays the role of that game's Ke1; cross-type roles: our Nb3 as their Ra8, our Bc1 as their Pa7; transferred move c2c3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kc3 (summed QAP objective 360, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



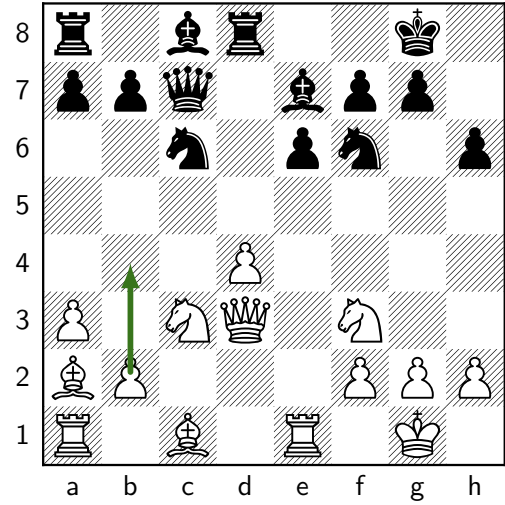
(a)



(b)

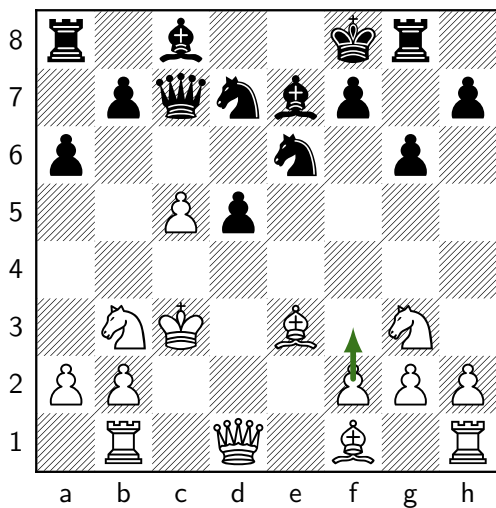


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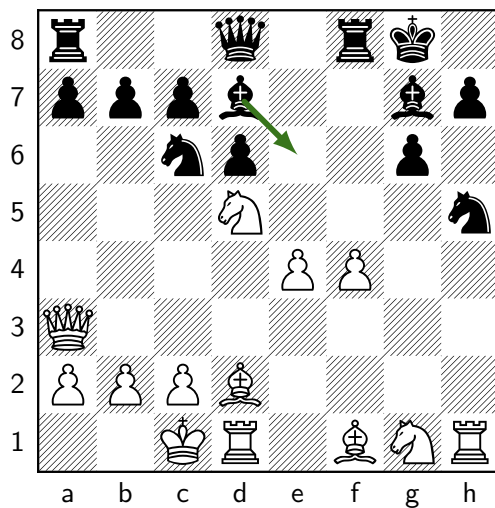


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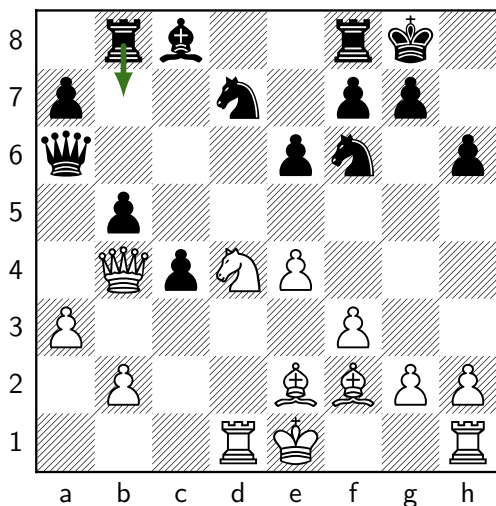
Figure 13: **Move 19, White: Be3.** (a) Our position; Melanie plays Be3 (c1e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1247` (ply 45) [structural distance $d=103$, QAP obj=78]: the master played Qb7; under μ our Bc1 plays the role of that game’s Qa1 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Re8, our Qd1 as their Bf8; transferred move c1e3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2348` (ply 30) [structural distance $d=97$, QAP obj=86]: the master played 0-0; under μ our Kc3 plays the role of that game’s Ke1; cross-type roles: our Pb2 as their Ra1, our Bc1 as their Rh1; transferred move c3c2. (d) Retrieved from `ccrl_top_p0000#1555` (ply 24) [structural distance $d=79$, QAP obj=84]: the master played b4; under μ our Nb3 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Ng3 as their Ra1, our qc7 as their Re1; transferred move b3a5. Synthesis: 1 retrieved analog transferred to Be3 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



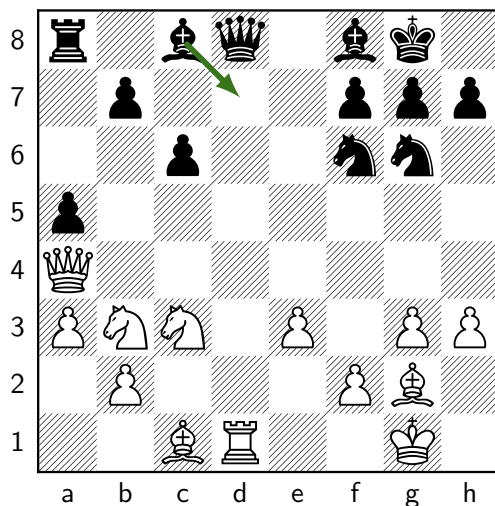
(a)



(b)

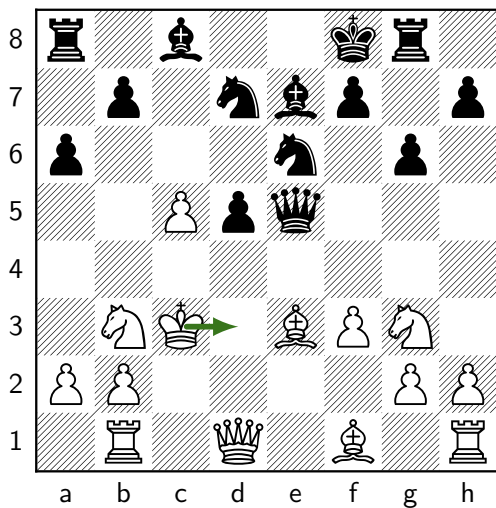


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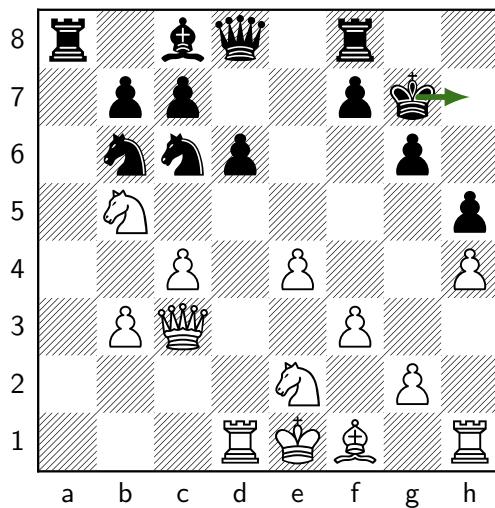


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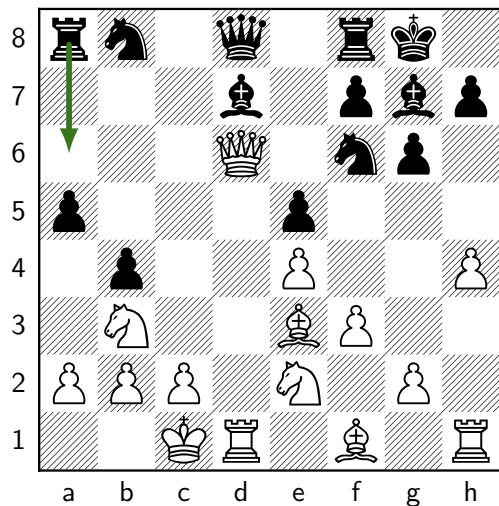
Figure 14: **Move 20, White: f3.** (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3159` (ply 27) [structural distance $d=103$, QAP obj=87]: the master played Bb6; under μ our Pf2 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our Be3 as their Ra8, our Pb2 as their Qd8; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1819` (ply 35) [structural distance $d=93$, QAP obj=88]: the master played Rb7; under μ our Nb3 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Rb1 as their Bc8, our Ng3 as their Rf8; transferred move b3c1. (d) Retrieved from `ccr1_top_p0003#176` (ply 39) [structural distance $d=103$, QAP obj=78]: the master played Bd7; under μ our Rh1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Nb3 as their Ra8, our Pc5 as their Qd8; transferred move h1g1. Synthesis: 1 retrieved analog transferred to f3 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

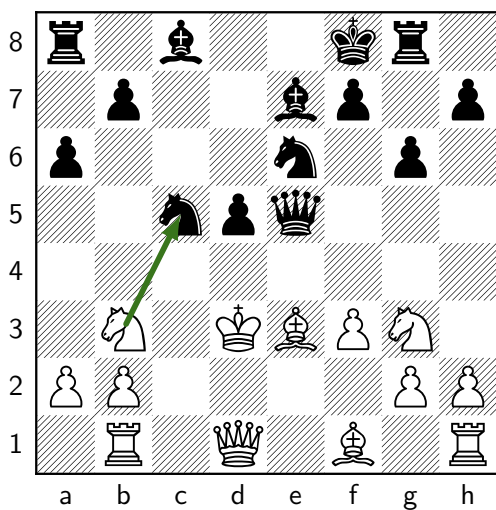


(b)

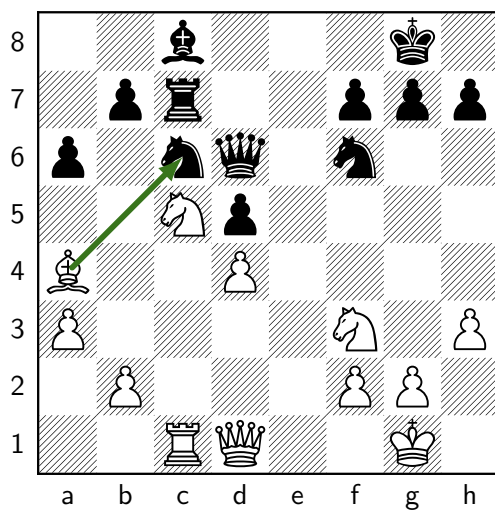


(c)

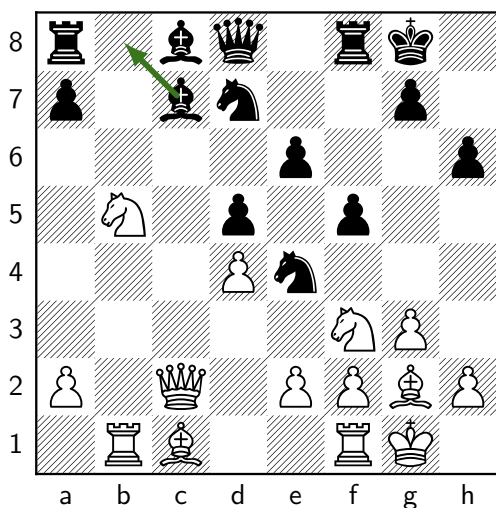
Figure 15: **Move 21, White: Kd3.** (a) Our position; Melanie plays Kd3 (c3d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0003#2666` (ply 31) [structural distance $d=95$, QAP obj=56]: the master played Kh7; under μ our *Kc3* plays the role of that game’s *Kg2*; cross-type roles: our *Ph2* as their *Ra8*, our *pg6* as their *Bc8*; transferred move *c3b4*. (c) Retrieved from `ccr1_top-p0001#1754` (ply 27) [structural distance $d=100$, QAP obj=5]: the master played Ra6; under μ our *Qd1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bf1* as their *Nb8*, our *Pg2* as their *Qd8*; transferred move *d1d4*. Synthesis: no retrieved analog transferred to Kd3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



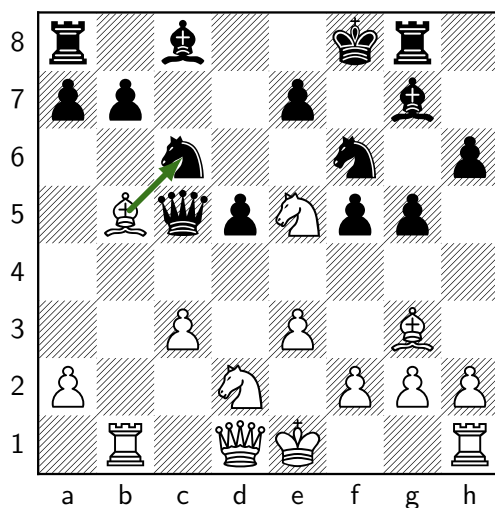
(a)



(b)

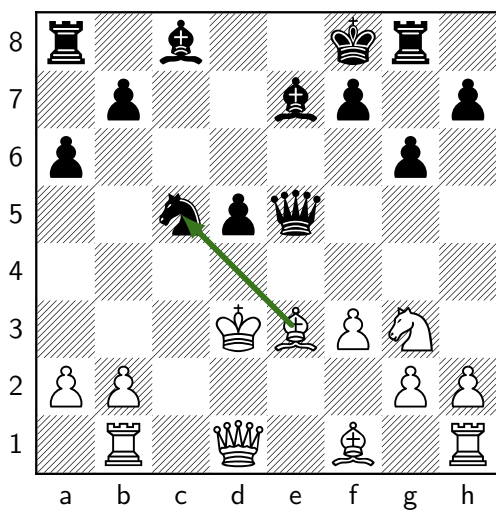


(c)

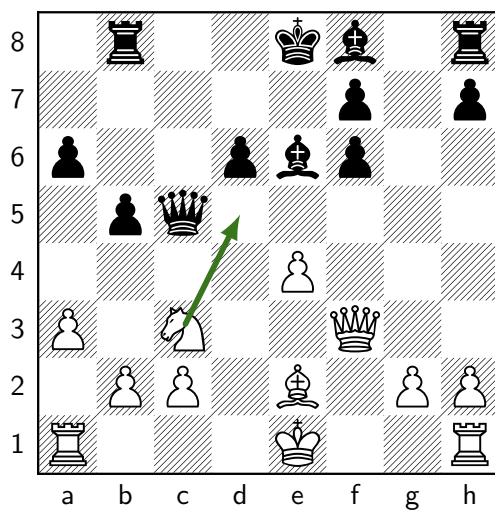


(d)

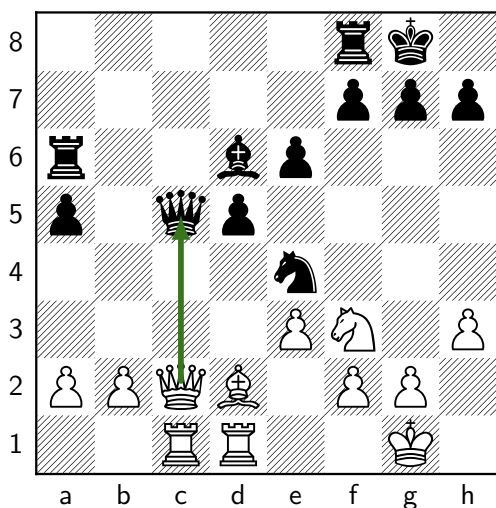
Figure 16: **Move 22, White: Nxc5**. (a) Our position; Melanie plays Nxc5 (b3c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1311 (ply 42) [structural distance $d=123$, QAP obj=7]: the master played Bxc6; under μ our Nb3 plays the role of that game’s Ba4 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pb2, our Be3 as their Ph3; transferred move b3c5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2317 (ply 27) [structural distance $d=124$, QAP obj=6]: the master played Bb8; under μ our Nb3 plays the role of that game’s Bc2 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Bc8, our Rh1 as their Qd8; transferred move b3c5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#1149 (ply 28) [structural distance $d=114$, QAP obj=5]: the master played Bxc6; under μ our Nb3 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Be3 as their Rb1, our Bf1 as their Rh1; transferred move b3c5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nxc5 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



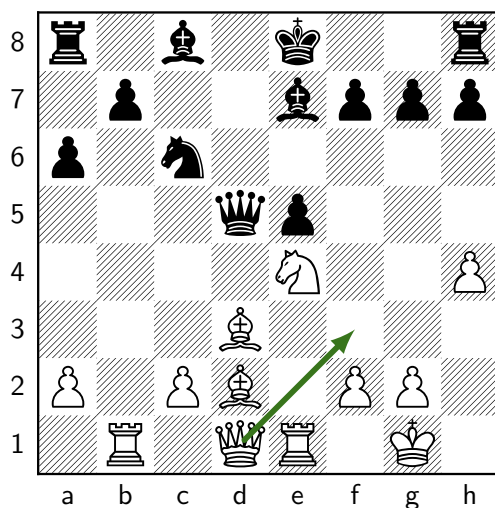
(a)



(b)

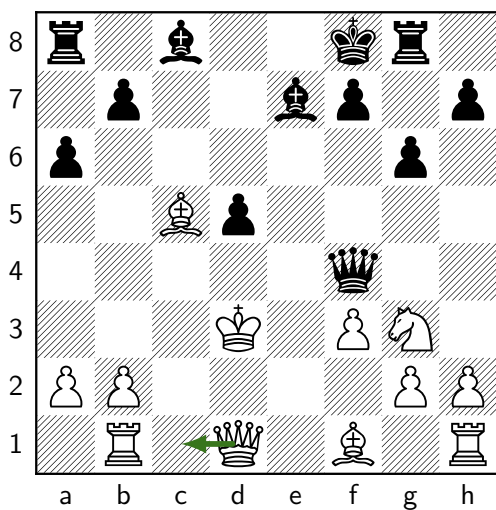


(c)

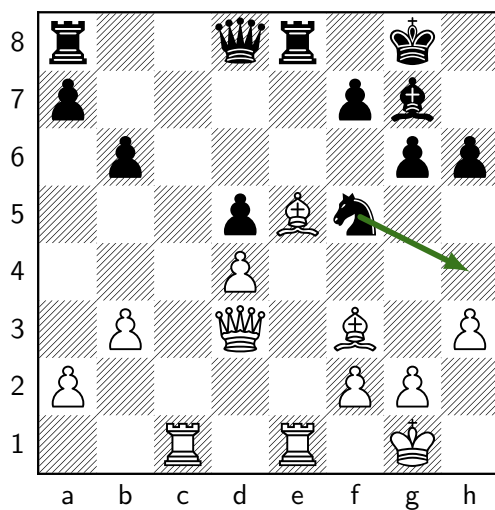


(d)

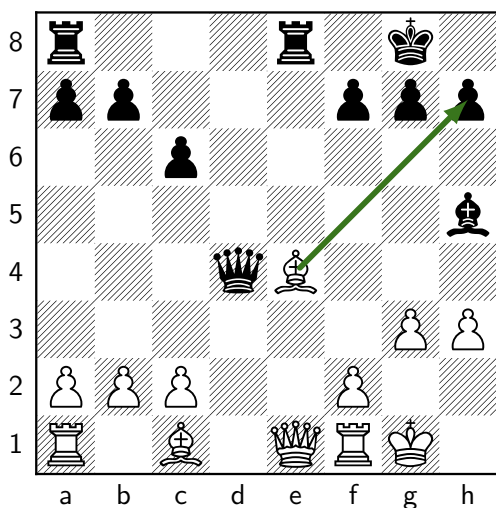
Figure 17: **Move 23, White: Bxc5.** (a) Our position; Melanie plays Bxc5 (e3c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1383` (ply 28) [structural distance $d=84$, QAP obj=6]: the master played Nd5; under μ our *Be3* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *Qd1* as their *Rh1*; transferred move e3c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2937` (ply 36) [structural distance $d=76$, QAP obj=6]: the master played Qxc5; under μ our *Be3* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Rd1*, our *Ng3* as their *Pb2*; transferred move e3c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1531` (ply 30) [structural distance $d=79$, QAP obj=3]: the master played Qf3; under μ our *Be3* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *nc5* as their *Rb1*, our *Qd1* as their *Re1*; transferred move e3c5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxc5 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



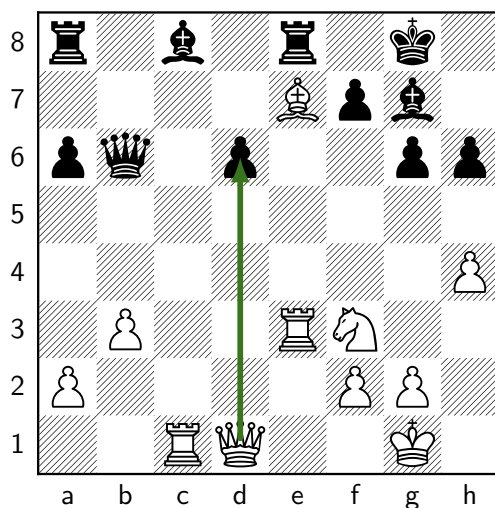
(a)



(b)

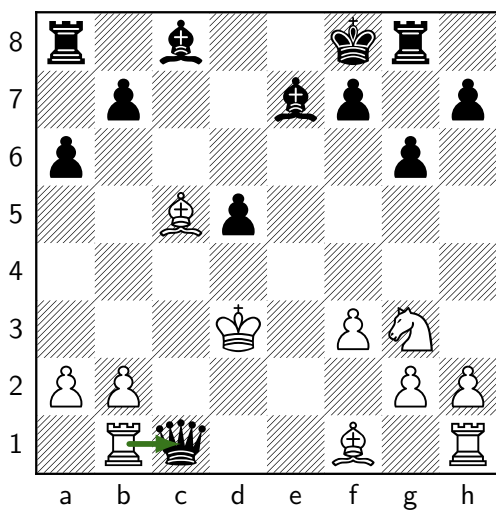


(c)

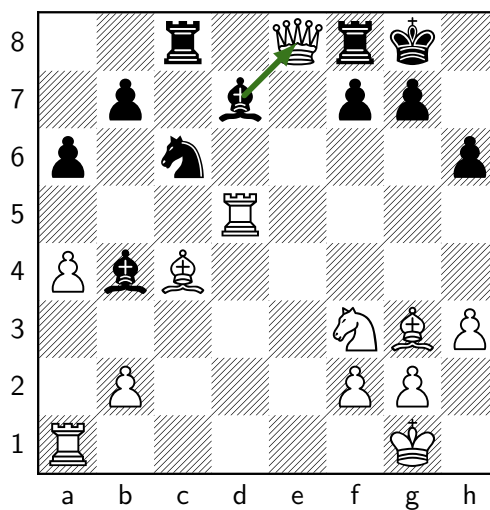


(d)

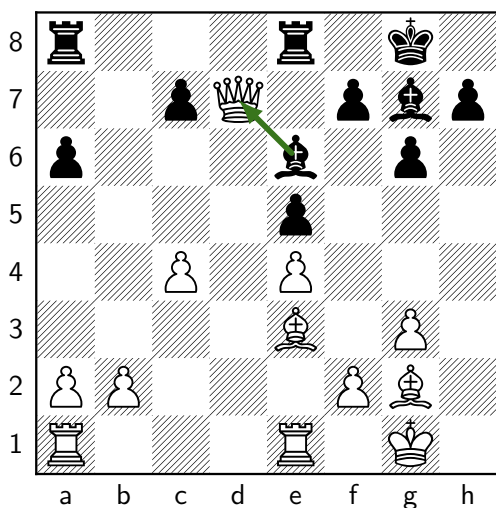
Figure 18: **Move 24, White: Qc1.** (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2434` (ply 41) [structural distance $d=96$, QAP obj=49]: the master played Nh4; under μ our Ng3 plays the role of that game’s Nf4; cross-type roles: our Pa2 as their Ra8, our pa6 as their Qd8; transferred move g3h5. (c) Retrieved from `ccrl_top_p0001#1197` (ply 32) [structural distance $d=93$, QAP obj=49]: the master played Bxh7+; under μ our Ng3 plays the role of that game’s Be4 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Rf1, our Rh1 as their Pa2; transferred move g3h5. (d) Retrieved from `ccrl_top_p0002#2899` (ply 40) [structural distance $d=96$, QAP obj=35]: the master played Qxd6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Ph2 as their Rc1, our Rb1 as their Pg2; transferred move d1a4. Synthesis: no retrieved analog transferred to Qc1 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



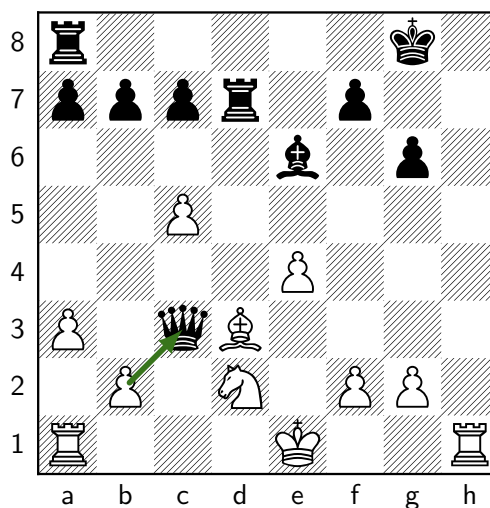
(a)



(b)

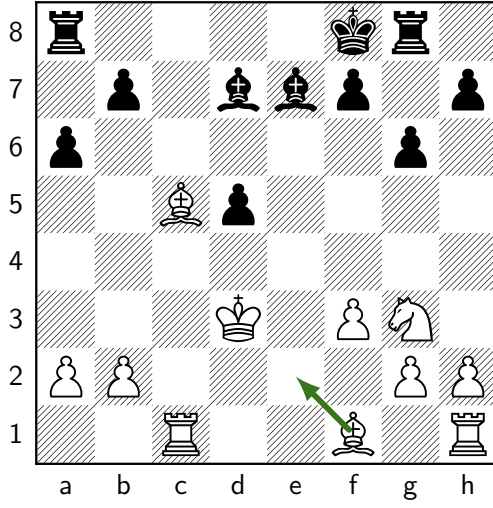


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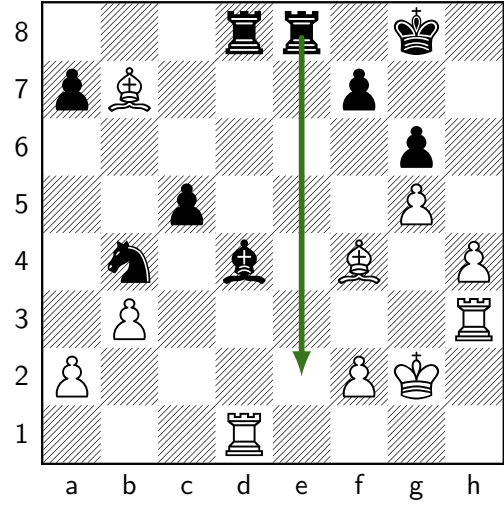


(d)

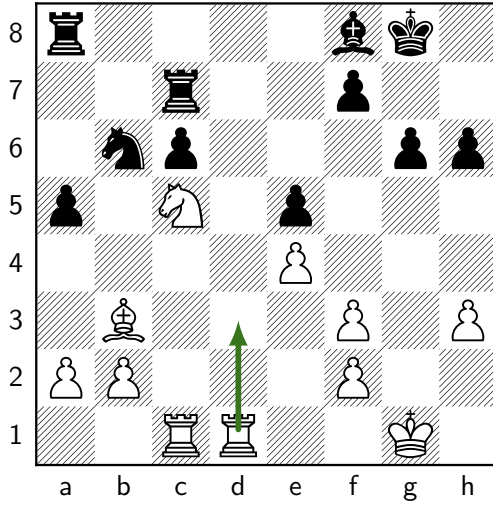
Figure 19: **Move 25, White: Rxc1.** (a) Our position; Melanie plays Rxc1 (b1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2551 (ply 37) [structural distance $d=103$, QAP obj=77]: the master played Bxe8; under μ our Rb1 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our bc8 as their Nc6, our pg6 as their rd5; transferred move b1c1 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1331 (ply 35) [structural distance $d=102$, QAP obj=60]: the master played Bxd7; under μ our Rb1 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Pf7, our Pf3 as their Bg7; transferred move b1c1 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3161 (ply 42) [structural distance $d=83$, QAP obj=47]: the master played bxc3; under μ our Rb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our pf7 as their Ra1, our pg6 as their Rh1; transferred move b1c1 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Rxc1 (summed QAP objective 476, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



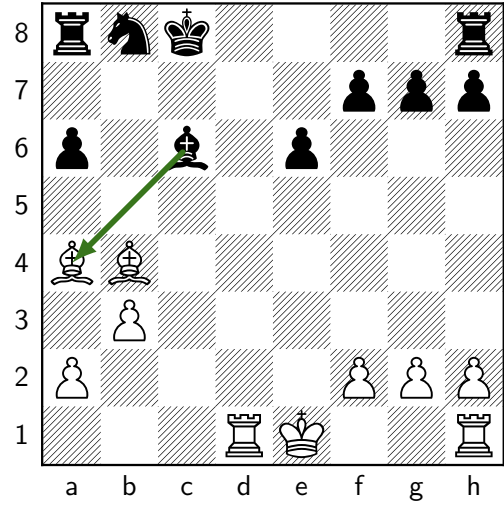
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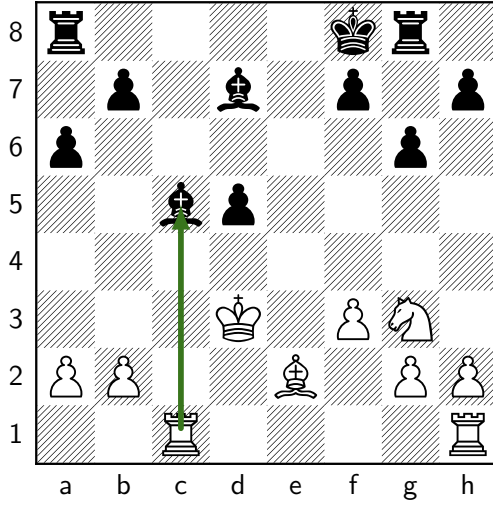


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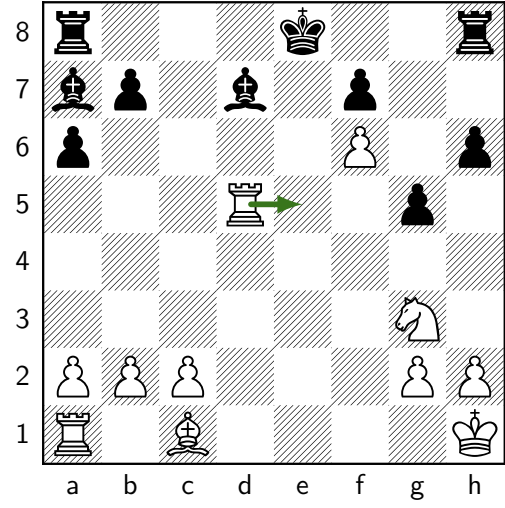


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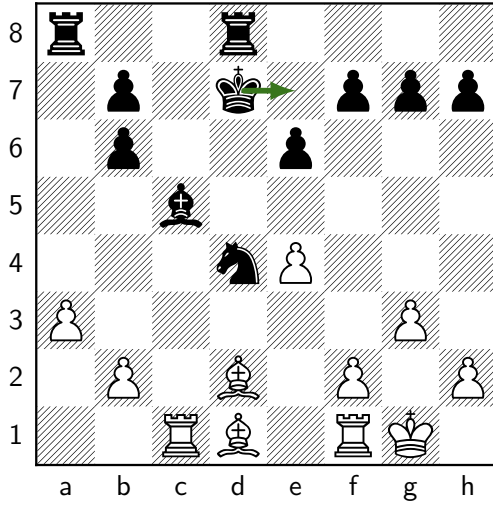
Figure 20: **Move 26, White: Be2.** (a) Our position; Melanie plays Be2 (**f1e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1018` (ply 65) [structural distance $d=85$, QAP obj=30]: the master played Re2; under μ our *Bf1* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *pb7* as their *Rd8*, our *pg6* as their *bb7*; transferred move **f1e2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#2134` (ply 42) [structural distance $d=85$, QAP obj=14]: the master played Rd3; under μ our *Bf1* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Rh1* as their *Pb2*, our *Ng3* as their *Bb3*; transferred move **f1e2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#3230` (ply 41) [structural distance $d=74$, QAP obj=41]: the master played Bxa4; under μ our *Bc5* plays the role of that game’s *Bc3*; cross-type roles: our *Pf3* as their *Ra8*, our *Rc1* as their *Nb8*; transferred move **c5b4**. Synthesis: 2 retrieved analogs transferred to Be2 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



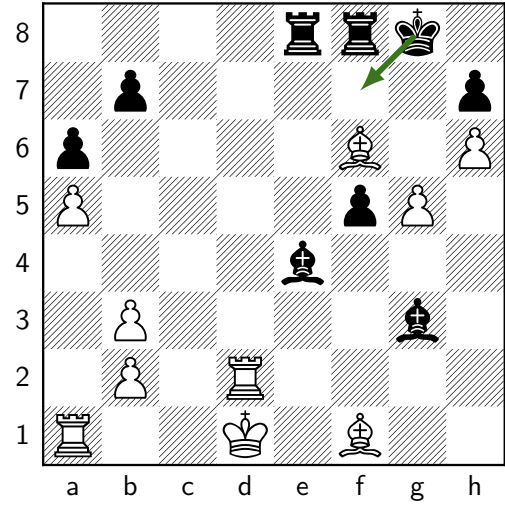
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(b)

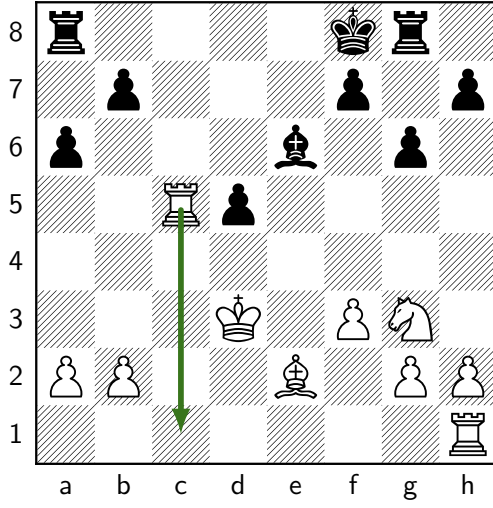


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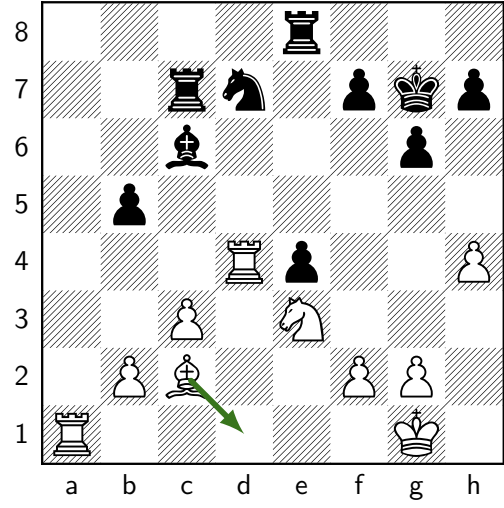


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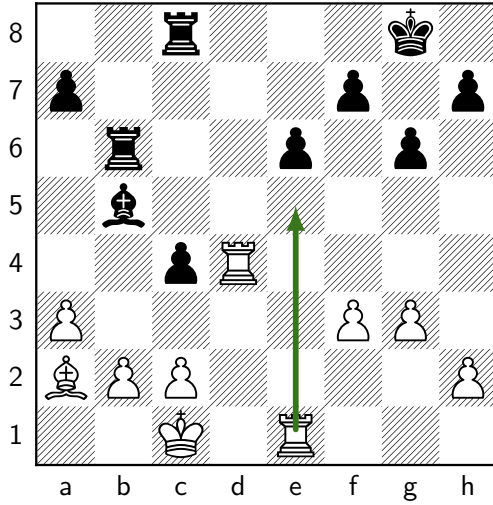
Figure 21: **Move 27, White: Rxc5.** (a) Our position; Melanie plays Rxc5 (c1c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2870` (ply 40) [structural distance $d=67$, QAP obj=12]: the master played Re5+; under μ our Rc1 plays the role of that game’s Rd5; cross-type roles: our Pf3 as their Bc1, our Be2 as their Pb2; transferred move c1c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3754` (ply 37) [structural distance $d=54$, QAP obj=22]: the master played Ke7; under μ our Kd3 plays the role of that game’s Kd2; cross-type roles: our pg6 as their Ra8, our Be2 as their Rd8; transferred move d3c2. (d) Retrieved from `ccr1_top_p0003#112` (ply 69) [structural distance $d=57$, QAP obj=14]: the master played Kf7; under μ our Kd3 plays the role of that game’s Kg1; cross-type roles: our Ng3 as their Pb7, our bd7 as their Ph7; transferred move d3c2. Synthesis: 1 retrieved analog transferred to Rxc5 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



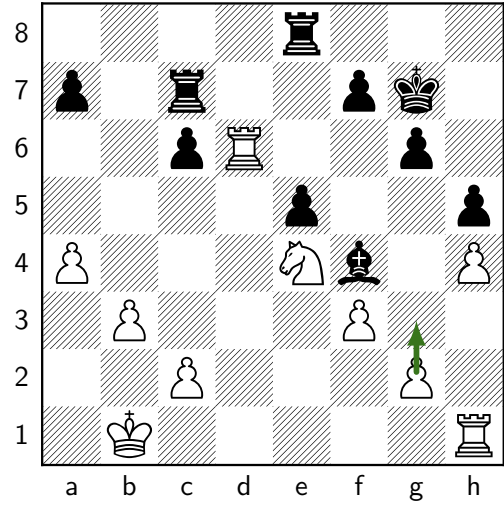
(a)



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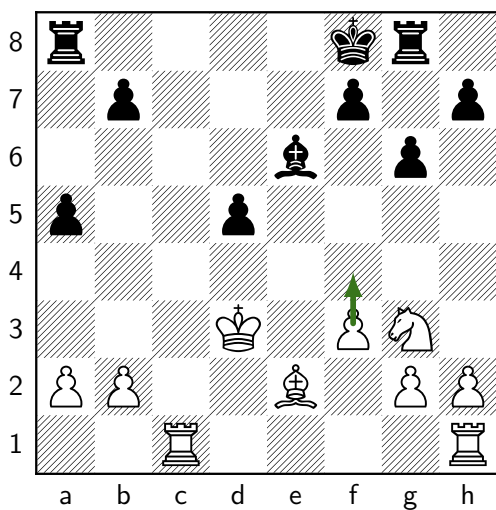


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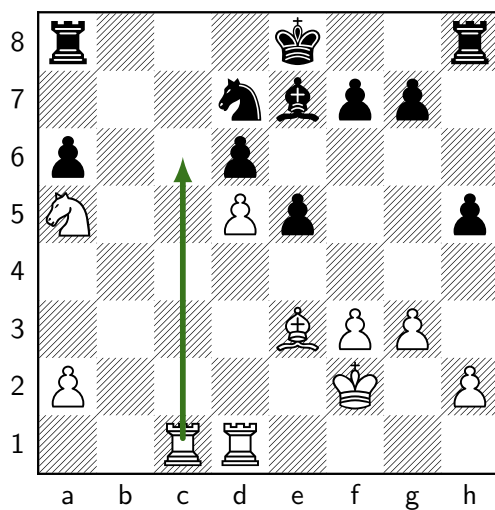


(d)

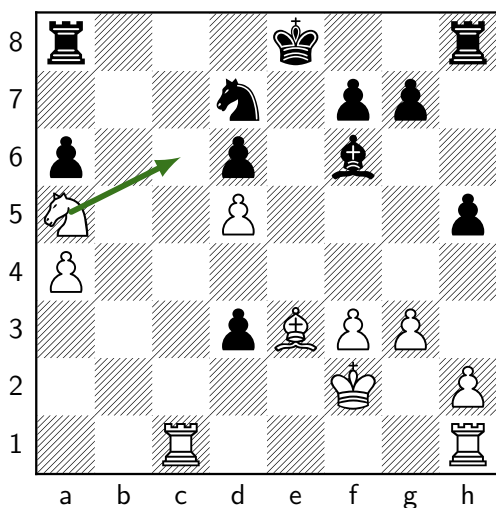
Figure 22: **Move 28, White: Rcc1.** (a) Our position; Melanie plays Rcc1 (c5c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#152 (ply 54) [structural distance $d=65$, QAP obj=41]: the master played Bd1; under μ our Rc5 plays the role of that game’s Bc2 (a **cross-type** correspondence); cross-type roles: our Be2 as their Ne3, our Ng3 as their Ph4; transferred move c5c1 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2342 (ply 52) [structural distance $d=61$, QAP obj=29]: the master played Re5; under μ our Be2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Rc5 as their Ba2, our Rh1 as their Pc2; transferred move e2d1. (d) Retrieved from ccr1_top_p0002#1376 (ply 54) [structural distance $d=64$, QAP obj=22]: the master played g3; under μ our Ng3 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pf7 as their Rh1, our Be2 as their Pf3; transferred move g3h5. Synthesis: 1 retrieved analog transferred to Rcc1 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



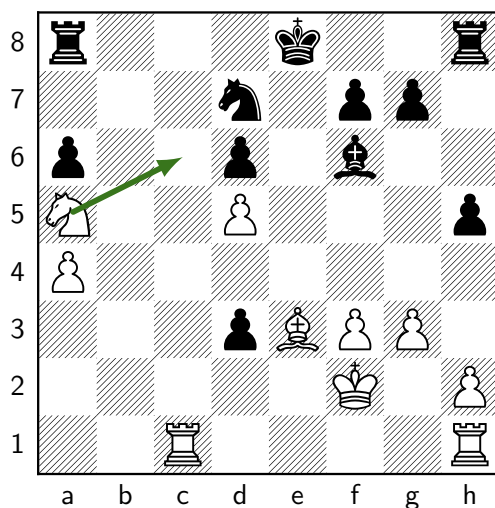
(a)



(b)

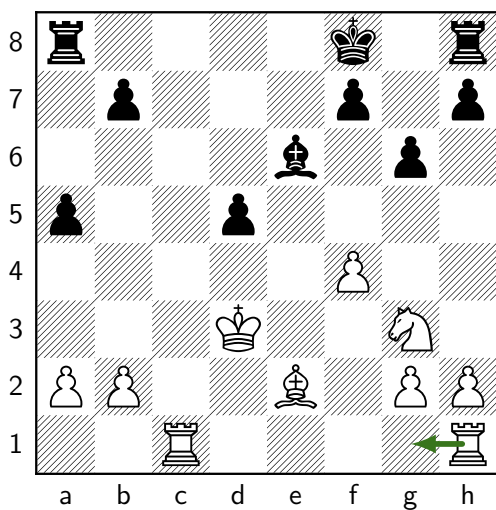


(c)

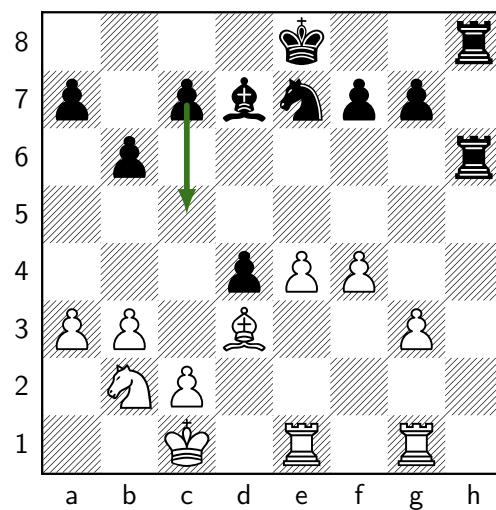


(d)

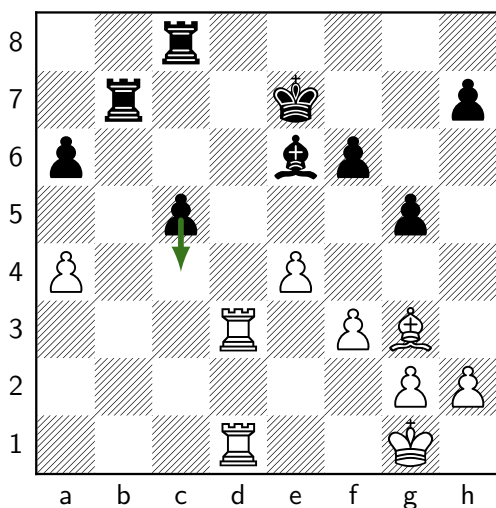
Figure 23: **Move 29, White: f4**. (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2918` (ply 42) [structural distance $d=58$, QAP obj=3]: the master played Rc6; under μ our Pf3 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pa2, our Ng3 as their Ph2; transferred move f3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2586` (ply 46) [structural distance $d=62$, QAP obj=3]: the master played Nc6; under μ our Pf3 plays the role of that game’s Na5 (a **cross-type** correspondence); cross-type roles: our be6 as their pa6, our rg8 as their bf6; transferred move f3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2316` (ply 46) [structural distance $d=62$, QAP obj=3]: the master played Nc6; under μ our Pf3 plays the role of that game’s Na5 (a **cross-type** correspondence); cross-type roles: our be6 as their pa6, our rg8 as their bf6; transferred move f3f4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f4 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



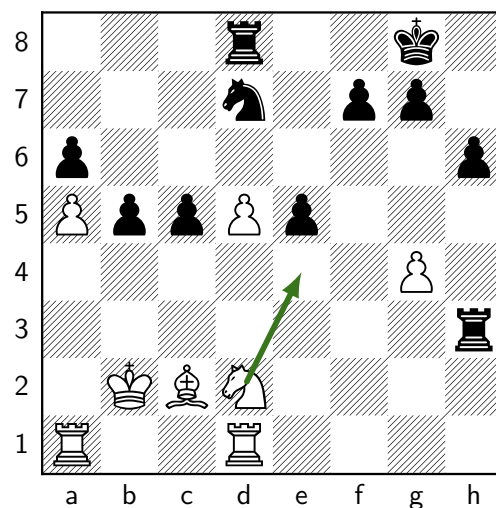
(a)



(b)

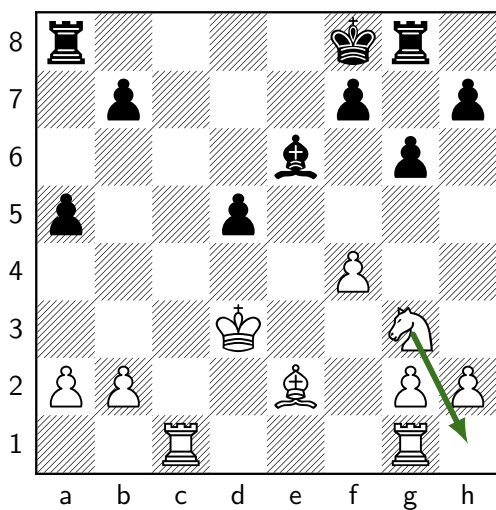


(c)

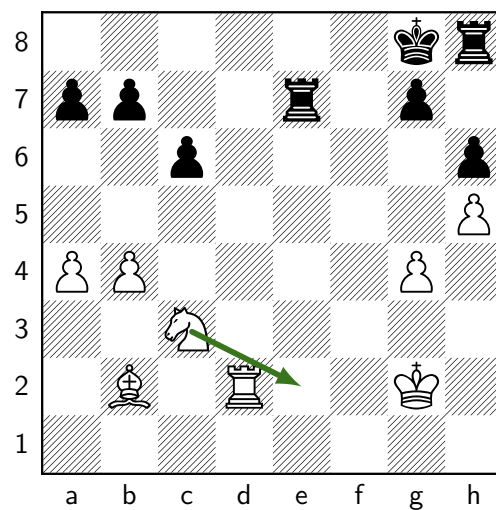


(d)

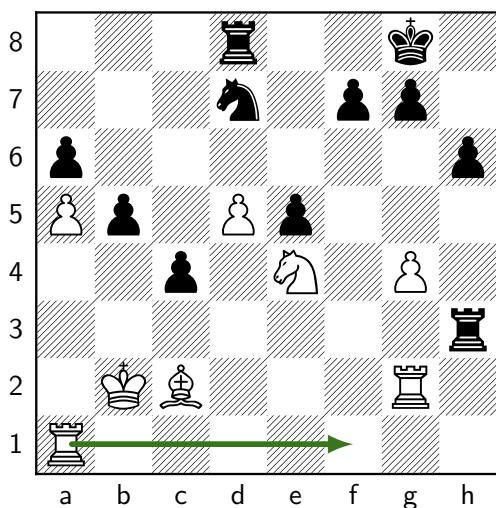
Figure 24: **Move 30, White: Rhg1.** (a) Our position; Melanie plays Rhg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1812` (ply 45) [structural distance $d=72$, QAP obj=3]: the master played c5; under μ our *Rh1* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *Pa2* as their *Ne7*, our *Ng3* as their *Pg7*; transferred move h1g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#367` (ply 63) [structural distance $d=66$, QAP obj=3]: the master played c4; under μ our *Rh1* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *be6* as their *Rc8*, our *Be2* as their *Rb7*; transferred move h1g1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1570` (ply 66) [structural distance $d=66$, QAP obj=2]: the master played Ne4; under μ our *Rh1* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *pg6* as their *Ra1*, our *pd5* as their *rh3*; transferred move h1g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rhg1 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



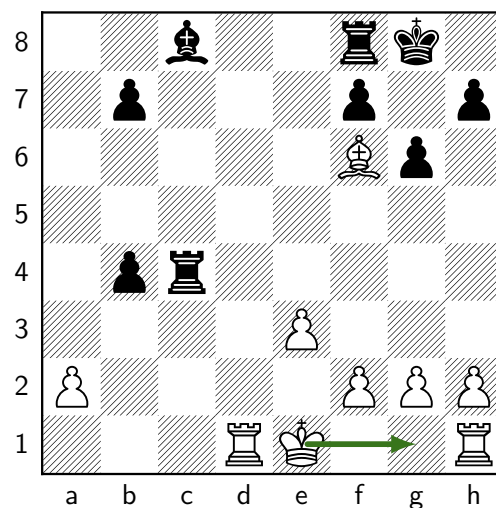
(a)



(b)

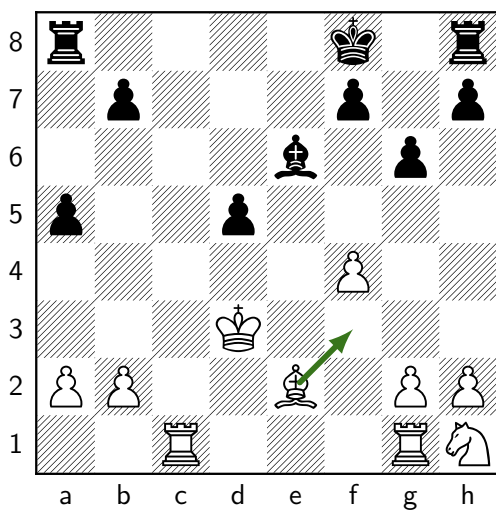


(c)

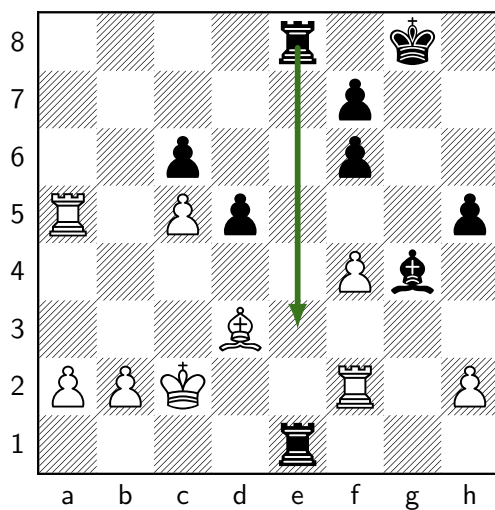


(d)

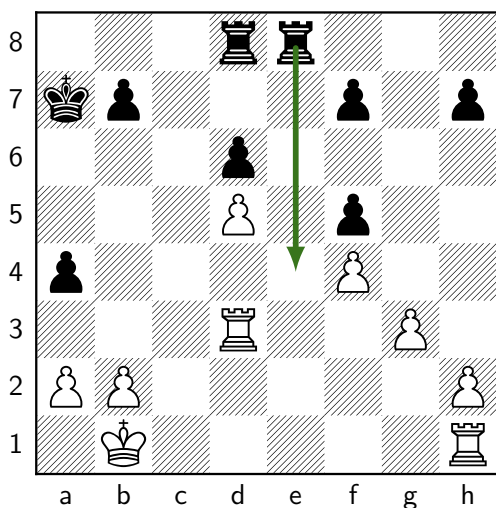
Figure 25: **Move 31, White: Nh1 (g3h1).** (a) Our position; Melanie plays Nh1 (g3h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2682` (ply 70) [structural distance $d=64$, QAP obj=4]: the master played Ne2; under μ our Ng3 plays the role of that game’s Nc3; cross-type roles: our Pg2 as their Bb2, our Be2 as their Rd2; transferred move g3h1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1570` (ply 72) [structural distance $d=57$, QAP obj=2]: the master played Rf1; under μ our Ng3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Bc2, our Pa2 as their rh3; transferred move g3h1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#339` (ply 46) [structural distance $d=62$, QAP obj=5]: the master played 0-0; under μ our Kd3 plays the role of that game’s Ke1; cross-type roles: our Be2 as their Rd1, our be6 as their Rh1; transferred move d3c2. Synthesis: 2 retrieved analogs transferred to Nh1 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



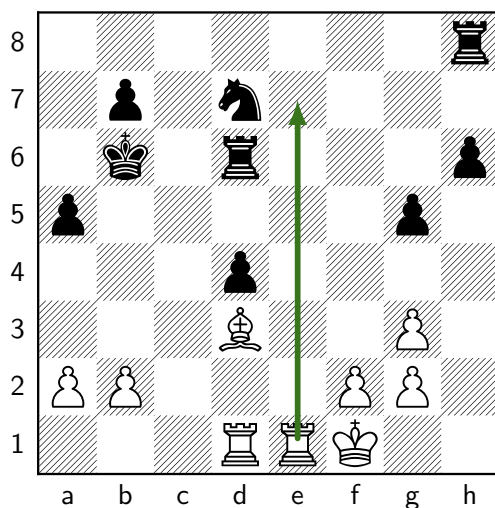
(a)



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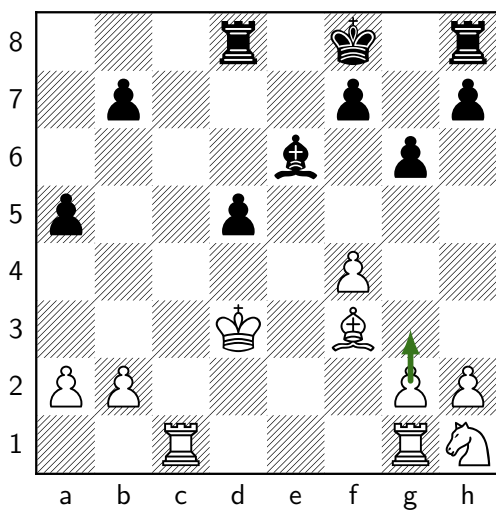


(c)

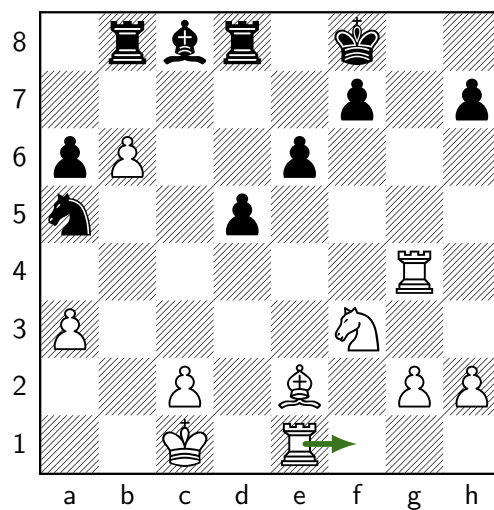


(d)

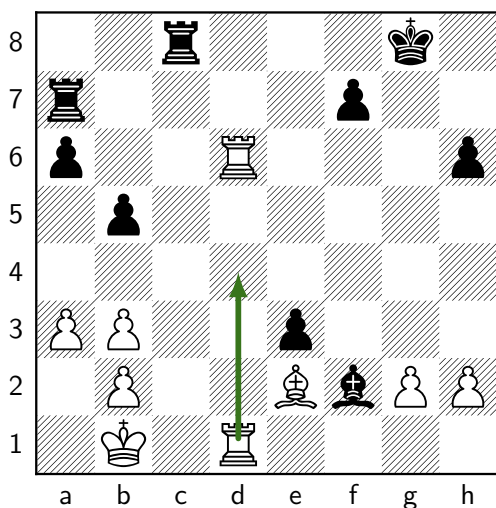
Figure 26: **Move 32, White: Bf3.** (a) Our position; Melanie plays Bf3 (e2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#66` (ply 55) [structural distance $d=65$, QAP obj=3]: the master played R8e3; under μ our *Be2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rc1* as their *Pc6*, our *Nh1* as their *Pd5*; transferred move e2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#802` (ply 59) [structural distance $d=66$, QAP obj=1]: the master played Re4; under μ our *Be2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Nh1* as their *Pf7*, our *Rc1* as their *pd5*; transferred move e2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2469` (ply 52) [structural distance $d=65$, QAP obj=4]: the master played Re7; under μ our *Be2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rg1* as their *Pa2*, our *pd5* as their *Bd3*; transferred move e2h5. Synthesis: 2 retrieved analogs transferred to Bf3 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



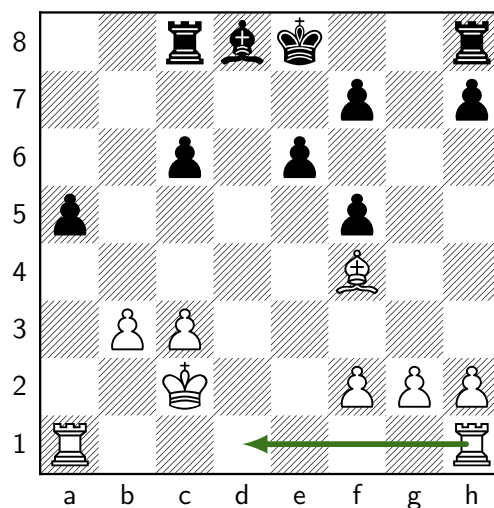
(a)



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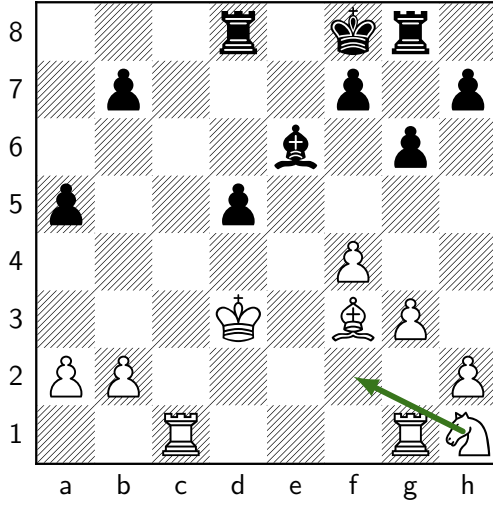


(c)

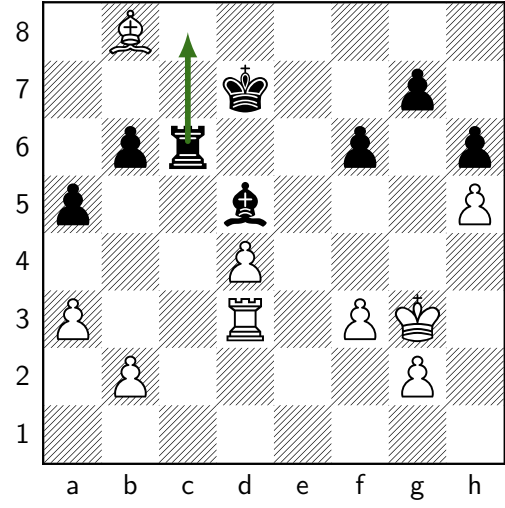


(d)

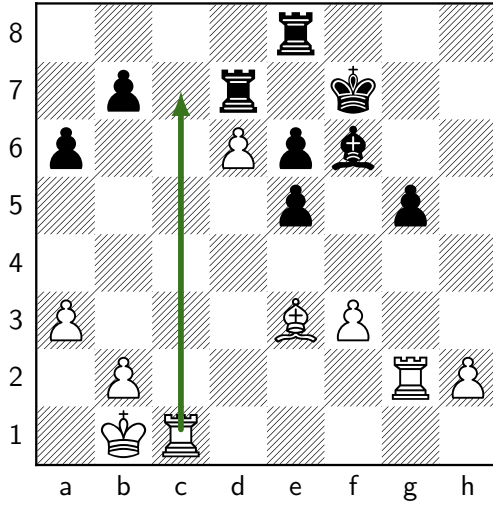
Figure 27: **Move 33, White: g3**. (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1328` (ply 58) [structural distance $d=48$, QAP obj=22]: the master played Rf1; under μ our Pg2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Pg2, our Rg1 as their Ph2; transferred move g2g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1703` (ply 58) [structural distance $d=54$, QAP obj=22]: the master played R1d4; under μ our Pg2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Be2, our Pf4 as their bf2; transferred move g2g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#375` (ply 48) [structural distance $d=49$, QAP obj=12]: the master played Rh1; under μ our Pg2 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Ra1, our Rg1 as their Pf2; transferred move g2g3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g3 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



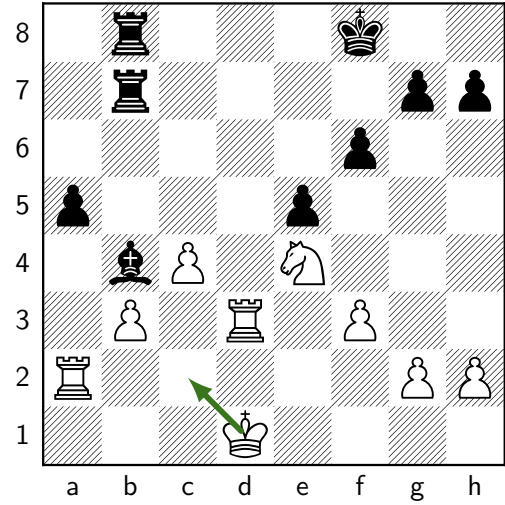
(a)



(b)

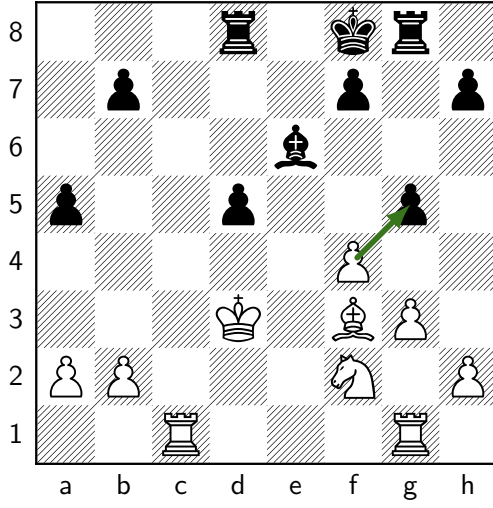


(c)

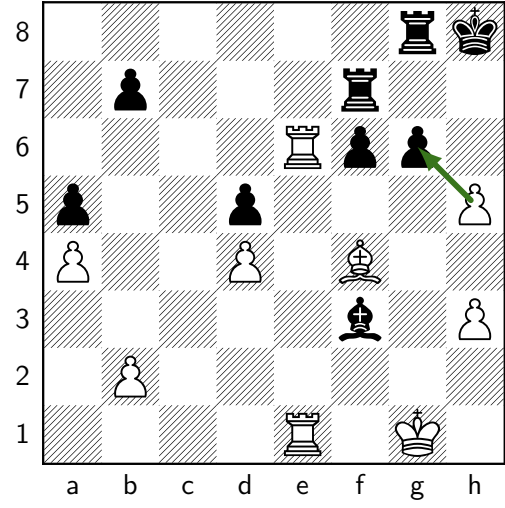


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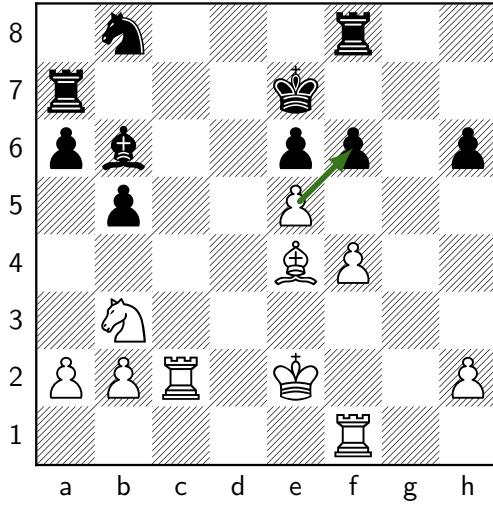
Figure 28: **Move 34, White: Nf2**. (a) Our position; Melanie plays Nf2 (h1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1311` (ply 83) [structural distance $d=56$, QAP obj=22]: the master played Rc8; under μ our Nh1 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Pf4 as their bb8, our Rc1 as their Ph6; transferred move h1f2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1602` (ply 66) [structural distance $d=57$, QAP obj=16]: the master played Rc7; under μ our Nh1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Rg2, our Rg1 as their Ph2; transferred move h1f2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#173` (ply 62) [structural distance $d=57$, QAP obj=25]: the master played Kc2; under μ our Kd3 plays the role of that game’s Kd1; cross-type roles: our Bf3 as their Ra2, our Nh1 as their Pg2; transferred move d3d2. Synthesis: 2 retrieved analogs transferred to Nf2 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



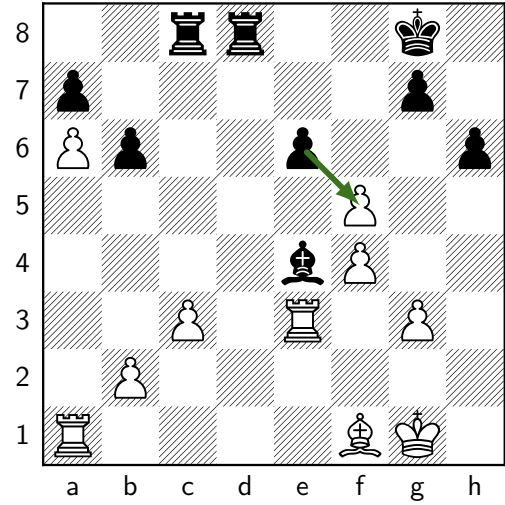
(a)



(b)

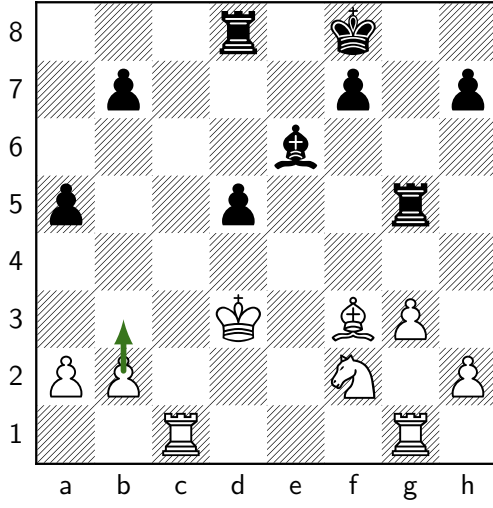


(c)

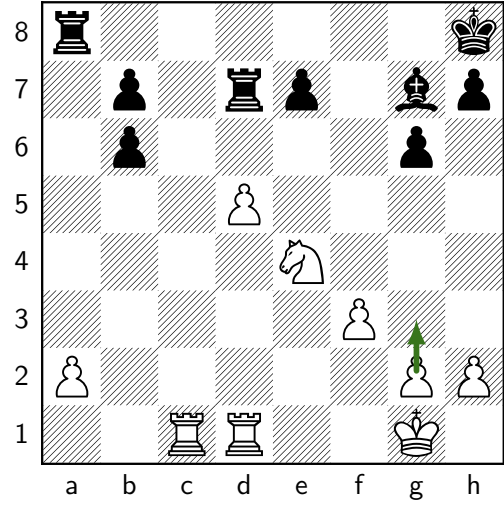


(d)

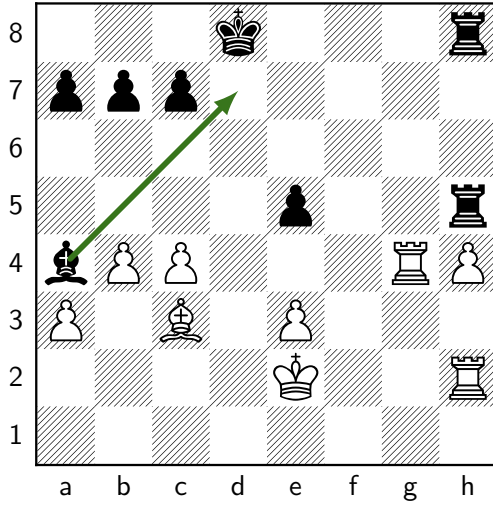
Figure 29: **Move 35, White: f4g5.** (a) Our position; Melanie plays f4g5 (f4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#463 (ply 70) [structural distance $d=44$, QAP obj=55]: the master played h5g6; under μ our Pf4 plays the role of that game’s Ph5; cross-type roles: our Nf2 as their Re1, our pg5 as their bf3; transferred move f4g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1574 (ply 52) [structural distance $d=38$, QAP obj=40]: the master played exf6+; under μ our Pf4 plays the role of that game’s Pe5; cross-type roles: our Nf2 as their Rf1, our Rc1 as their Pa2; transferred move f4g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1319 (ply 55) [structural distance $d=45$, QAP obj=18]: the master played exf5; under μ our Pf4 plays the role of that game’s Pe3; cross-type roles: our Bf3 as their Rc8, our Nf2 as their Rd8; transferred move f4g5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f4g5 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



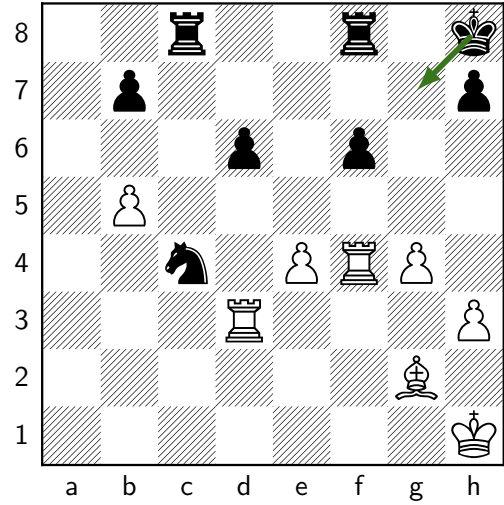
(a)



(b)



(c)



(d)

Figure 30: **Move 36, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3624` (ply 56) [structural distance $d=47$, QAP obj=47]: the master played g3; under μ our *Pb2* plays the role of that game’s *Pg2*; cross-type roles: our *be6* as their *Rc1*, our *Nf2* as their *Ph2*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1240` (ply 81) [structural distance $d=34$, QAP obj=34]: the master played Bd7; under μ our *Pb2* plays the role of that game’s *Ba5* (a **cross-type** correspondence); cross-type roles: our *Nf2* as their *Rh8*, our *be6* as their *pc4*; transferred move b2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3532` (ply 83) [structural distance $d=43$, QAP obj=55]: the master played Kg7; under μ our *Kd3* plays the role of that game’s *Kh1*; cross-type roles: our *Nf2* as their *Ph7*, our *Pb2* as their *Nc4*; transferred move d3c2. Synthesis: 2 retrieved analogs transferred to b3 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

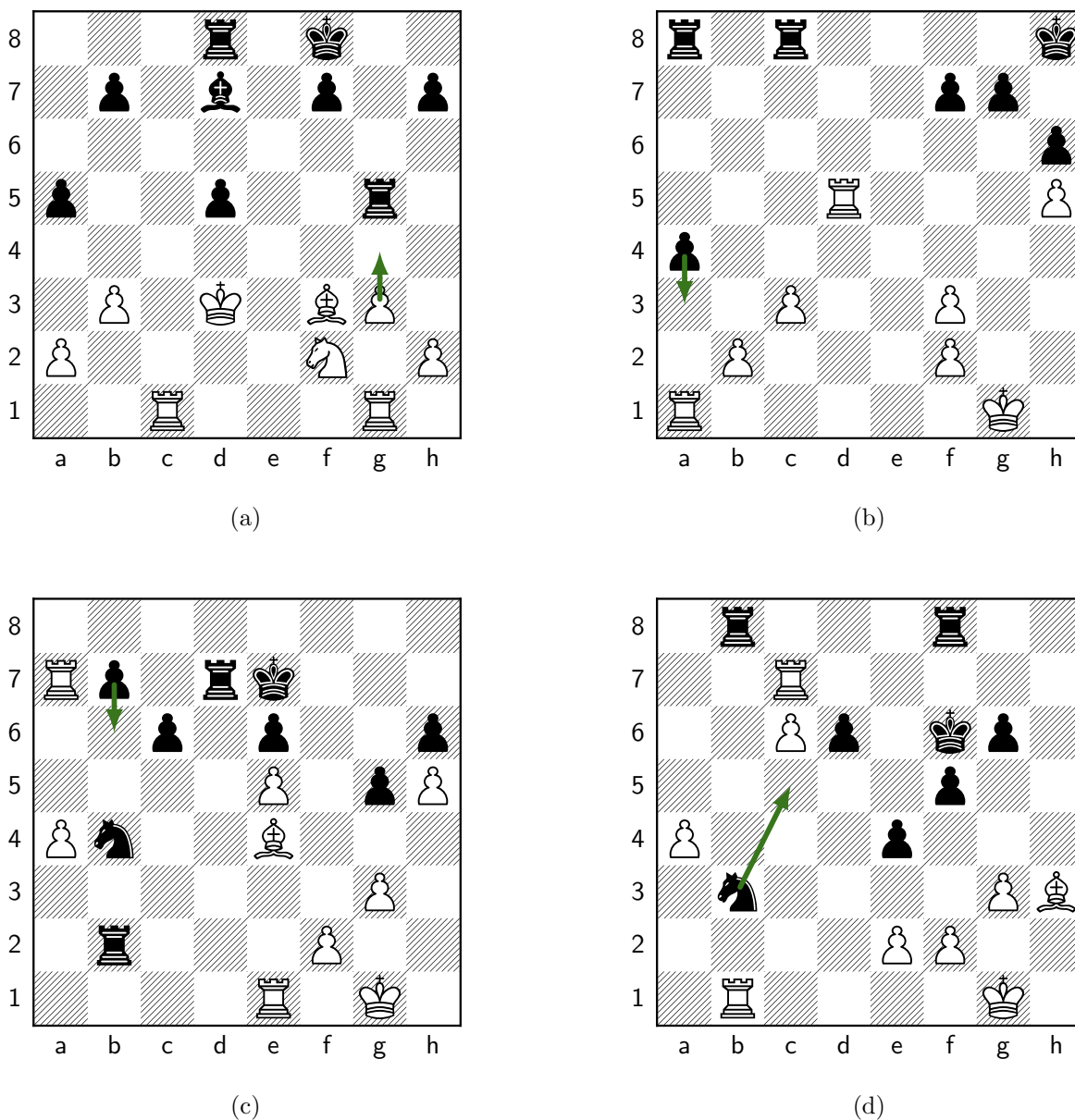


Figure 31: **Move 37, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1976` (ply 69) [structural distance $d=41$, QAP obj=35]: the master played a3; under μ our $Pg3$ plays the role of that game’s $Pa5$; cross-type roles: our $Bf3$ as their $Rc8$, our $bd7$ as their $Ph6$; transferred move g3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1317` (ply 59) [structural distance $d=39$, QAP obj=32]: the master played b6; under μ our $Pg3$ plays the role of that game’s $Pb2$; cross-type roles: our $Rc1$ as their $Ph6$, our $bd7$ as their $pe5$; transferred move g3g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2175` (ply 67) [structural distance $d=34$, QAP obj=22]: the master played Nc5; under μ our $Pg3$ plays the role of that game’s $Nb6$ (a **cross-type** correspondence); cross-type roles: our $bd7$ as their $Rb8$, our $Nf2$ as their $Rf8$; transferred move g3g4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g4 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

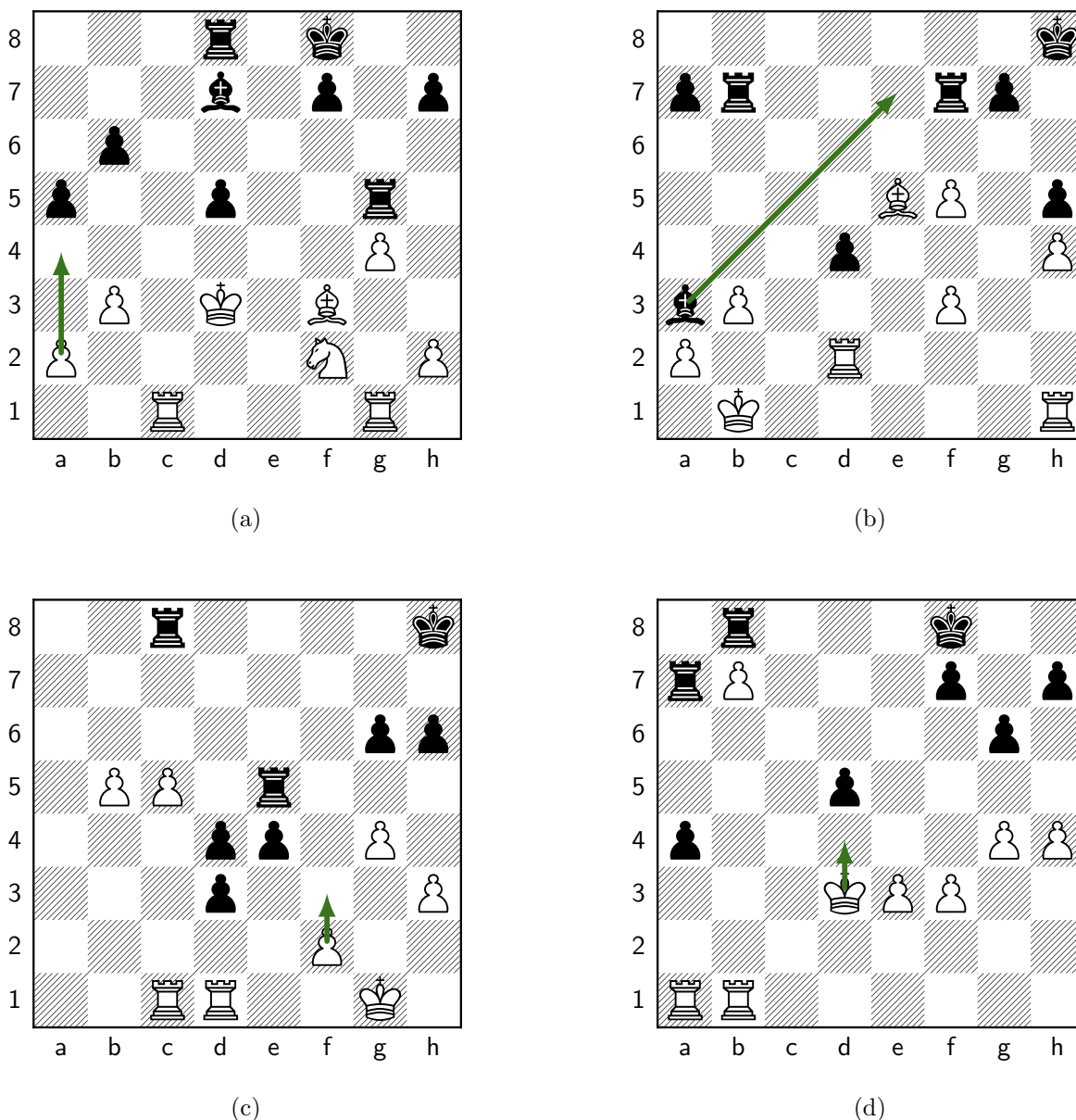
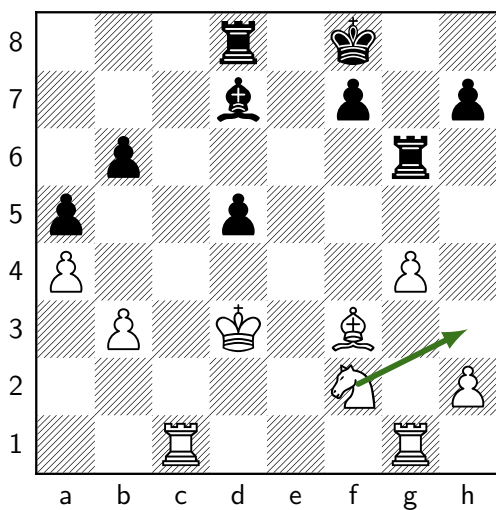
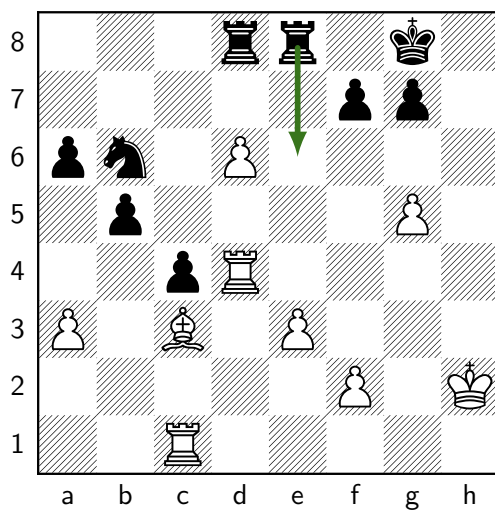


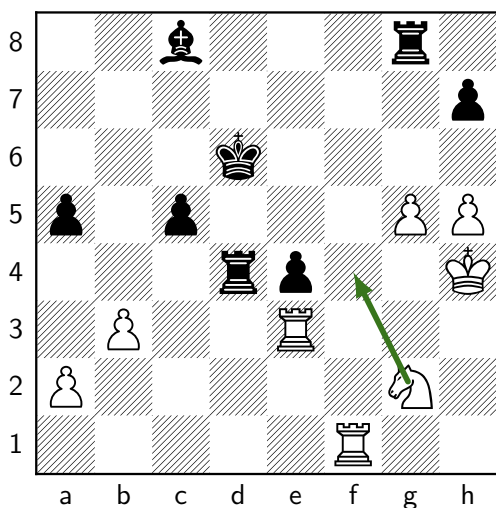
Figure 32: **Move 38, White: a4**. (a) Our position; Melanie plays **a4** (a2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#602` (ply 59) [structural distance $d=45$, QAP obj=53]: the master played **Be7**; under μ our *Pa2* plays the role of that game's *Ba6* (a **cross-type** correspondence); cross-type roles: our *Rc1* as their *Pa7*, our *Bf3* as their *Rf7*; transferred move **a2a4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2745` (ply 82) [structural distance $d=46$, QAP obj=36]: the master played **f3**; under μ our *Pa2* plays the role of that game's *Pf2*; cross-type roles: our *Bf3* as their *Rd1*, our *Nf2* as their *Pb5*; transferred move **a2a4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#536` (ply 64) [structural distance $d=45$, QAP obj=62]: the master played **Kd4**; under μ our *Kd3* plays the role of that game's *Kd3*; cross-type roles: our *Bf3* as their *Ra1*, our *Nf2* as their *Pe3*; transferred move **d3d4**. Synthesis: 2 retrieved analogs transferred to **a4** (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



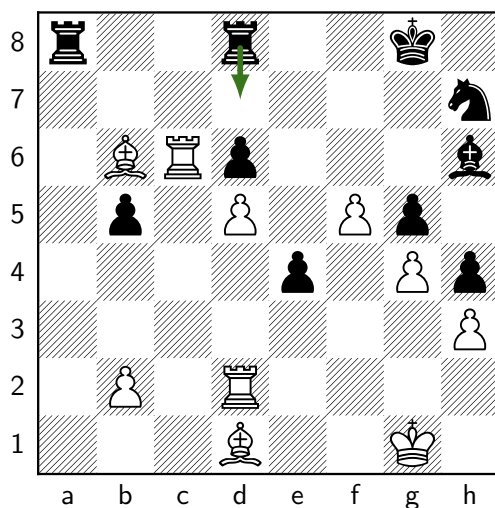
(a)



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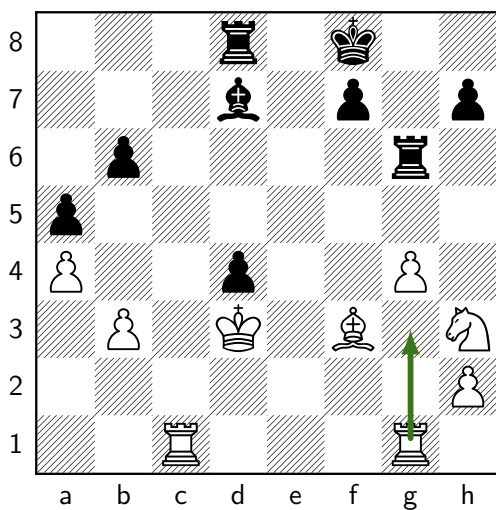


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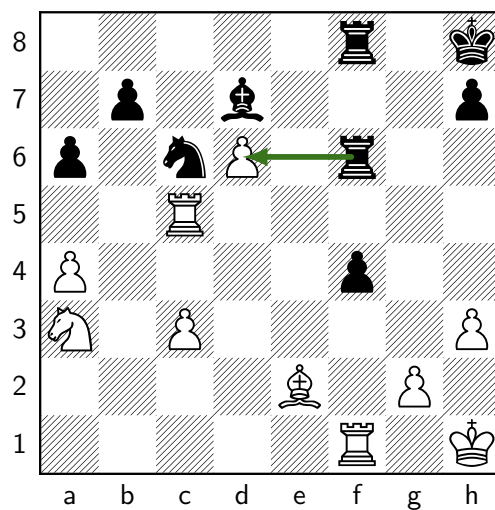


(d)

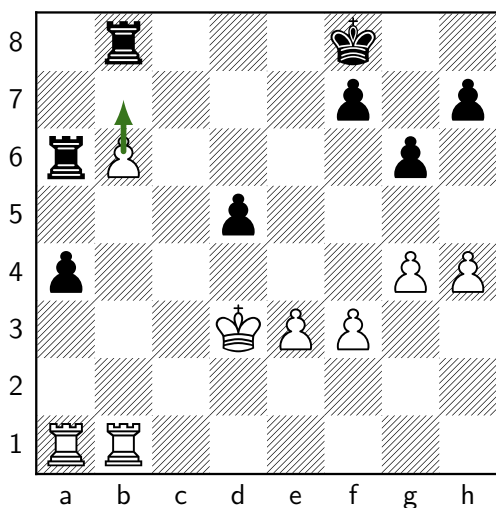
Figure 33: **Move 39, White: Nh3.** (a) Our position; Melanie plays Nh3 (f2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3001` (ply 69) [structural distance $d=53$, QAP obj=44]: the master played Re6; under μ our Nf2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Pf7, our Rg1 as their Pa6; transferred move f2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#336` (ply 88) [structural distance $d=52$, QAP obj=35]: the master played Nf4; under μ our Nf2 plays the role of that game’s Ng2; cross-type roles: our Bf3 as their Re3, our Rg1 as their pc5; transferred move f2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2311` (ply 69) [structural distance $d=51$, QAP obj=15]: the master played Rd7; under μ our Nf2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our pd5 as their Ra8, our pa5 as their Nh7; transferred move f2h3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nh3 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



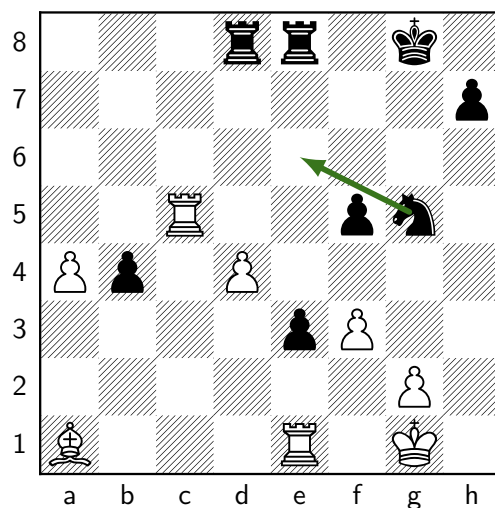
(a)



(b)



(c)



(d)

Figure 34: **Move 40, White: Rg3.** (a) Our position; Melanie plays Rg3 (g1g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1221` (ply 65) [structural distance $d=51$, QAP obj=21]: the master played Rxd6; under μ our Rg1 plays the role of that game’s Rf3; cross-type roles: our pf7 as their Rf8, our Pb3 as their Bd7; transferred move g1g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#536` (ply 62) [structural distance $d=49$, QAP obj=53]: the master played b7; under μ our Pg4 plays the role of that game’s Pb6; cross-type roles: our bd7 as their Ra1, our Rc1 as their Pe3; transferred move g4g5. (d) Retrieved from `ccr1_top_p0001#3860` (ply 77) [structural distance $d=46$, QAP obj=53]: the master played Ne6; under μ our Kd3 plays the role of that game’s Ng4 (a **cross-type** correspondence); cross-type roles: our bd7 as their rc5, our rd8 as their pg2; transferred move d3e4. Synthesis: 1 retrieved analog transferred to Rg3 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

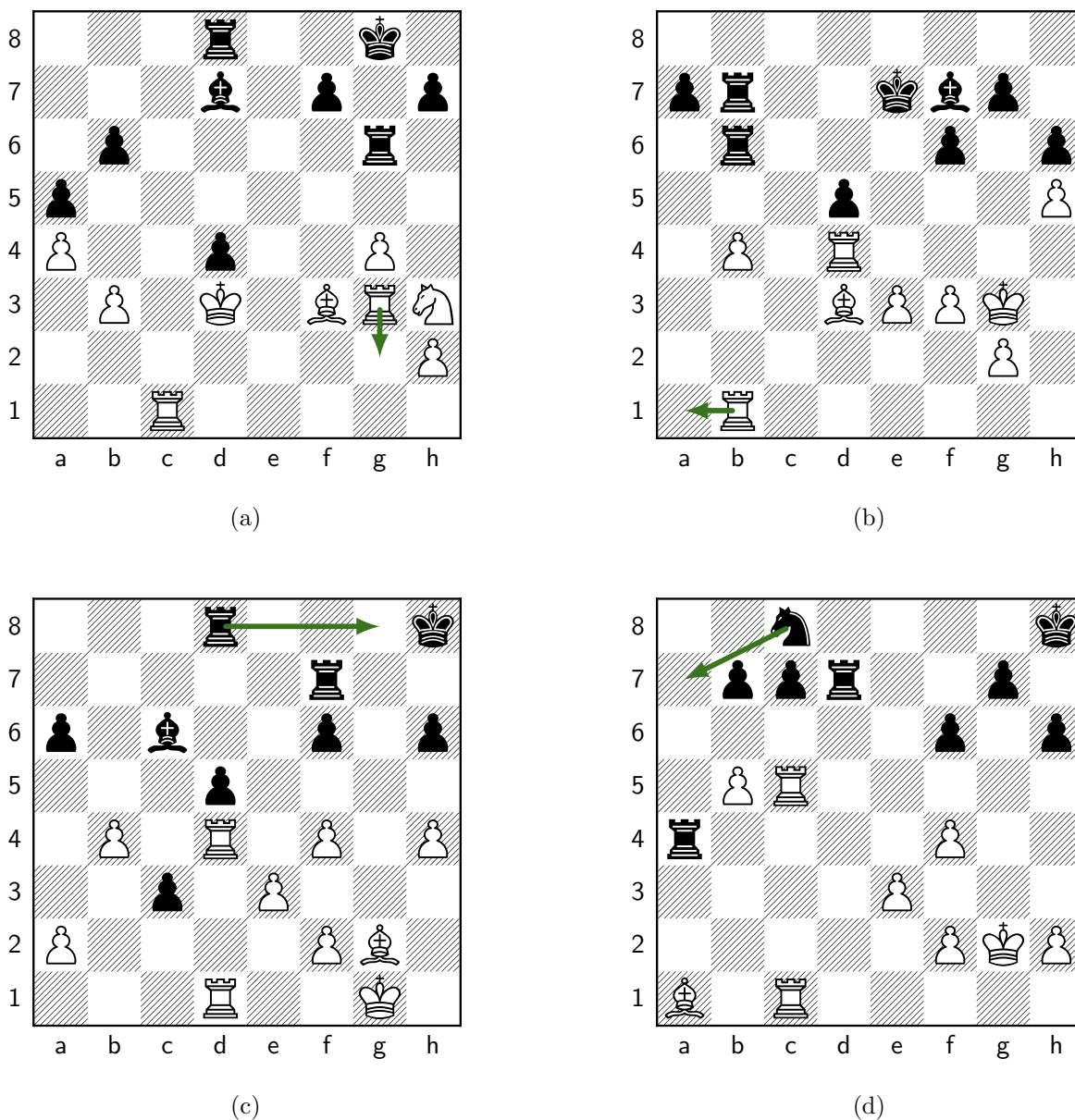
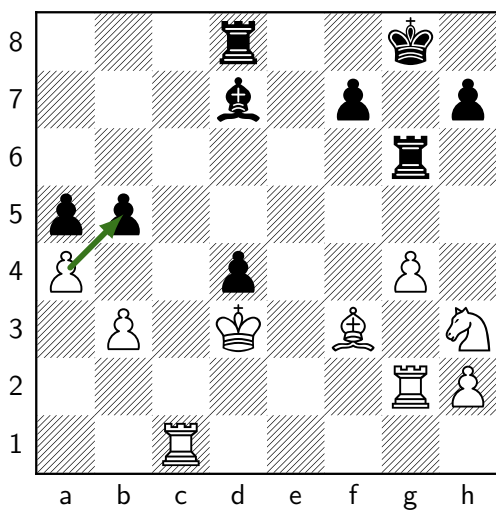
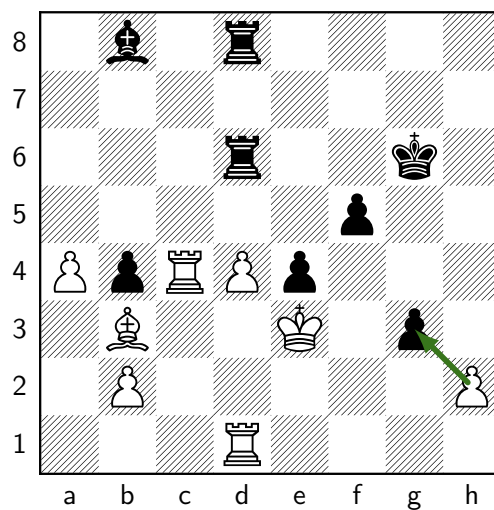


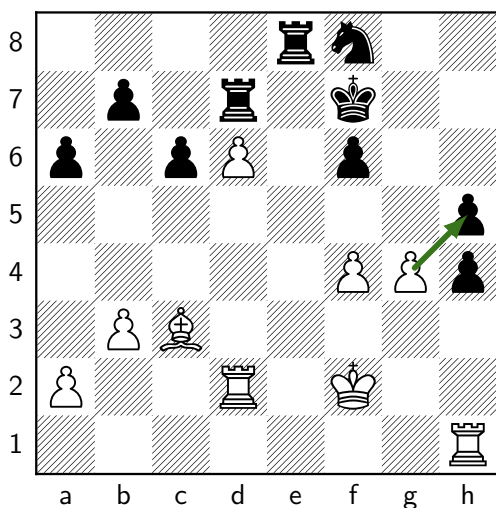
Figure 35: **Move 41, White: Rg2.** (a) Our position; Melanie plays Rg2 (g3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2145` (ply 66) [structural distance $d=58$, QAP obj=58]: the master played Ra1; under μ our Rg3 plays the role of that game’s Rb1; cross-type roles: our Rc1 as their Pe3, our Nh3 as their Pf3; transferred move g3g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1677` (ply 83) [structural distance $d=49$, QAP obj=56]: the master played Rg8; under μ our Rg3 plays the role of that game’s Rd1; cross-type roles: our Nh3 as their pb4; transferred move g3g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#917` (ply 67) [structural distance $d=59$, QAP obj=50]: the master played Na7; under μ our Rg3 plays the role of that game’s Nc1 (a **cross-type** correspondence); cross-type roles: our Nh3 as their Pb7, our Bf3 as their Rd7; transferred move g3g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rg2 (summed QAP objective 209, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



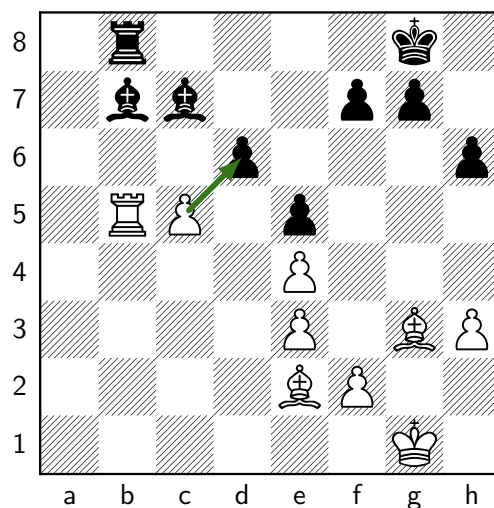
(a)



(b)

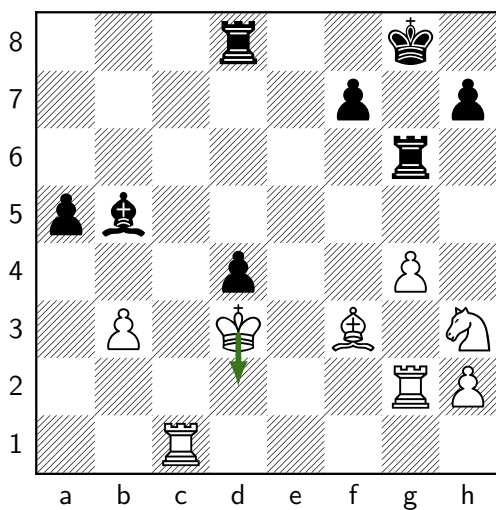


(c)

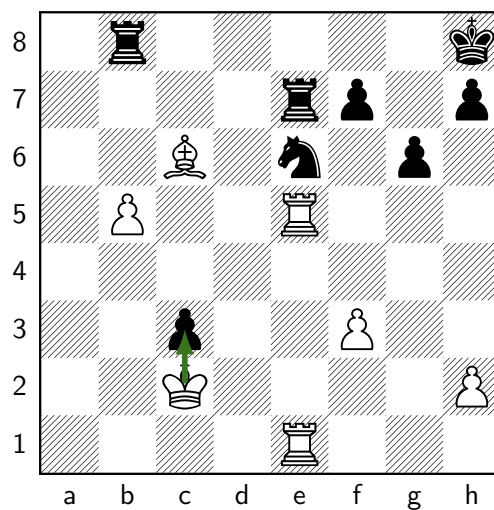


(d)

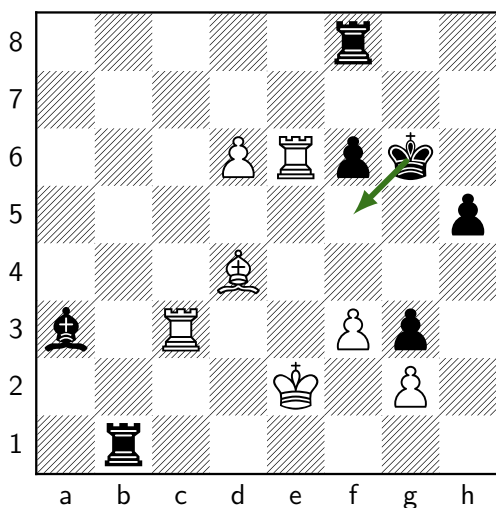
Figure 36: **Move 42, White: axb5.** (a) Our position; Melanie plays axb5 (a4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2366 (ply 94) [structural distance $d=52$, QAP obj=102]: the master played hxg3; under μ our $Pa4$ plays the role of that game’s $Ph2$; cross-type roles: our $pb5$ as their $Rc4$, our $pf7$ as their $bb8$; transferred move a4b5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1625 (ply 68) [structural distance $d=55$, QAP obj=65]: the master played gxh5; under μ our $Pa4$ plays the role of that game’s $Pg4$; cross-type roles: our $Kd3$ as their $Rh1$, our $Nh3$ as their $Bc3$; transferred move a4b5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3195 (ply 80) [structural distance $d=53$, QAP obj=42]: the master played cxd6; under μ our $Pa4$ plays the role of that game’s $Pc5$; cross-type roles: our $rd8$ as their $Be2$, our $Nh3$ as their $Pf2$; transferred move a4b5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to axb5 (summed QAP objective 276, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



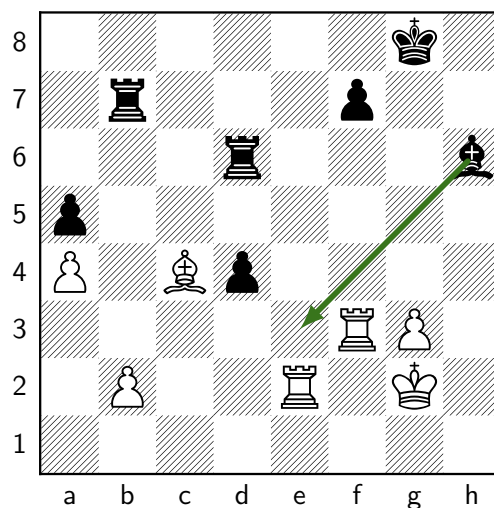
(a)



(b)

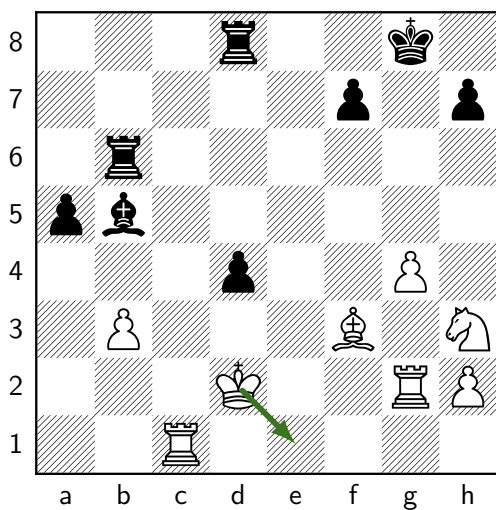


(c)

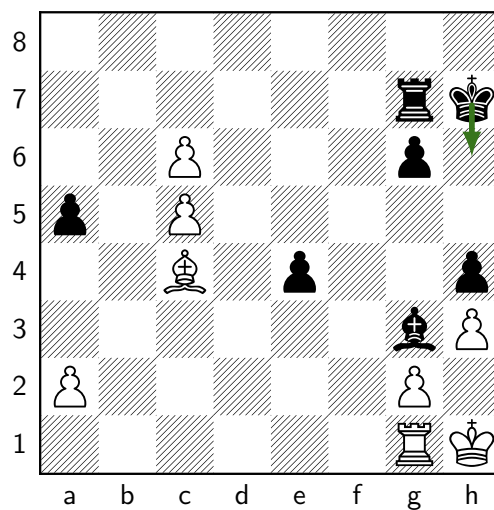


(d)

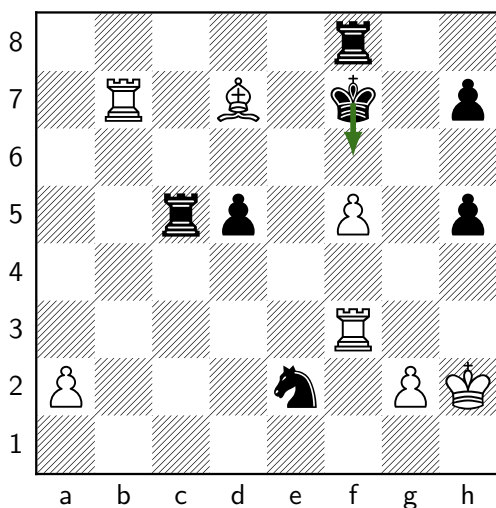
Figure 37: **Move 43, White: Kd2.** (a) Our position; Melanie plays Kd2 (d3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#138` (ply 70) [structural distance $d=45$, QAP obj=5]: the master played Kxc3; under μ our *Kd3* plays the role of that game’s *Kc2*; cross-type roles: our *pf7* as their *Re1*, our *Bf3* as their *Pf3*; transferred move d3d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1629` (ply 91) [structural distance $d=50$, QAP obj=26]: the master played Kf5; under μ our *Kd3* plays the role of that game’s *Kg3*; cross-type roles: our *Bf3* as their *Rf8*, our *Rc1* as their *bd4*; transferred move d3e4. (d) Retrieved from `ccrl_top_p0002#3380` (ply 99) [structural distance $d=49$, QAP obj=18]: the master played Be3; under μ our *Rc1* plays the role of that game’s *Bh3* (a **cross-type** correspondence); cross-type roles: our *Bf3* as their *Rd6*, our *rg6* as their *bc4*; transferred move c1c4. Synthesis: 1 retrieved analog transferred to Kd2 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



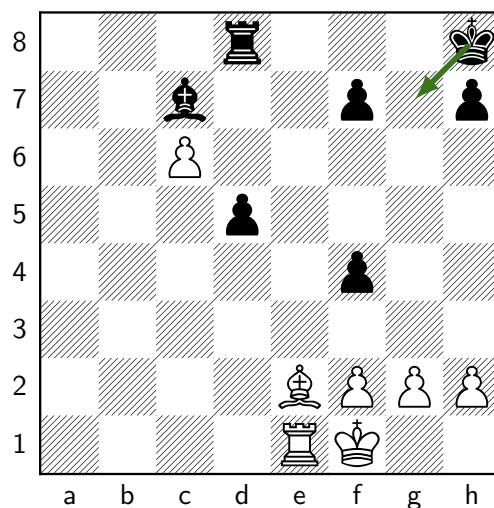
(a)



(b)



(c)



(d)

Figure 38: **Move 44, White: Ke1.** (a) Our position; Melanie plays **Ke1** (**d2e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1133` (ply 87) [structural distance $d=47$, QAP obj=2]: the master played **Kh6**; under μ our *Kd2* plays the role of that game's *Kh2*; cross-type roles: our *Rc1* as their *Pa5*, our *Bf3* as their *pc5*; transferred move **d2e1** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2781` (ply 75) [structural distance $d=39$, QAP obj=2]: the master played **Kf6**; under μ our *Kd2* plays the role of that game's *Kf2*; cross-type roles: our *bb5* as their *rb7*, our *rd8* as their *bd7*; transferred move **d2e1** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#964` (ply 71) [structural distance $d=42$, QAP obj=2]: the master played **Kg7**; under μ our *Kd2* plays the role of that game's *Kh1*; cross-type roles: our *Nh3* as their *Bc7*, our *Rc1* as their *Pf7*; transferred move **d2e1** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Ke1** (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

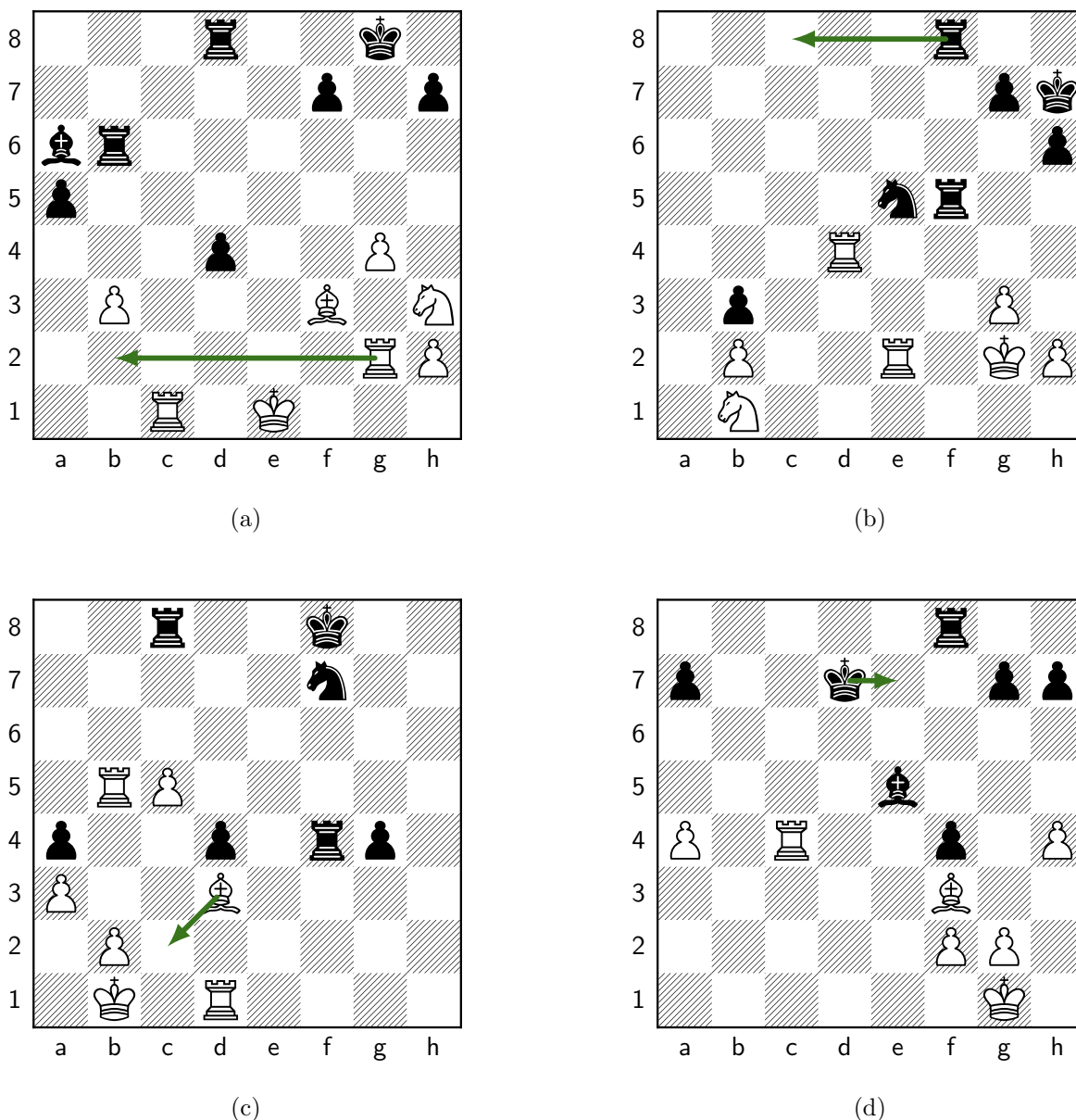
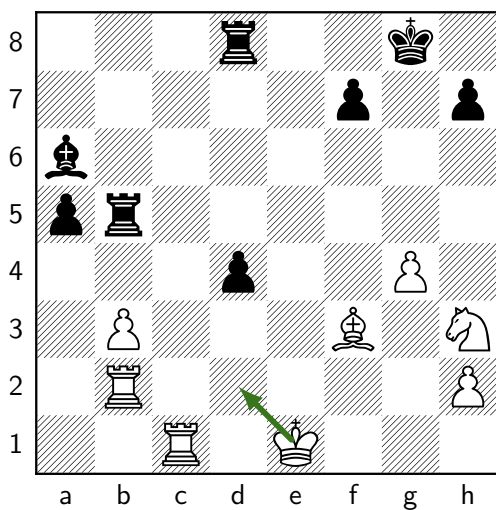
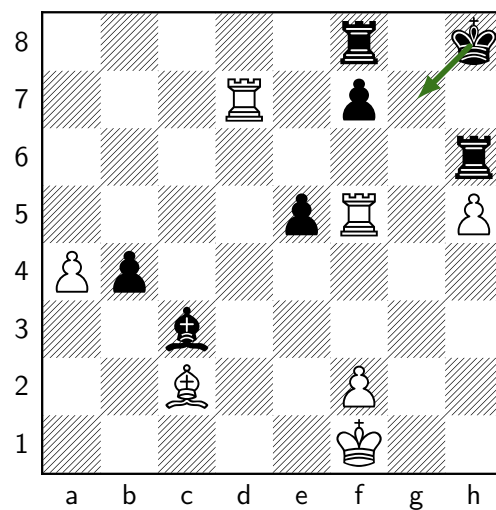


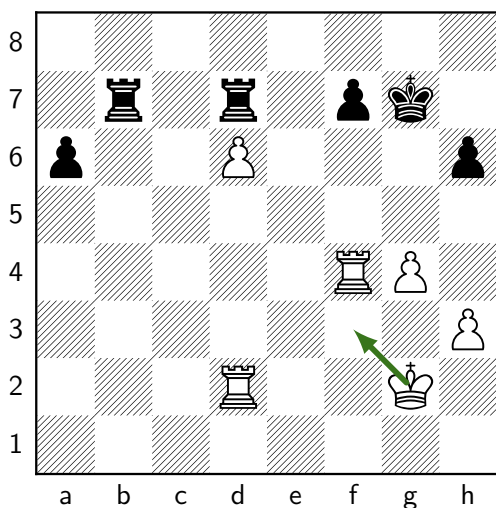
Figure 39: **Move 45, White: Rb2.** (a) Our position; Melanie plays Rb2 (g2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3063` (ply 73) [structural distance $d=46$, QAP obj=8]: the master played Rc8; under μ our Rg2 plays the role of that game’s Rf1; cross-type roles: our ba6 as their Pg7, our Rc1 as their Ph6; transferred move g2b2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1580` (ply 74) [structural distance $d=46$, QAP obj=22]: the master played Bc2; under μ our Bf3 plays the role of that game’s Bd3; cross-type roles: our ba6 as their pd4, our Rg2 as their pg4; transferred move f3e2. (d) Retrieved from `ccrl_top_p0003#1541` (ply 73) [structural distance $d=42$, QAP obj=20]: the master played Ke7; under μ our Ke1 plays the role of that game’s Kd2; cross-type roles: our Bf3 as their Rf8, our Rg2 as their Pa7; transferred move e1d2. Synthesis: 1 retrieved analog transferred to Rb2 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



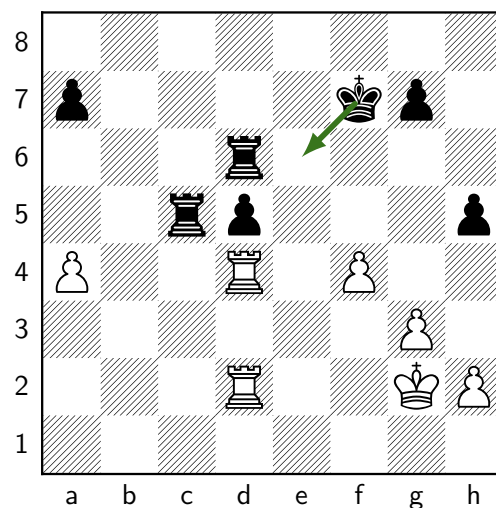
(a)



(b)

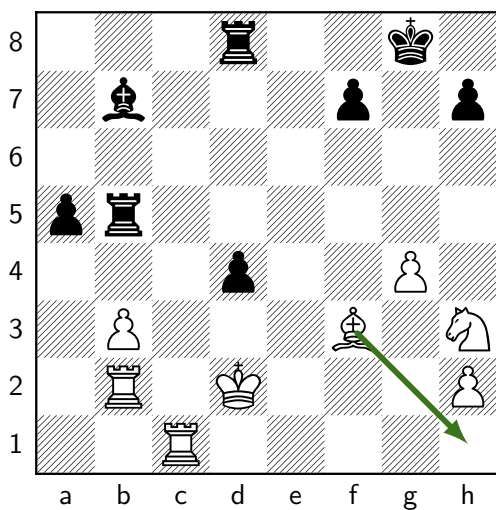


(c)

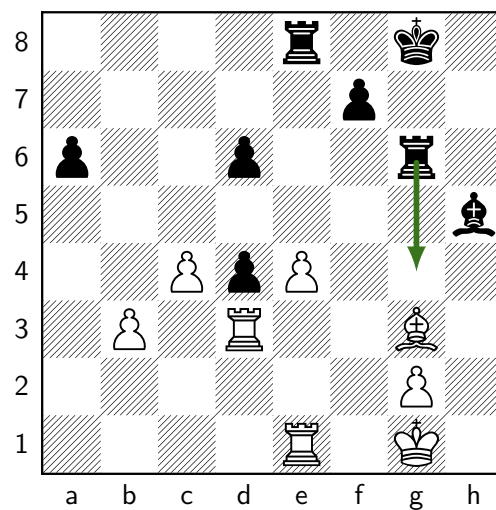


(d)

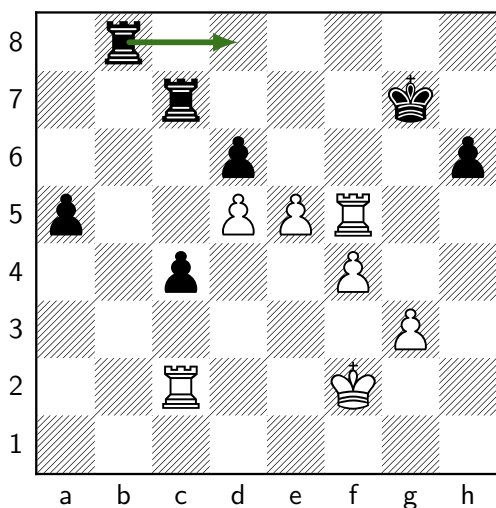
Figure 40: **Move 46, White: Kd2.** (a) Our position; Melanie plays Kd2 (e1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2025` (ply 77) [structural distance $d=43$, QAP obj=53]: the master played Kg7; under μ our *Ke1* plays the role of that game’s *Kh1*; cross-type roles: our *Bf3* as their *pa4*, our *Nh3* as their *Bc3*; transferred move e1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1039` (ply 82) [structural distance $d=47$, QAP obj=26]: the master played Kf3; under μ our *Ke1* plays the role of that game’s *Kg2*; cross-type roles: our *Nh3* as their *Pg4*, our *Bf3* as their *pa6*; transferred move e1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3768` (ply 93) [structural distance $d=45$, QAP obj=25]: the master played Ke6; under μ our *Ke1* plays the role of that game’s *Kf2*; cross-type roles: our *Rc1* as their *Pa7*, our *Rb2* as their *Pg7*; transferred move e1d2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kd2 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



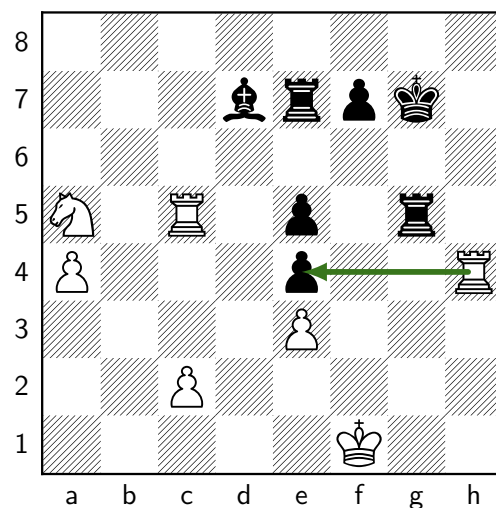
(a)



(b)

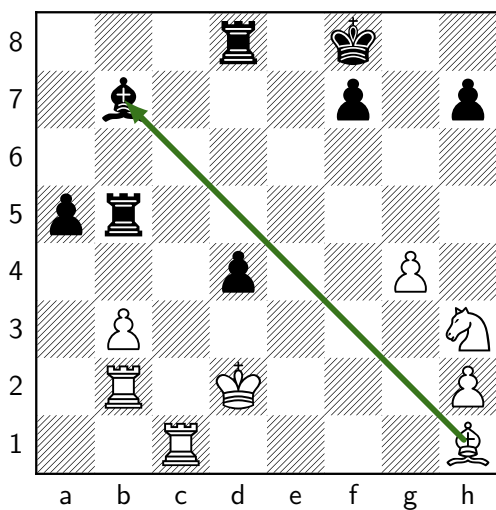


(c)

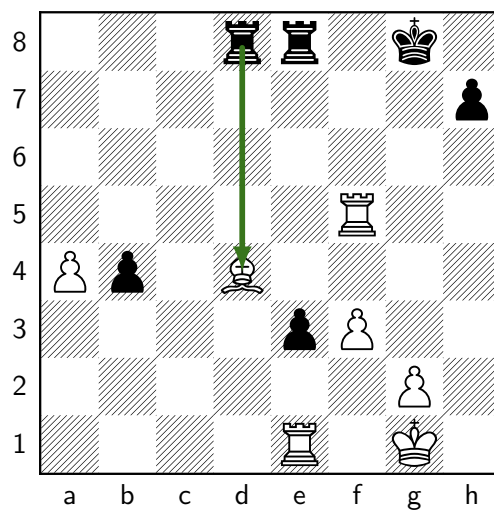


(d)

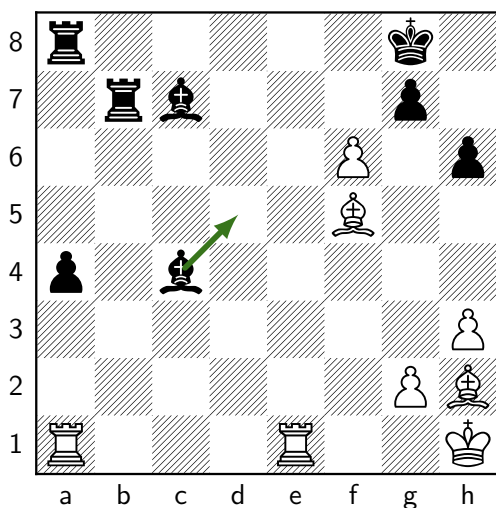
Figure 41: **Move 47, White: Bh1**. (a) Our position; Melanie plays Bh1 (f3h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#819` (ply 85) [structural distance $d=40$, QAP obj=36]: the master played Rg4; under μ our Bf3 plays the role of that game’s Rg3 (a **cross-type** correspondence); cross-type roles: our rd8 as their Pf7, our Nh3 as their Pa6; transferred move f3h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3387` (ply 107) [structural distance $d=46$, QAP obj=60]: the master played Rd8; under μ our Rc1 plays the role of that game’s Rb1; cross-type roles: our Bf3 as their Pd6, our Nh3 as their pd5; transferred move c1g1. (d) Retrieved from `ccr1_top_p0002#2491` (ply 96) [structural distance $d=50$, QAP obj=45]: the master played Rxe4; under μ our Rc1 plays the role of that game’s Rh4; cross-type roles: our Nh3 as their Pe3, our bb7 as their Pa4; transferred move c1c4. Synthesis: 1 retrieved analog transferred to Bh1 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



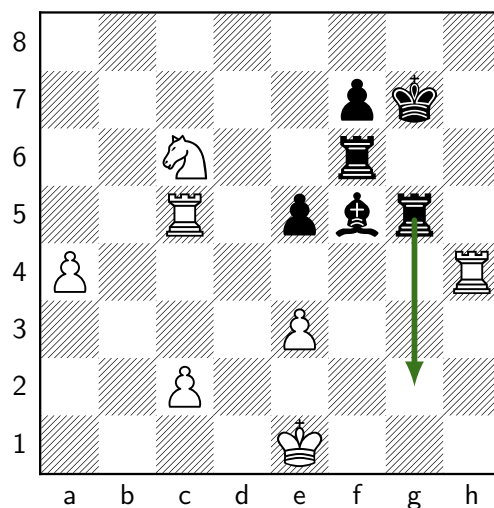
(a)



(b)

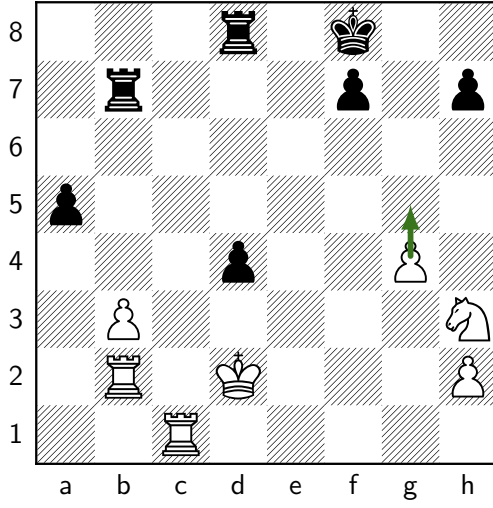


(c)

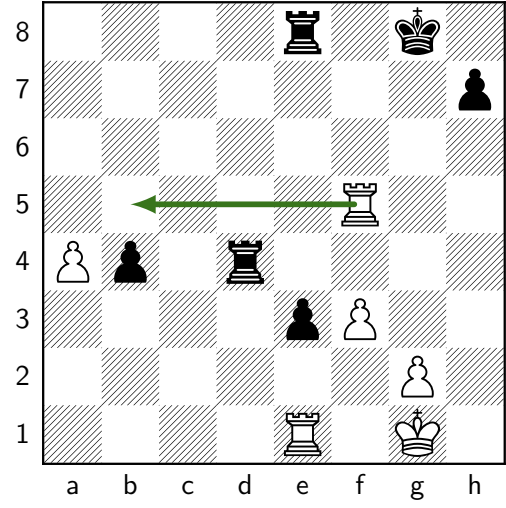


(d)

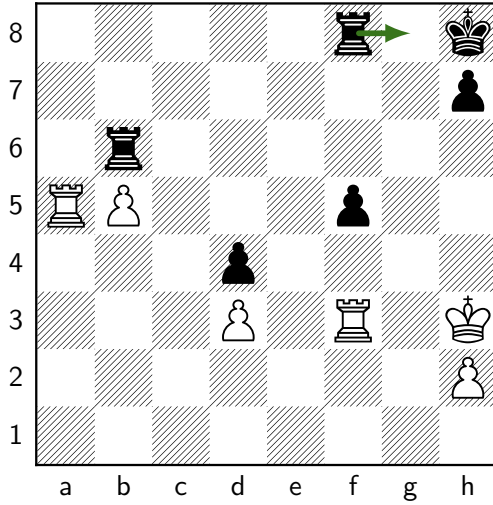
Figure 42: **Move 48, White: Bxb7**. (a) Our position; Melanie plays Bxb7 (h1b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3860` (ply 81) [structural distance $d=48$, QAP obj=42]: the master played Rxd4; under μ our *Bh1* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *pd4* as their *rf5*, our *Rc1* as their *Pb4*; transferred move h1b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#253` (ply 79) [structural distance $d=49$, QAP obj=50]: the master played Bd5; under μ our *Rb2* plays the role of that game’s *Bc5* (a **cross-type** correspondence); cross-type roles: our *pf7* as their *Ra8*, our *pd4* as their *Rb7*; transferred move b2a2. (d) Retrieved from `ccr1_top_p0002#2491` (ply 103) [structural distance $d=52$, QAP obj=46]: the master played Rg2; under μ our *Rb2* plays the role of that game’s *Rg4*; cross-type roles: our *rb5* as their *nc6*, our *bb7* as their *pc2*; transferred move b2a2. Synthesis: 1 retrieved analog transferred to Bxb7 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



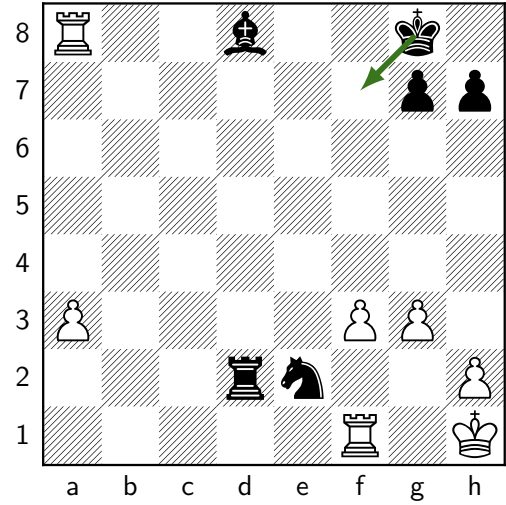
(a)



(b)

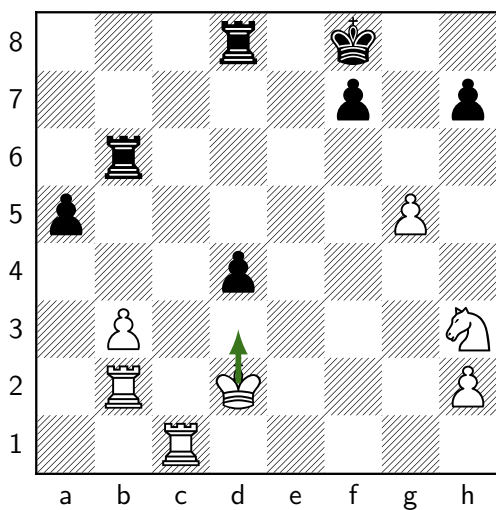


(c)

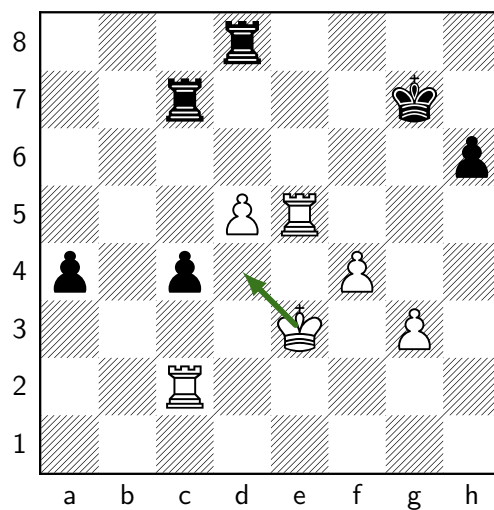


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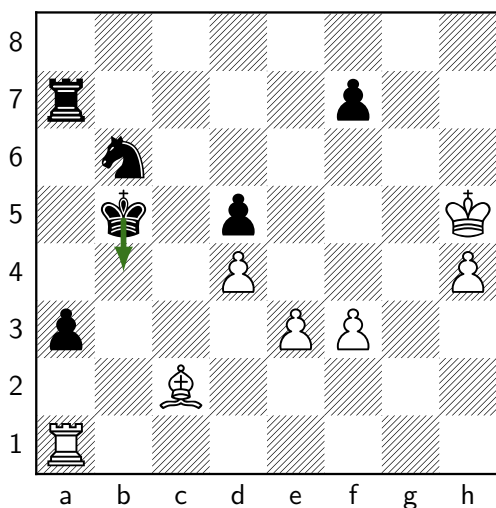
Figure 43: **Move 49, White: g5.** (a) Our position; Melanie plays g5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3860` (ply 82) [structural distance $d=29$, QAP obj=22]: the master played Rb5; under μ our Pg4 plays the role of that game’s Rf5 (a **cross-type** correspondence); cross-type roles: our Nh3 as their Pf3; transferred move g4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1248` (ply 81) [structural distance $d=22$, QAP obj=20]: the master played Rg8; under μ our Rb2 plays the role of that game’s Rf1; cross-type roles: our Ph2 as their Rb6, our Pg4 as their ra5; transferred move b2c2. (d) Retrieved from `ccr1_top_p0003#401` (ply 83) [structural distance $d=32$, QAP obj=16]: the master played Kf7; under μ our Kd2 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Bd8, our Rc1 as their Pg7; transferred move d2e2. Synthesis: 1 retrieved analog transferred to g5 (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



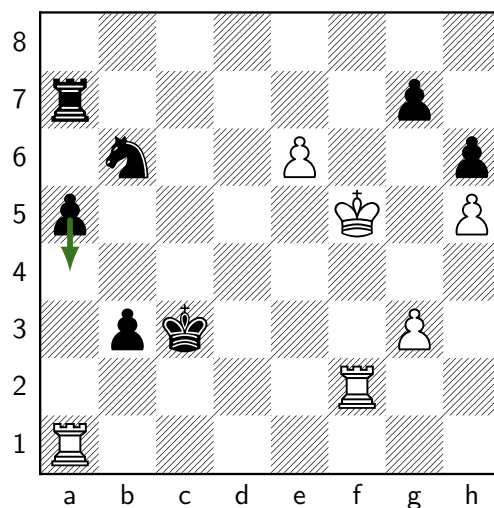
(a)



(b)



(c)



(d)

Figure 44: **Move 50, White: Kd3.** (a) Our position; Melanie plays Kd3 (d2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3387` (ply 112) [structural distance $d=29$, QAP obj=33]: the master played Kd4; under μ our *Kd2* plays the role of that game’s *Ke3*; cross-type roles: our *Rc1* as their *pa4*; transferred move d2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3012` (ply 93) [structural distance $d=29$, QAP obj=30]: the master played Kb4; under μ our *Kd2* plays the role of that game’s *Kb4*; cross-type roles: our *Rc1* as their *Nb6*, our *rd8* as their *ph4*; transferred move d2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3318` (ply 107) [structural distance $d=28$, QAP obj=34]: the master played a4; under μ our *Pb3* plays the role of that game’s *Pa4*; cross-type roles: our *Rc1* as their *Nb6*, our *Nh3* as their *pg3*; transferred move b3b4. Synthesis: 2 retrieved analogs transferred to Kd3 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

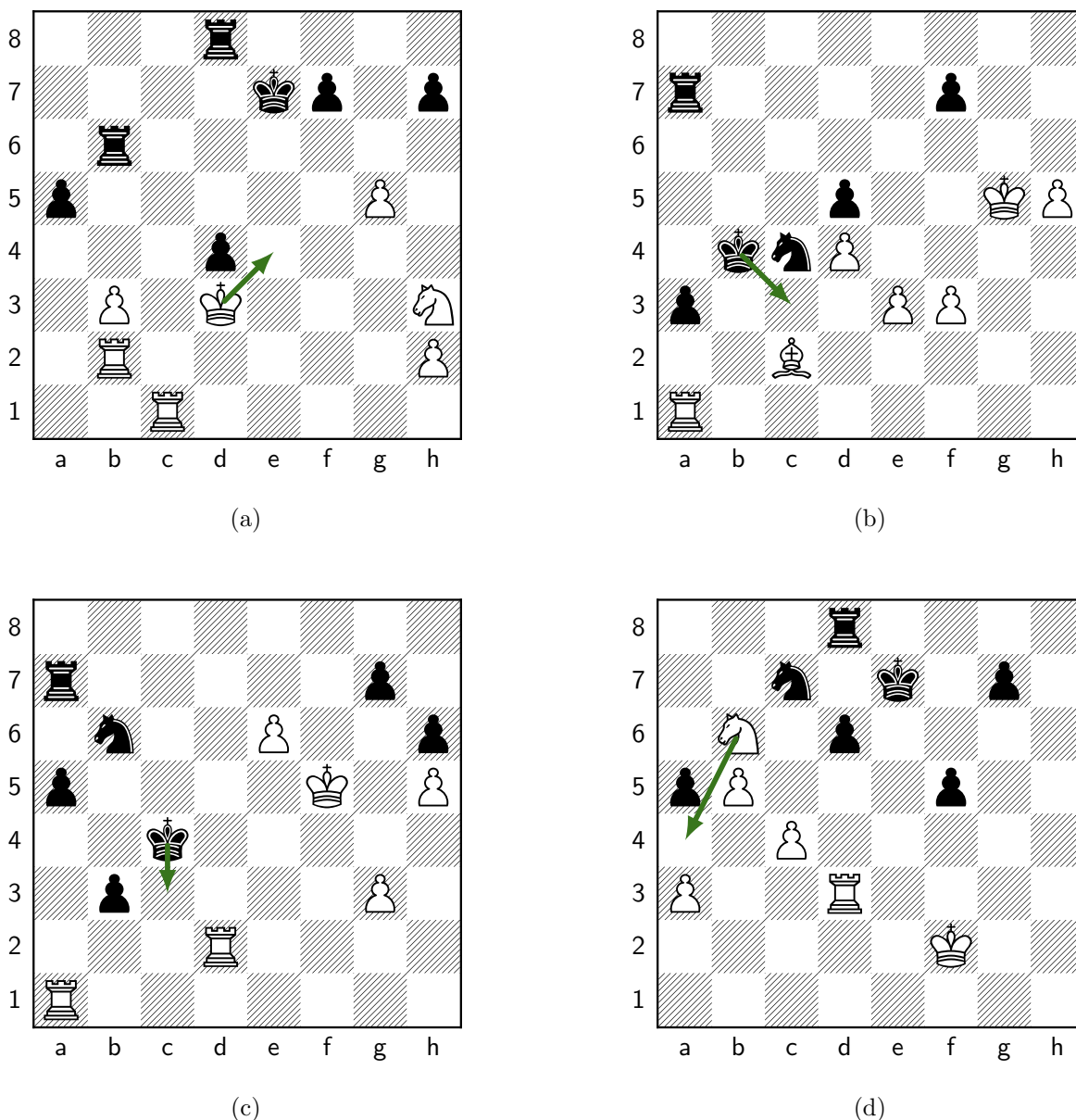
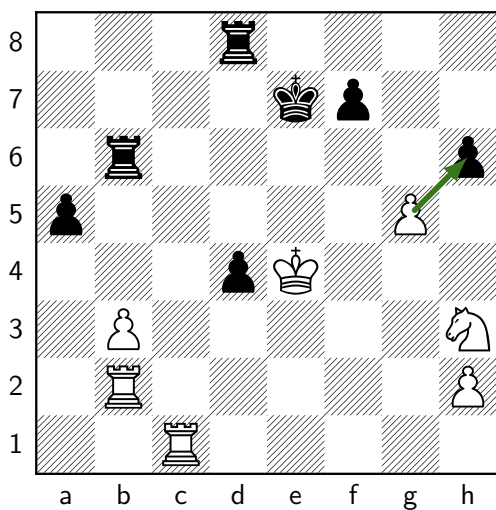
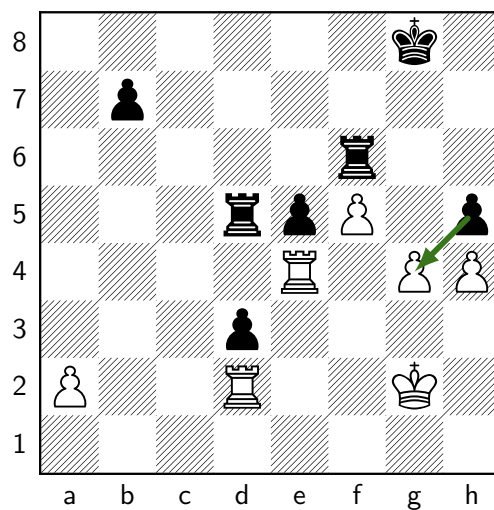


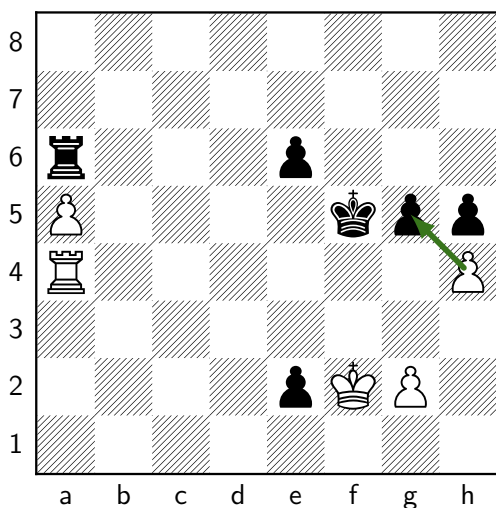
Figure 45: **Move 51, White: Ke4.** (a) Our position; Melanie plays Ke4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3012` (ply 97) [structural distance $d=30$, QAP obj=26]: the master played Kc3; under μ our Kd3 plays the role of that game’s Kb5; cross-type roles: our Rc1 as their Pd5, our Kd3 as their Nc4; transferred move d3e4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3318` (ply 105) [structural distance $d=27$, QAP obj=23]: the master played Kc3; under μ our Kd3 plays the role of that game’s Kc5; cross-type roles: our Rc1 as their Nb6, our Nh3 as their pg3; transferred move d3e4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3894` (ply 88) [structural distance $d=29$, QAP obj=34]: the master played Na4; under μ our Rc1 plays the role of that game’s Nb6 (a **cross-type** correspondence); cross-type roles: our Nh3 as their pa5, our Kd3 as their nc7; transferred move c1f1. Synthesis: 2 retrieved analogs transferred to Ke4 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



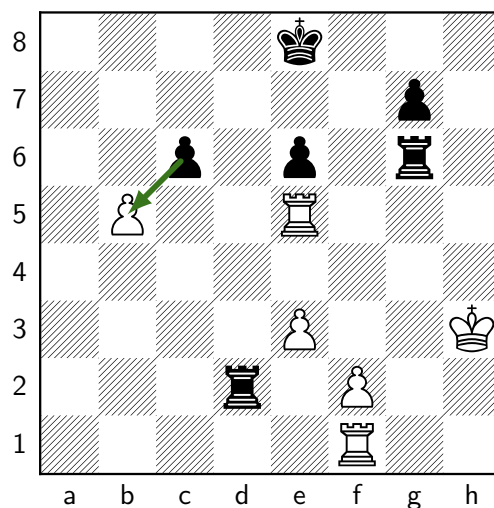
(a)



(b)

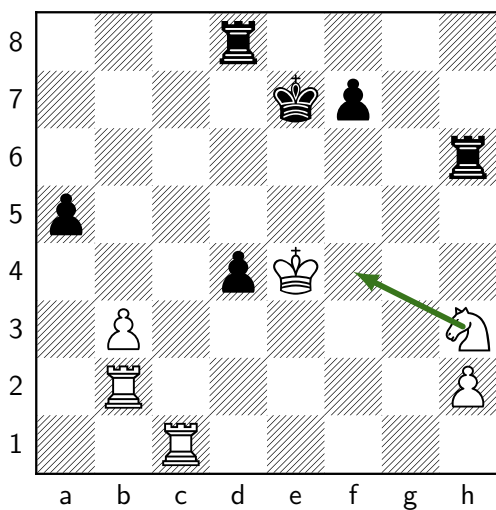


(c)

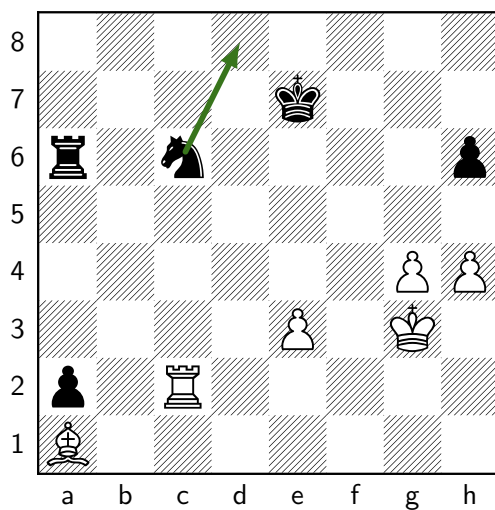


(d)

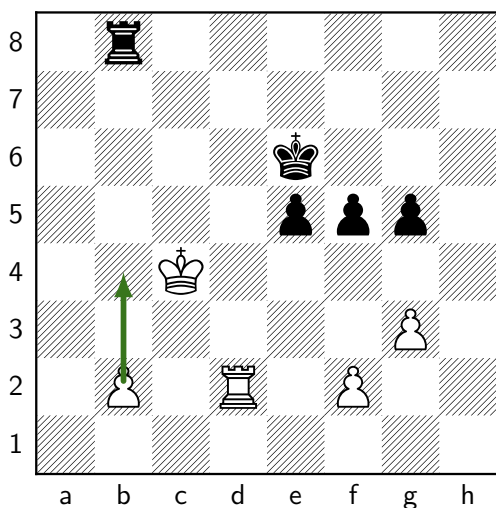
Figure 46: **Move 52, White: gxh6.** (a) Our position; Melanie plays gxh6 (g5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2310 (ply 87) [structural distance $d=35$, QAP obj=75]: the master played hxg4; under μ our Pg5 plays the role of that game’s Ph4; cross-type roles: our Rc1 as their Pb7, our Pg5 as their Rf6; transferred move g5h6 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2848 (ply 86) [structural distance $d=35$, QAP obj=70]: the master played hxg5; under μ our Pg5 plays the role of that game’s Ph4; cross-type roles: our Rc1 as their Pg2; transferred move g5h6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3602 (ply 89) [structural distance $d=36$, QAP obj=58]: the master played cxb5; under μ our Pg5 plays the role of that game’s Pc3; cross-type roles: our Nh3 as their Pg7, our Rc1 as their pe3; transferred move g5h6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to gxh6 (summed QAP objective 259, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



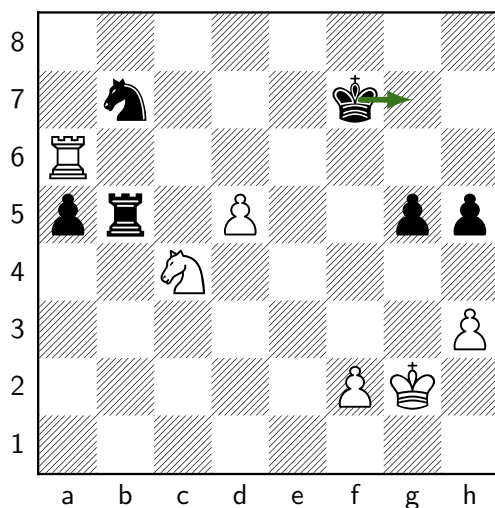
(a)



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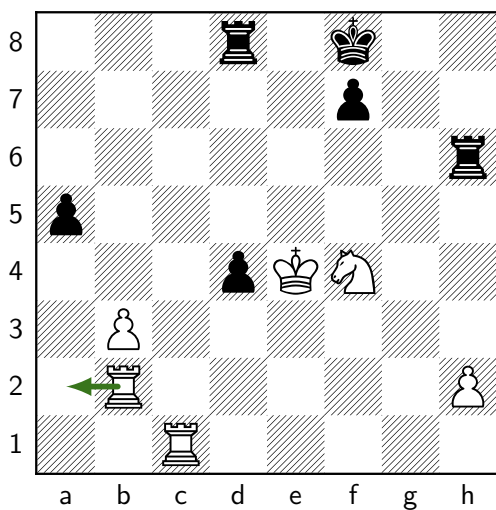


(c)

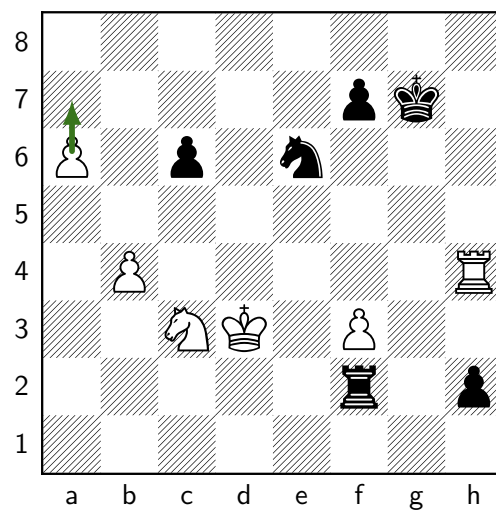


(d)

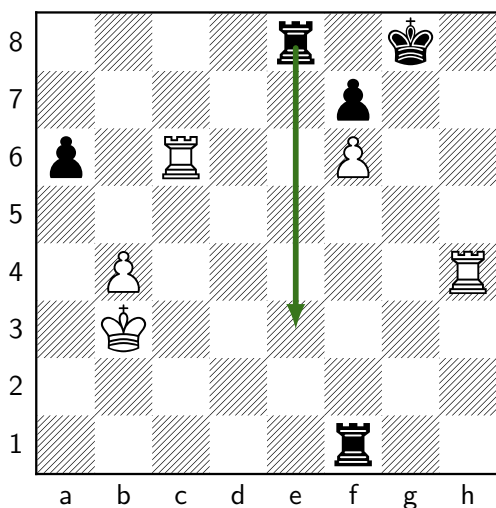
Figure 47: **Move 53, White: Nf4**. (a) Our position; Melanie plays Nf4 (h3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#818` (ply 105) [structural distance $d=24$, QAP obj=19]: the master played Nd8; under μ our Nh3 plays the role of that game’s Nc3; cross-type roles: our Rb2 as their pg4, our Nh3 as their Pa2; transferred move h3f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1204` (ply 88) [structural distance $d=25$, QAP obj=2]: the master played b4; under μ our Nh3 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Rb2 as their Pf2; transferred move h3f4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2921` (ply 103) [structural distance $d=25$, QAP obj=28]: the master played Kg7; under μ our Ke4 plays the role of that game’s Kf2; cross-type roles: our Ph2 as their Nb7, our Nh3 as their Pa5; transferred move e4e5. Synthesis: 2 retrieved analogs transferred to Nf4 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



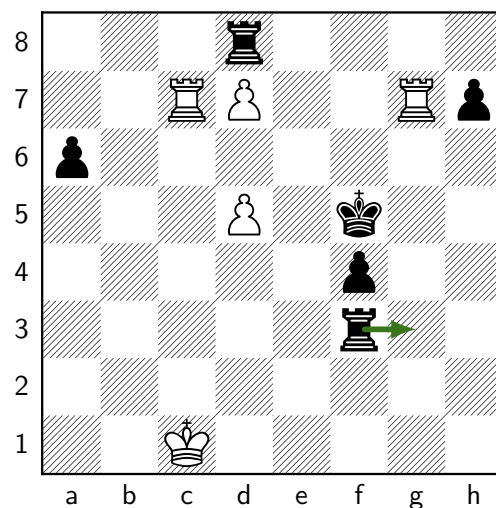
(a)



(b)

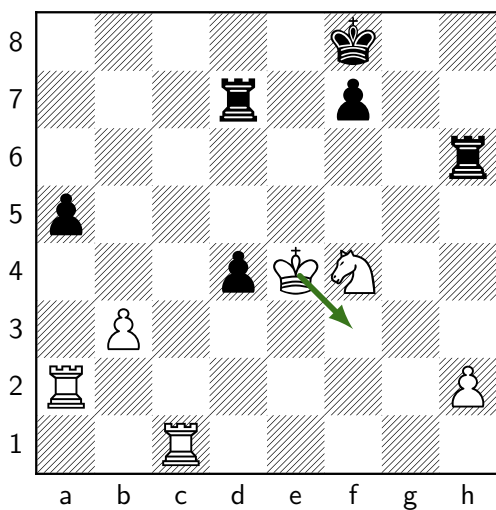


(c)

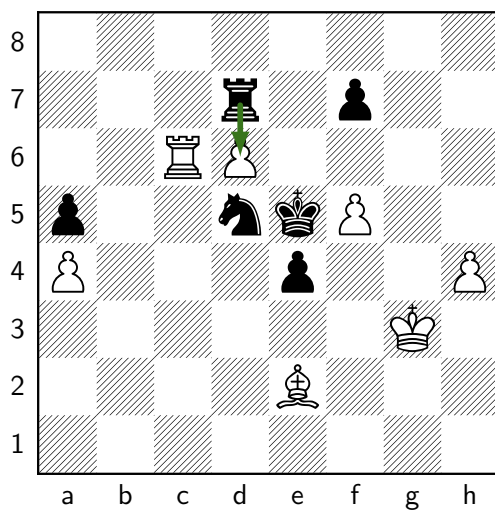


(d)

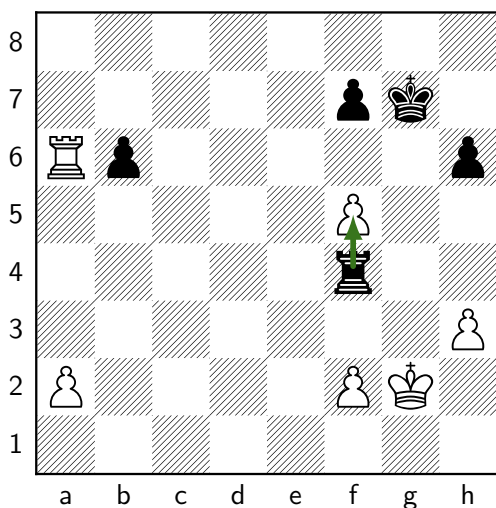
Figure 48: **Move 54, White: Ra2.** (a) Our position; Melanie plays Ra2 (b2a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2859` (ply 90) [structural distance $d=22$, QAP obj=20]: the master played a7; under μ our *Rb2* plays the role of that game's *Pa6* (a **cross-type** correspondence); cross-type roles: our *Ke4* as their *rf2*, our *Rc1* as their *Pb4*; transferred move b2a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3851` (ply 115) [structural distance $d=25$, QAP obj=7]: the master played Re3+; under μ our *Rb2* plays the role of that game's *Re1*; cross-type roles: our *Nf4* as their *Pf7*; transferred move b2a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1709` (ply 111) [structural distance $d=23$, QAP obj=24]: the master played Rg3; under μ our *Nf4* plays the role of that game's *Rf6* (a **cross-type** correspondence); cross-type roles: our *Ke4* as their *Rd8*, our *Ke4* as their *rg7*; transferred move f4h5. Synthesis: 2 retrieved analogs transferred to Ra2 (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



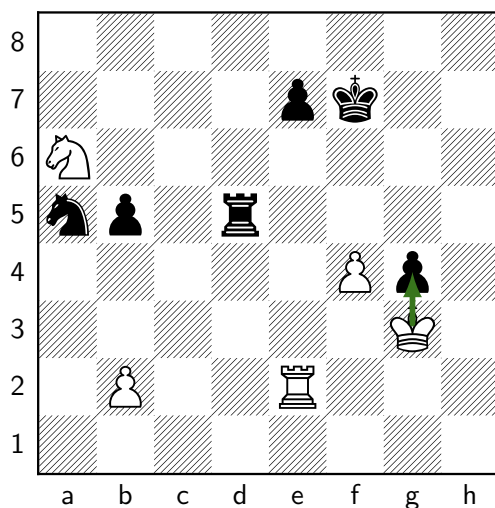
(a)



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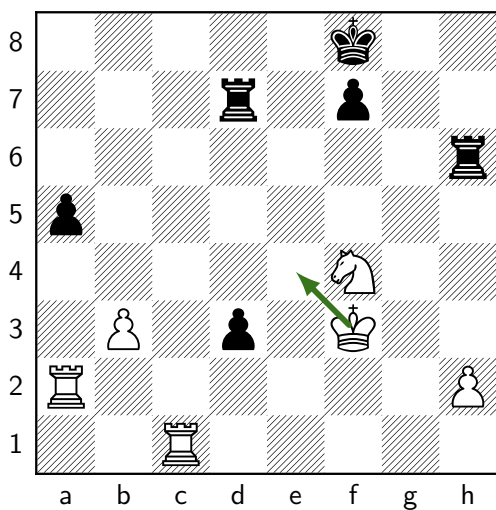


(c)

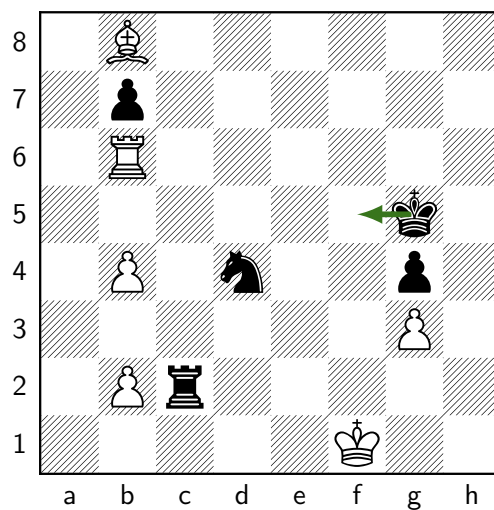


(d)

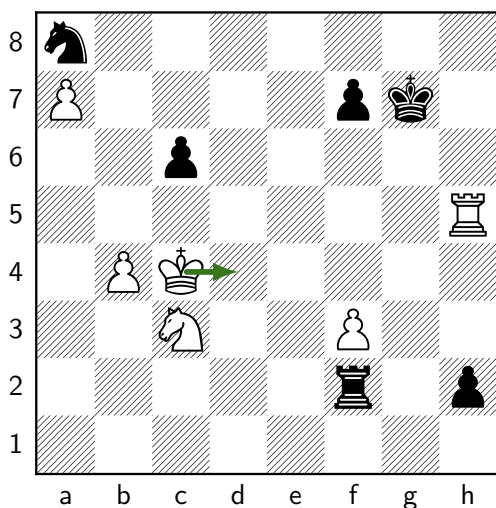
Figure 49: **Move 55, White: Kf3**. (a) Our position; Melanie plays Kf3 (e4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2062` (ply 101) [structural distance $d=25$, QAP obj=52]: the master played Rxd6; under μ our Ke4 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Pf7, our rh6 as their be2; transferred move e4f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2047` (ply 111) [structural distance $d=26$, QAP obj=28]: the master played Rxf5; under μ our Ke4 plays the role of that game’s Rf5 (a **cross-type** correspondence); cross-type roles: our Nf4 as their Pf7, our Ke4 as their ra6; transferred move e4f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1447` (ply 124) [structural distance $d=25$, QAP obj=21]: the master played Kxg4; under μ our Ke4 plays the role of that game’s Kg3; cross-type roles: our Nf4 as their Pf4, our pa5 as their na5; transferred move e4f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf3 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



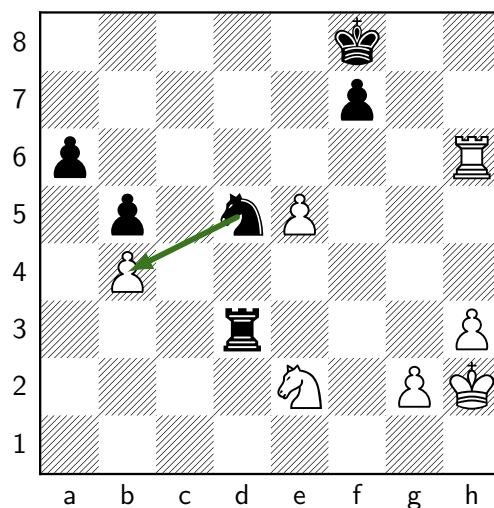
(a)



(b)

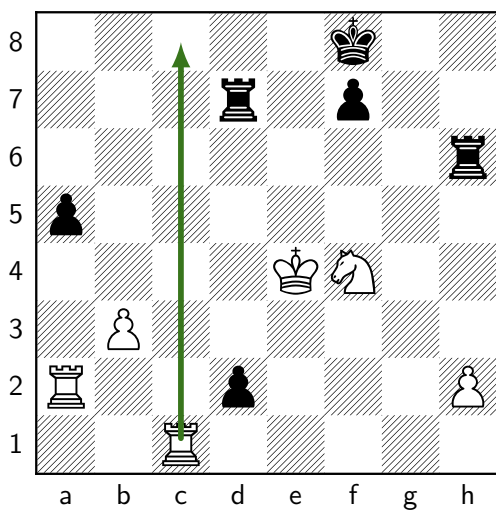


(c)

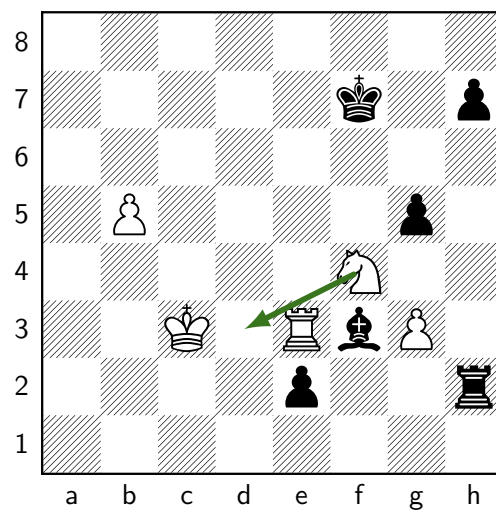


(d)

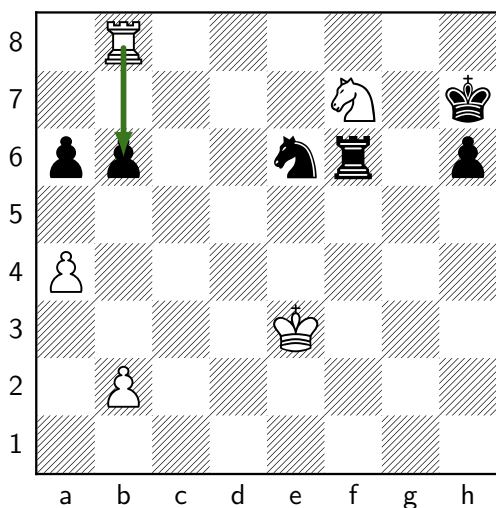
Figure 50: **Move 56, White: Ke4 (f3e4).** (a) Our position; Melanie plays Ke4 (f3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2497` (ply 105) [structural distance $d=28$, QAP obj=16]: the master played Kf5; under μ our Kf3 plays the role of that game’s Kg4; cross-type roles: our rh6 as their bb8, our Rc1 as their Nd4; transferred move f3e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2859` (ply 104) [structural distance $d=20$, QAP obj=16]: the master played Kd4; under μ our Kf3 plays the role of that game’s Kc4; cross-type roles: our Nf4 as their rf2, our rh6 as their Nc3; transferred move f3e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1993` (ply 99) [structural distance $d=23$, QAP obj=34]: the master played Nxb4; under μ our Nf4 plays the role of that game’s Nd4; cross-type roles: our Rc1 as their Pb5, our rd7 as their ne2; transferred move f4h3. Synthesis: 2 retrieved analogs transferred to Ke4 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



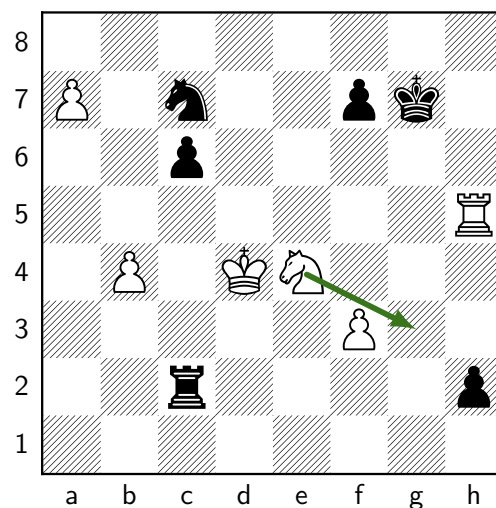
(a)



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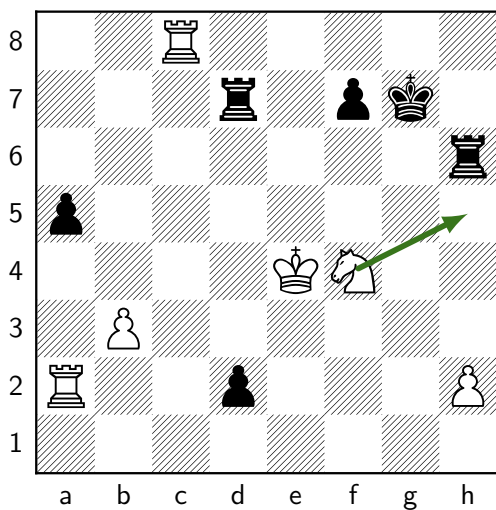


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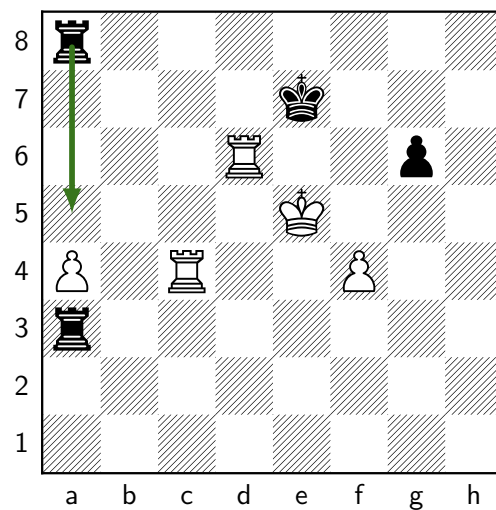


(d)

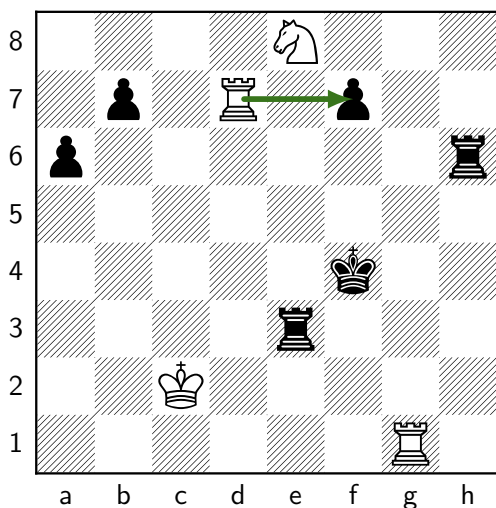
Figure 51: **Move 57, White: Rc8+.** (a) Our position; Melanie plays Rc8+ (c1c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top.p0003#2150 (ply 102) [structural distance $d=32$, QAP obj=13]: the master played Nd3; under μ our Rc1 plays the role of that game's Nf4 (a **cross-type** correspondence); cross-type roles: our pd2 as their Re3, our Rc1 as their bf3; transferred move c1c4. (c) Retrieved from ccr1_top.p0000#538 (ply 98) [structural distance $d=31$, QAP obj=12]: the master played Rxb6; under μ our Nf4 plays the role of that game's Rb8 (a **cross-type** correspondence); cross-type roles: our Ra2 as their pa6, our rd7 as their ne6; transferred move f4h5. (d) Retrieved from ccr1_top.p0000#2859 (ply 108) [structural distance $d=26$, QAP obj=5]: the master played Ng3; under μ our Nf4 plays the role of that game's Ne4; cross-type roles: our rh6 as their Pf3, our Rc1 as their Pa7; transferred move f4e2. Synthesis: no retrieved analog transferred to Rc8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



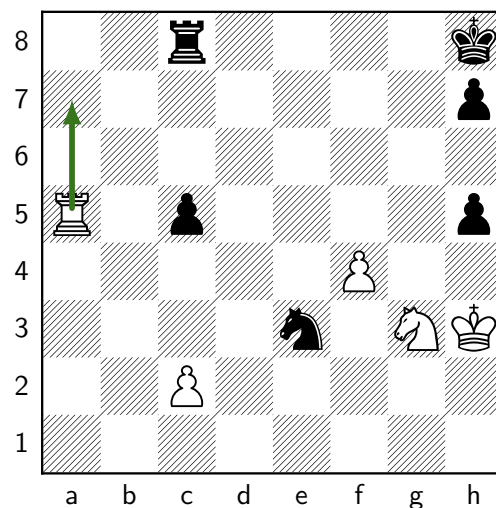
(a)



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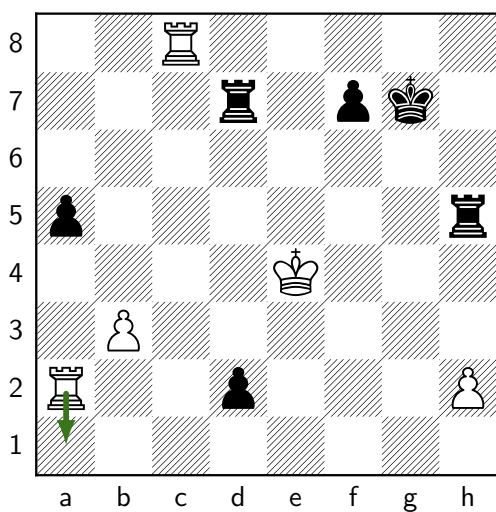


(c)

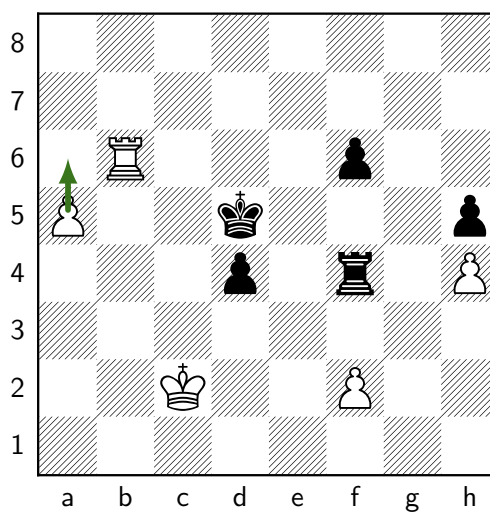


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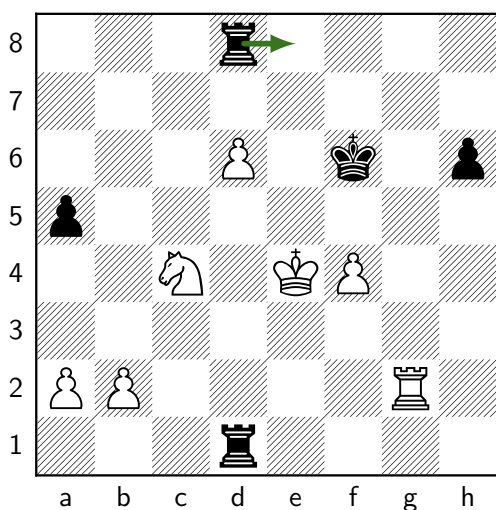
Figure 52: **Move 58, White: Nh5+ (f4h5).** (a) Our position; Melanie plays Nh5+ (f4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2053` (ply 109) [structural distance $d=29$, QAP obj=2]: the master played Ra5+; under μ our Nf4 plays the role of that game’s Ra1 (a **cross-type** correspondence); transferred move f4h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#95` (ply 104) [structural distance $d=28$, QAP obj=31]: the master played Rxf7+; under μ our Ra2 plays the role of that game’s Rd7; cross-type roles: our Ph2 as their Ne8; transferred move a2a5. (d) Retrieved from `ccr1_top_p0000#2079` (ply 114) [structural distance $d=30$, QAP obj=30]: the master played Ra7; under μ our Ra2 plays the role of that game’s Ra5; cross-type roles: our rh6 as their ne3, our Ra2 as their Ng3; transferred move a2a5. Synthesis: 1 retrieved analog transferred to Nh5+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



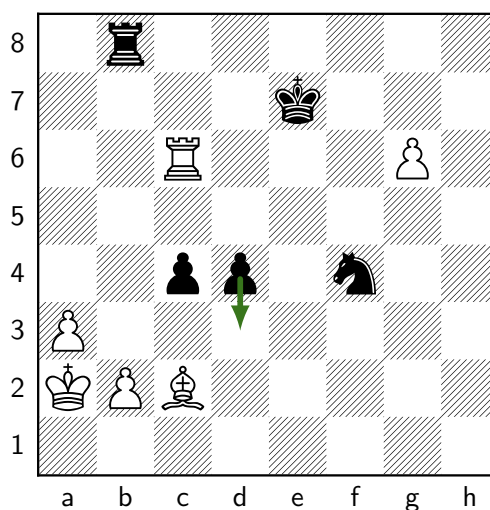
(a)



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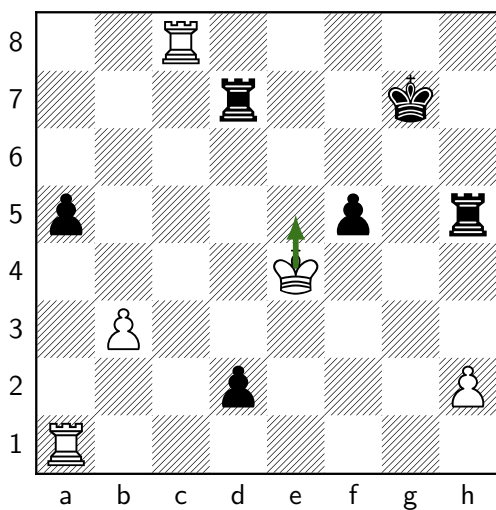


(c)

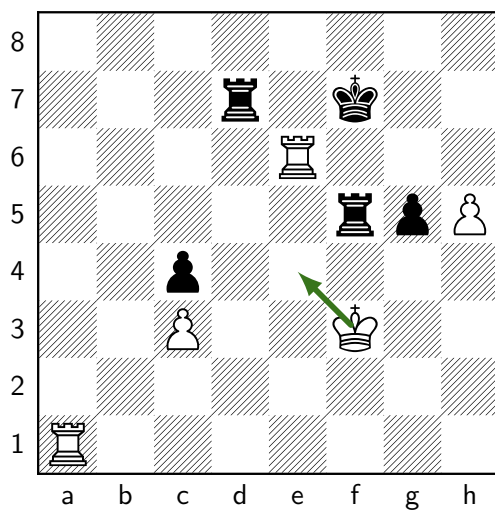


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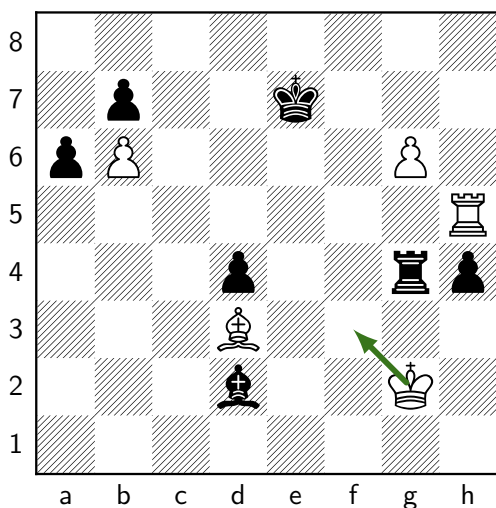
Figure 53: **Move 59, White: Ra1.** (a) Our position; Melanie plays Ra1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3190` (ply 102) [structural distance $d=21$, QAP obj=40]: the master played a6; under μ our *Rc8* plays the role of that game’s *Pa5* (a **cross-type** correspondence); transferred move c8b8. (c) Retrieved from `ccr1_top_p0002#997` (ply 101) [structural distance $d=27$, QAP obj=32]: the master played Re8+; under μ our *Ra2* plays the role of that game’s *Rd1*; cross-type roles: our *rh5* as their *nc4*, our *Rc8* as their *pb2*; transferred move a2b2. (d) Retrieved from `ccr1_top_p0002#2449` (ply 103) [structural distance $d=29$, QAP obj=31]: the master played d3; under μ our *Rc8* plays the role of that game’s *Pd5* (a **cross-type** correspondence); cross-type roles: our *Ra2* as their *Nf4*, our *rd7* as their *bc2*; transferred move c8c5. Synthesis: no retrieved analog transferred to Ra1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



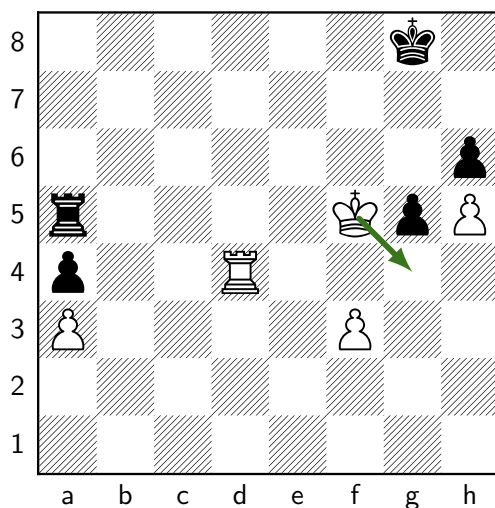
(a)



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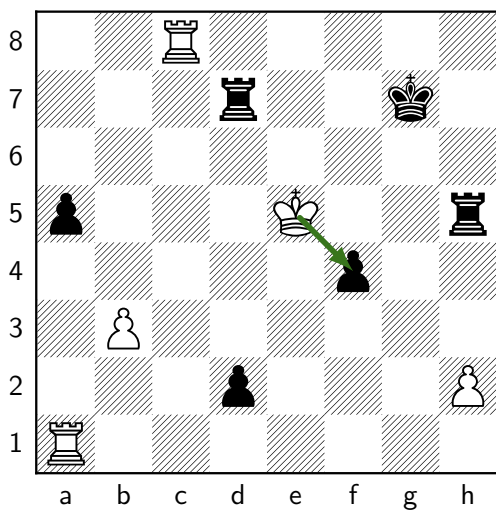


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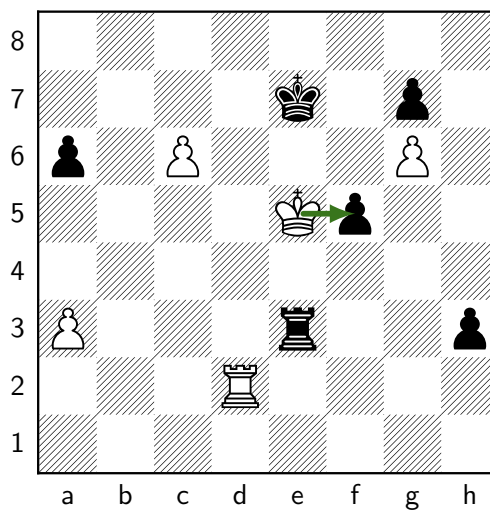


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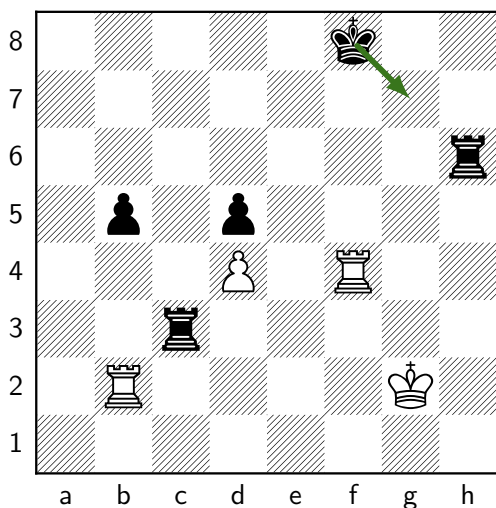
Figure 54: **Move 60, White: Ke5.** (a) Our position; Melanie plays Ke5 (e4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#835` (ply 130) [structural distance $d=27$, QAP obj=35]: the master played Ke4; under μ our Ke4 plays the role of that game's Kf3; cross-type roles: our pd2 as their Ra1, our pf5 as their rf5; transferred move e4e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#749` (ply 120) [structural distance $d=26$, QAP obj=34]: the master played Kf3; under μ our Ke4 plays the role of that game's Kg2; cross-type roles: our Ph2 as their bd2, our Rc8 as their Bd3; transferred move e4f3. (d) Retrieved from `ccrl_top_p0001#2663` (ply 128) [structural distance $d=28$, QAP obj=15]: the master played Kg4; under μ our Ke4 plays the role of that game's Kf5; cross-type roles: our pf5 as their ra5, our rd7 as their pg5; transferred move e4e3. Synthesis: 1 retrieved analog transferred to Ke5 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



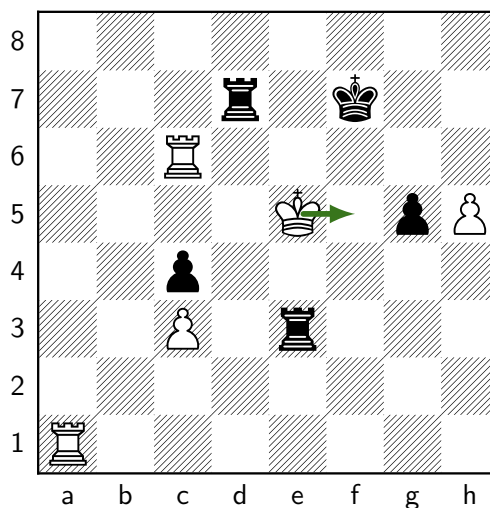
(a)



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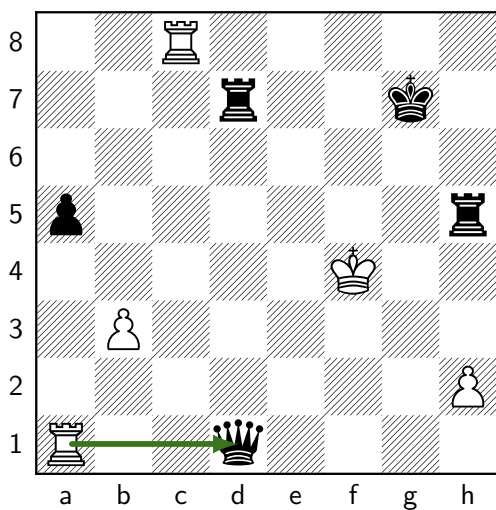


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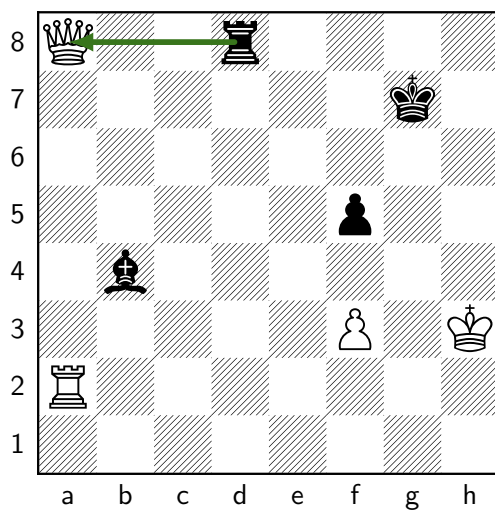


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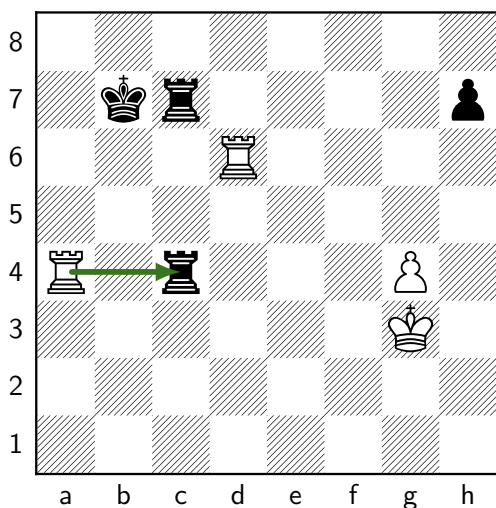
Figure 55: **Move 61, White: Kxf4.** (a) Our position; Melanie plays Kxf4 (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1334` (ply 110) [structural distance $d=17$, QAP obj=61]: the master played Kxf5; under μ our *Ke5* plays the role of that game's *Ke5*; cross-type roles: our *Rc8* as their *Pc6*, our *rd7* as their *pg7*; transferred move e5f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2134` (ply 137) [structural distance $d=25$, QAP obj=51]: the master played Kg7; under μ our *Ke5* plays the role of that game's *Kf1*; transferred move e5f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#835` (ply 136) [structural distance $d=18$, QAP obj=48]: the master played Kf5; under μ our *Ke5* plays the role of that game's *Ke5*; transferred move e5f4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kxf4 (summed QAP objective 277, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



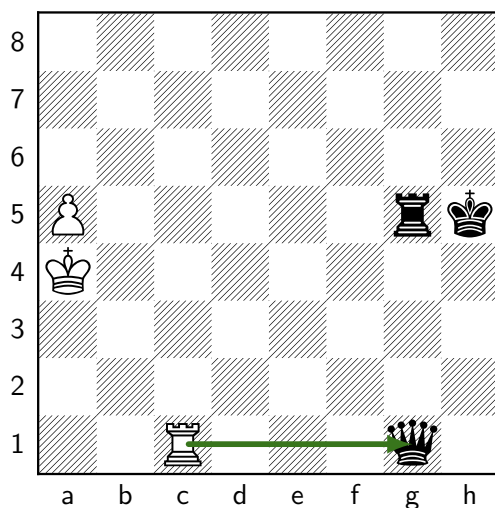
(a)



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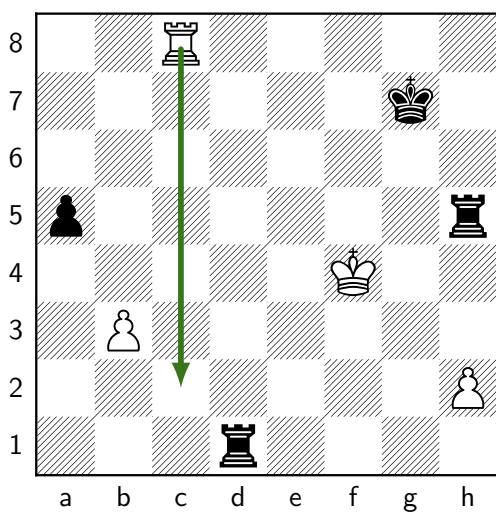


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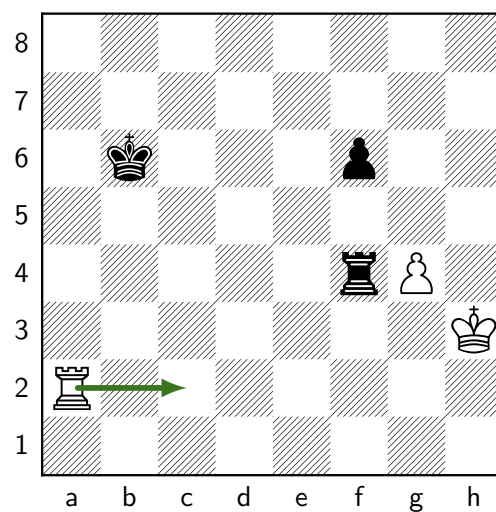


(d)

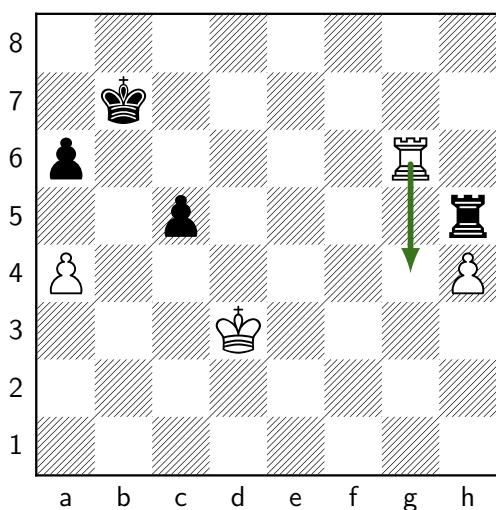
Figure 56: **Move 62, White: Rxd1.** (a) Our position; Melanie plays Rxd1 (a1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3067` (ply 109) [structural distance $d=42$, QAP obj=37]: the master played Rxa8; under μ our *Ra1* plays the role of that game's *Rd1*; cross-type roles: our *Rc8* as their *Bb4*, our *rh5* as their *pf3*; transferred move a1d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2460` (ply 134) [structural distance $d=41$, QAP obj=30]: the master played Rxc4; under μ our *Ra1* plays the role of that game's *Ra4*; cross-type roles: our *qd1* as their *rc4*; transferred move a1d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1464` (ply 116) [structural distance $d=42$, QAP obj=26]: the master played Rxc1; under μ our *Ra1* plays the role of that game's *Rc1*; cross-type roles: our *rh5* as their *Pa5*; transferred move a1d1 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Rxd1 (summed QAP objective 239, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



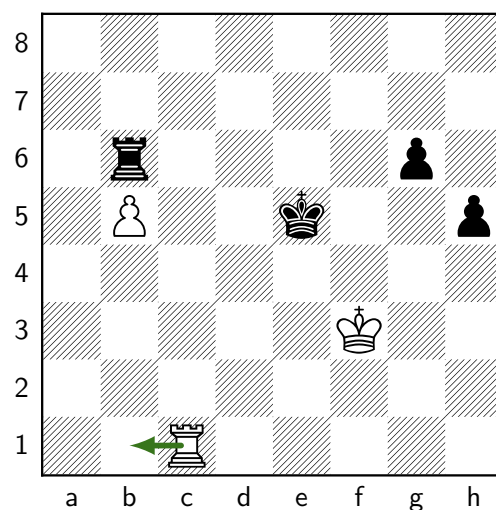
(a)



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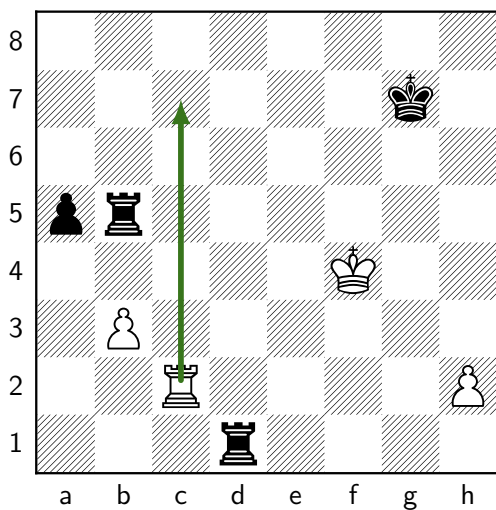


(c)

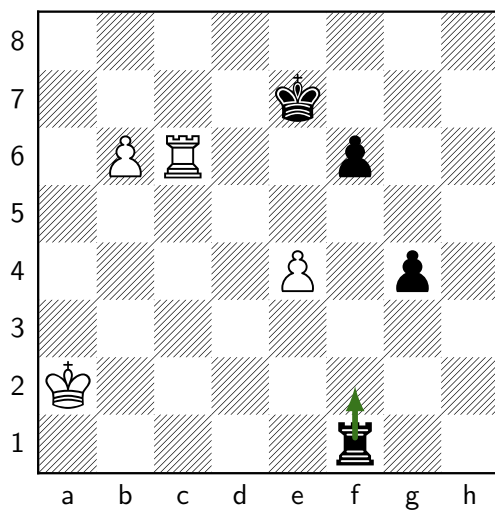


(d)

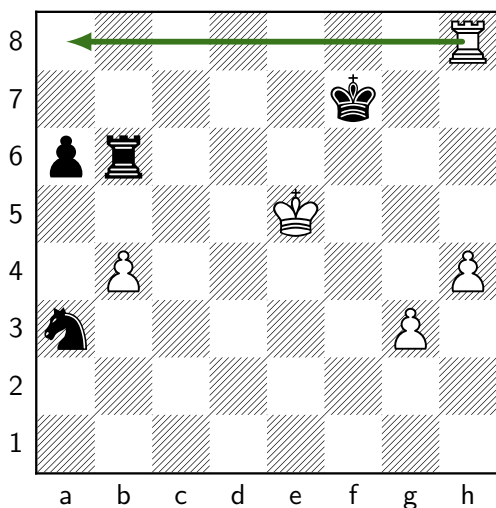
Figure 57: **Move 63, White: Rc2.** (a) Our position; Melanie plays Rc2 (c8c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#50` (ply 116) [structural distance $d=8$, QAP obj=9]: the master played Rc2; under μ our *Rc8* plays the role of that game's *Ra2*; transferred move c8c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3772` (ply 112) [structural distance $d=9$, QAP obj=8]: the master played Rg4; under μ our *Rc8* plays the role of that game's *Rg6*; cross-type roles: our *rd1* as their *Pa4*; transferred move c8c2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#971` (ply 114) [structural distance $d=7$, QAP obj=8]: the master played Rb1; under μ our *Rc8* plays the role of that game's *Rc1*; transferred move c8c2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rc2 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



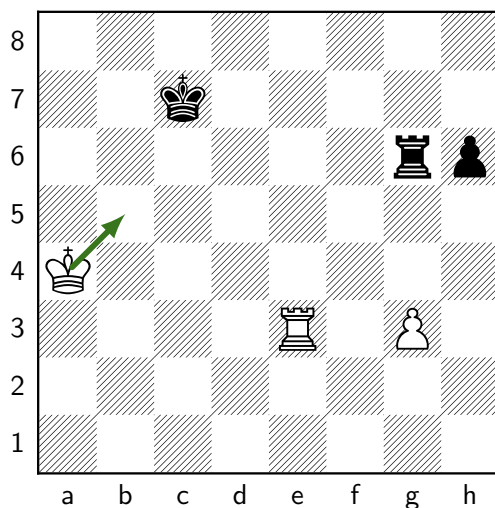
(a)



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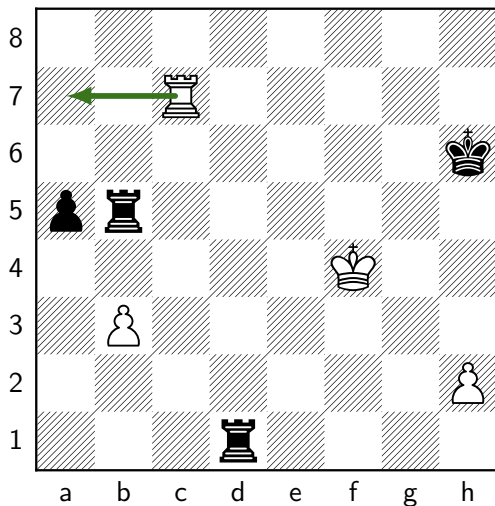


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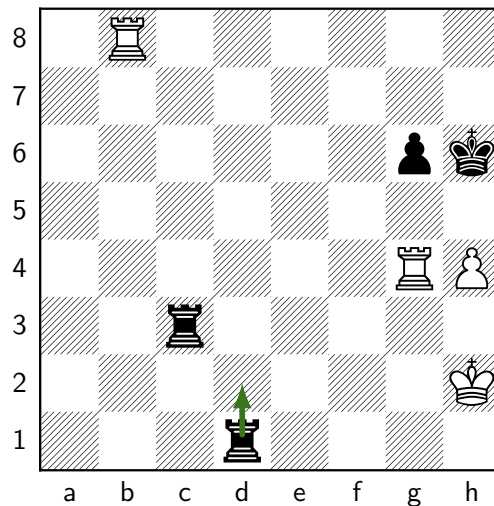


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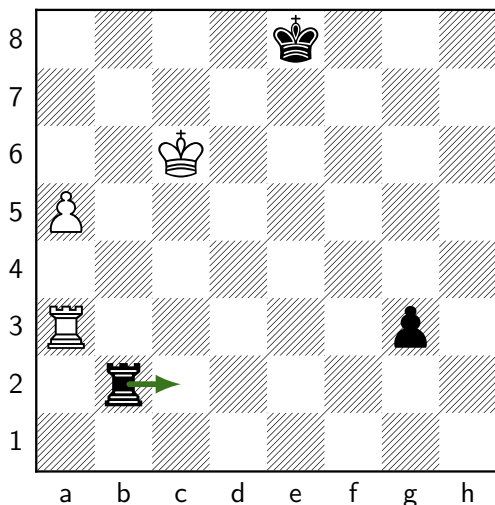
Figure 58: **Move 64, White: Rc7+.** (a) Our position; Melanie plays Rc7+ (c2c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1885` (ply 113) [structural distance $d=5$, QAP obj=16]: the master played Rf2+; under μ our Rc2 plays the role of that game’s Rf8; cross-type roles: our rd1 as their pe4; transferred move c2c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1898` (ply 124) [structural distance $d=7$, QAP obj=24]: the master played Ra8; under μ our Rc2 plays the role of that game’s Rh8; cross-type roles: our rd1 as their na3; transferred move c2c8. (d) Retrieved from `ccr1_top_p0000#2782` (ply 138) [structural distance $d=7$, QAP obj=16]: the master played Kb5; under μ our Kf4 plays the role of that game’s Ka4; transferred move f4g4. Synthesis: 1 retrieved analog transferred to Rc7+ (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



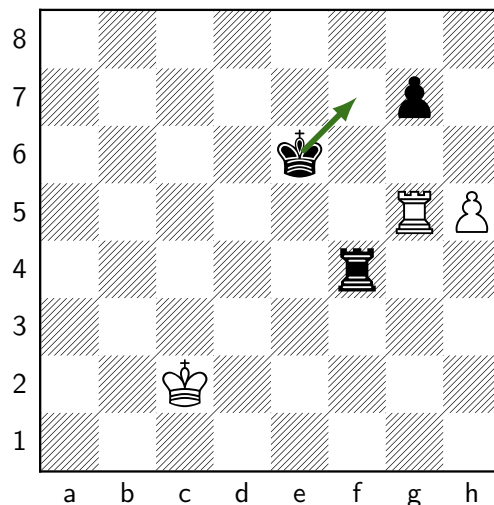
(a)



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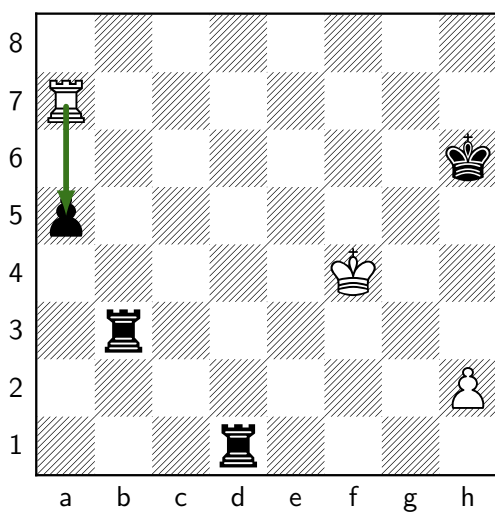


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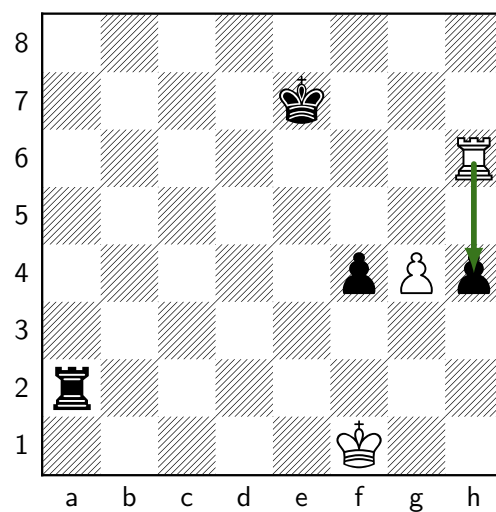


(d)

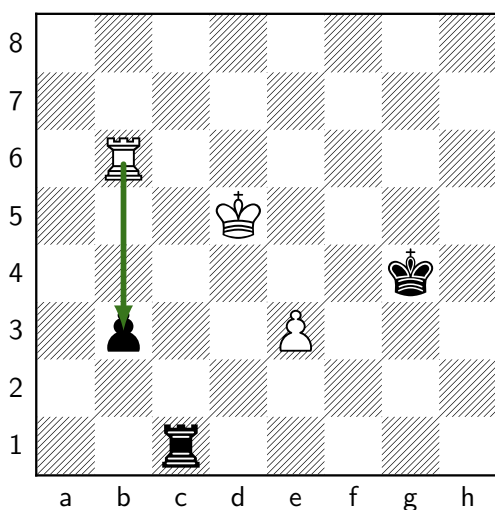
Figure 59: **Move 65, White: Ra7.** (a) Our position; Melanie plays Ra7 (c7a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3262` (ply 143) [structural distance $d=7$, QAP obj=23]: the master played Rd2+; under μ our Rc7 plays the role of that game’s Rd8; cross-type roles: our Ph2 as their Rc3; transferred move c7a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#421` (ply 129) [structural distance $d=8$, QAP obj=9]: the master played Rc2+; under μ our Rc7 plays the role of that game’s Rb7; transferred move c7a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1005` (ply 121) [structural distance $d=7$, QAP obj=24]: the master played Kf7; under μ our Kf4 plays the role of that game’s Ke3; transferred move f4g3. Synthesis: 2 retrieved analogs transferred to Ra7 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



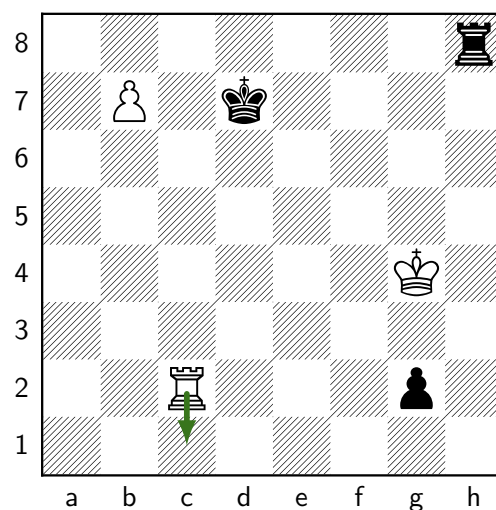
(a)



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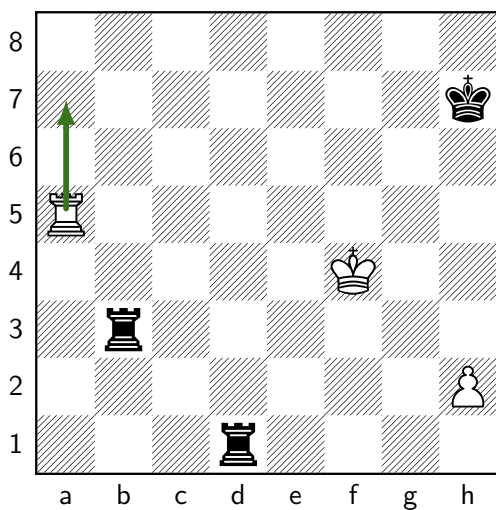


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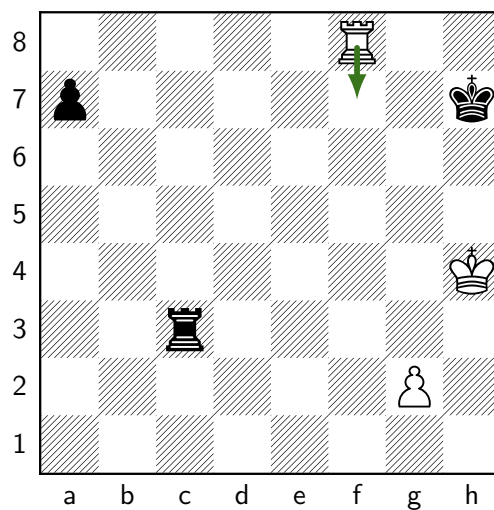


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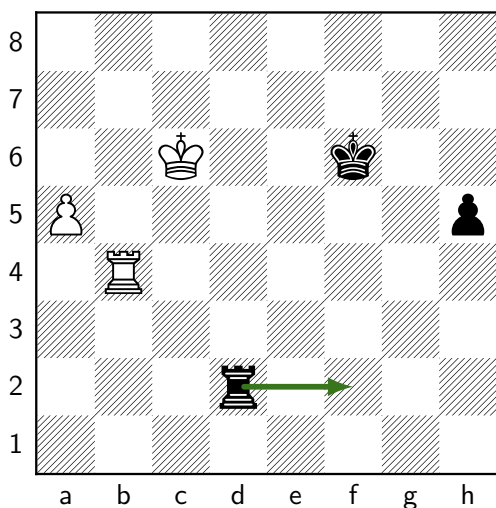
Figure 60: **Move 66, White: Rxa5.** (a) Our position; Melanie plays Rxa5 (a7a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3425` (ply 114) [structural distance $d=4$, QAP obj=19]: the master played Rxb4; under μ our *Ra7* plays the role of that game’s *Rh6*; cross-type roles: our *rd1* as their *pf4*; transferred move a7a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2899` (ply 128) [structural distance $d=4$, QAP obj=13]: the master played Rxb3; under μ our *Ra7* plays the role of that game’s *Rb6*; cross-type roles: our *rd1* as their *Pe3*; transferred move a7a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2138` (ply 118) [structural distance $d=2$, QAP obj=10]: the master played Rc1; under μ our *Ra7* plays the role of that game’s *Rc2*; transferred move a7a5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxa5 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



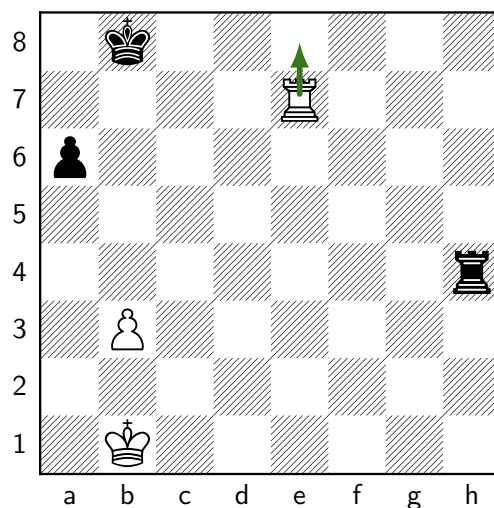
(a)



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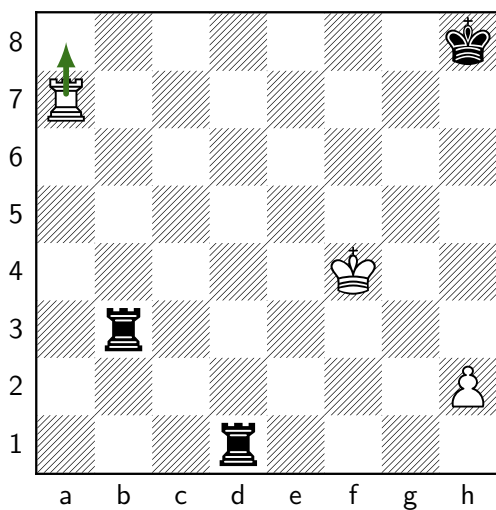


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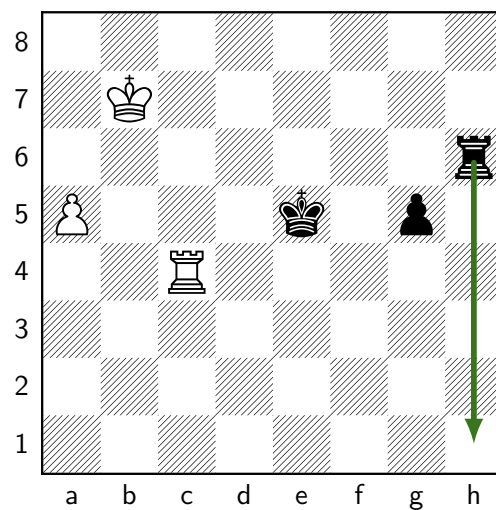


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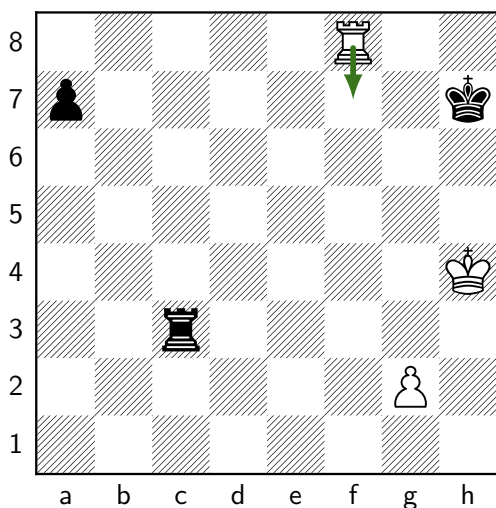
Figure 61: **Move 67, White: Ra7+.** (a) Our position; Melanie plays Ra7+ (a5a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1796` (ply 126) [structural distance $d=2$, QAP obj=1]: the master played Rf7+; under μ our Ra5 plays the role of that game’s Rf8; cross-type roles: our rd1 as their pa7; transferred move a5a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1100` (ply 117) [structural distance $d=3$, QAP obj=0]: the master played Rf2; under μ our Ra5 plays the role of that game’s Rd7; cross-type roles: our rd1 as their pa5; transferred move a5a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2413` (ply 118) [structural distance $d=3$, QAP obj=0]: the master played Re8+; under μ our Ra5 plays the role of that game’s Re7; cross-type roles: our rd1 as their pa6; transferred move a5a7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ra7+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



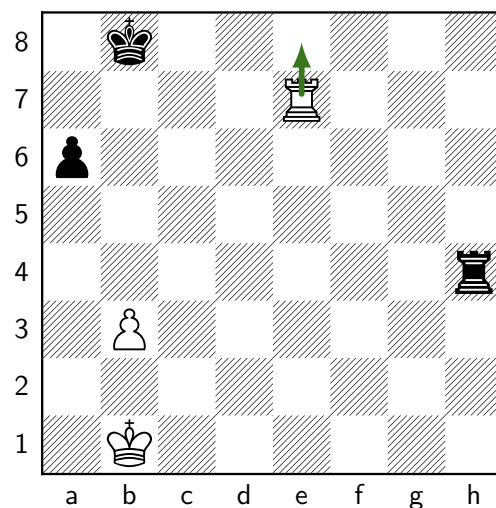
(a)



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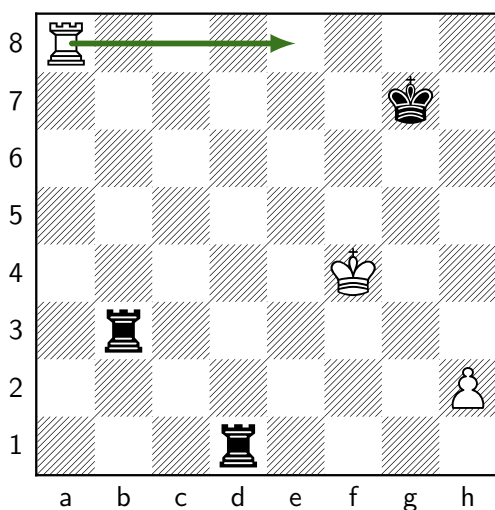


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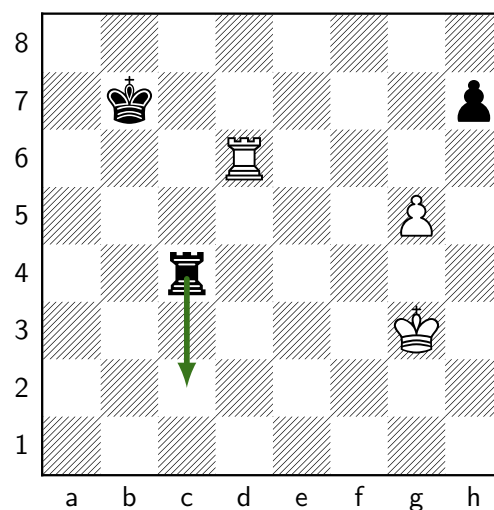


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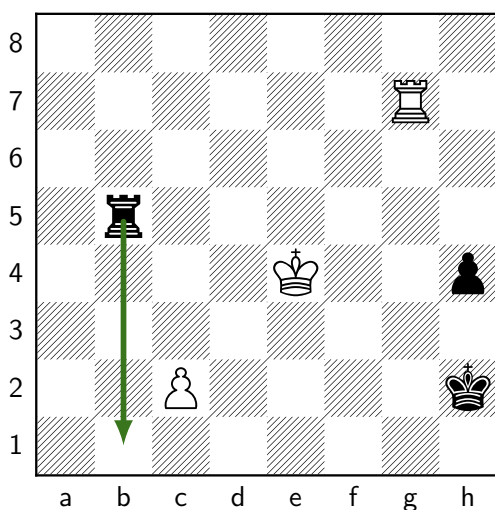
Figure 62: **Move 68, White: Ra8+ (a7a8)**. (a) Our position; Melanie plays Ra8+ (a7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3141` (ply 127) [structural distance $d=3$, QAP obj=1]: the master played Rh1; under μ our Ra7 plays the role of that game’s Rh3; cross-type roles: our rd1 as their pa5; transferred move a7a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1796` (ply 126) [structural distance $d=2$, QAP obj=1]: the master played Rf7+; under μ our Ra7 plays the role of that game’s Rf8; cross-type roles: our rd1 as their pa7; transferred move a7a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2413` (ply 118) [structural distance $d=3$, QAP obj=1]: the master played Re8+; under μ our Ra7 plays the role of that game’s Re7; cross-type roles: our rd1 as their pa6; transferred move a7a8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ra8+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



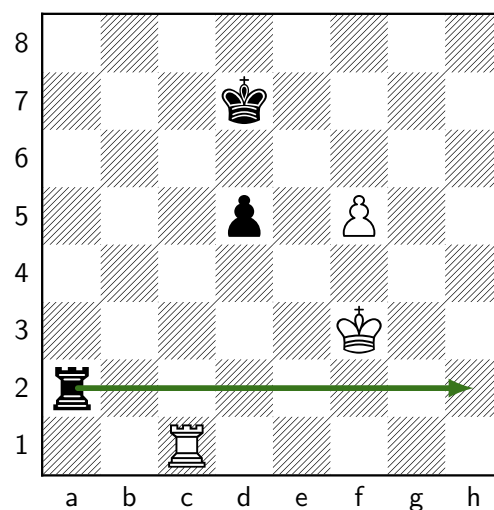
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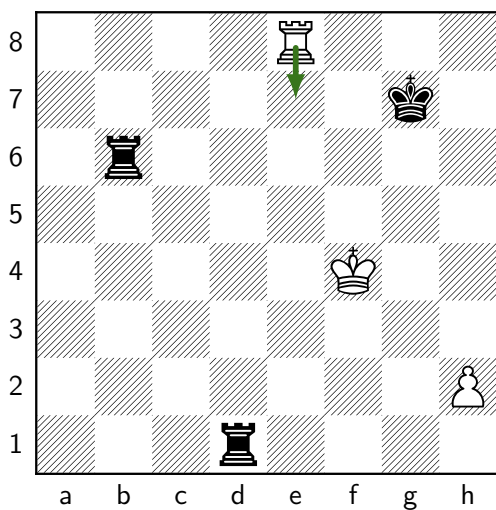


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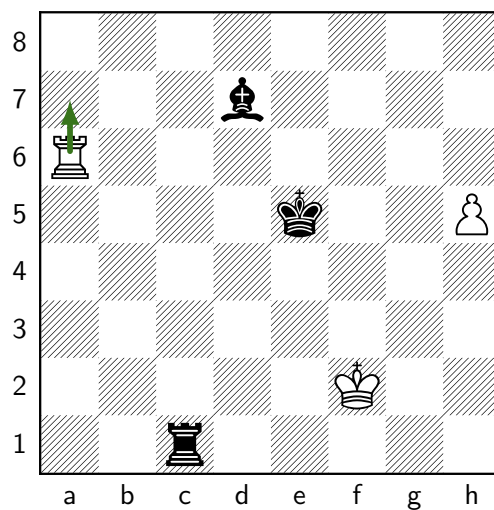


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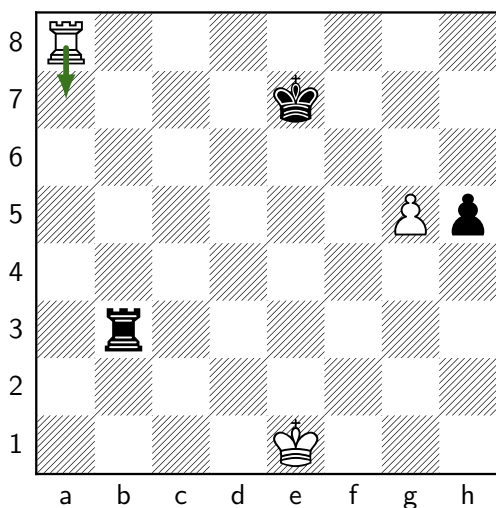
Figure 63: **Move 69, White: Re8**. (a) Our position; Melanie plays Re8 (a8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2460 (ply 137) [structural distance $d=3$, QAP obj=1]: the master played Rc2; under μ our *Ra8* plays the role of that game’s *Rc5*; cross-type roles: our *rd1* as their *pg5*; transferred move a8e8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1727 (ply 141) [structural distance $d=3$, QAP obj=1]: the master played Rb1; under μ our *Ra8* plays the role of that game’s *Rb4*; cross-type roles: our *rd1* as their *pc2*; transferred move a8e8 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#863 (ply 149) [structural distance $d=3$, QAP obj=1]: the master played Rh2; under μ our *Ra8* plays the role of that game’s *Ra7*; cross-type roles: our *rd1* as their *pf5*; transferred move a8e8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re8 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



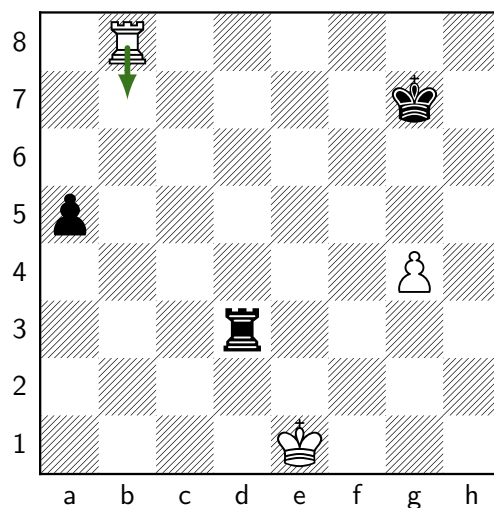
(a)



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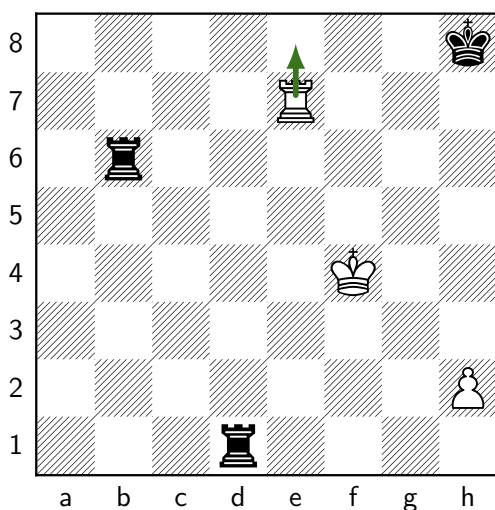


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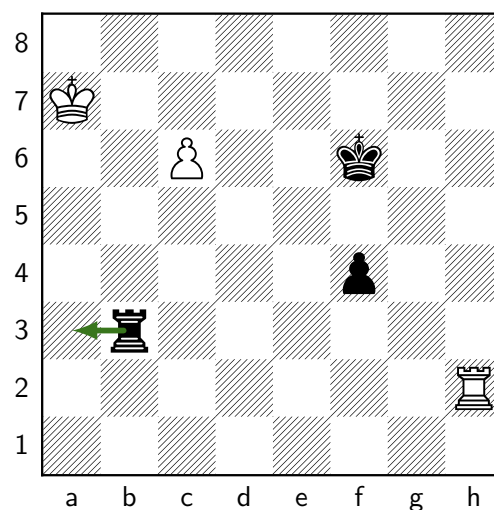


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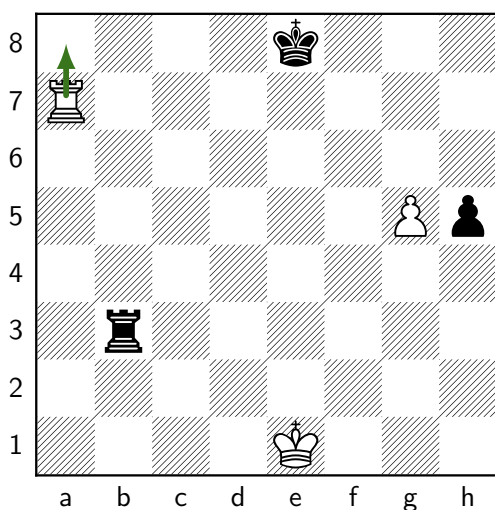
Figure 64: **Move 70, White: Re7+ (e8e7)**. (a) Our position; Melanie plays Re7+ (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3987` (ply 142) [structural distance $d=2$, QAP obj=1]: the master played Ra7; under μ our Re8 plays the role of that game’s Ra6; cross-type roles: our rb6 as their bd7; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3466` (ply 140) [structural distance $d=2$, QAP obj=1]: the master played Ra7+; under μ our Re8 plays the role of that game’s Ra8; cross-type roles: our rd1 as their Pg5; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1203` (ply 122) [structural distance $d=3$, QAP obj=1]: the master played Rb7+; under μ our Re8 plays the role of that game’s Rb8; cross-type roles: our rb6 as their pa5; transferred move e8e7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Re7+ (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



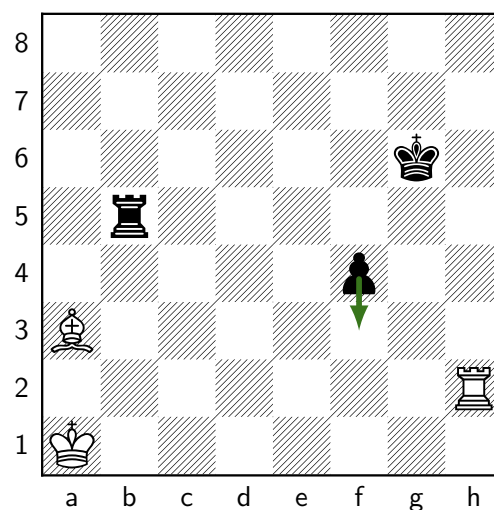
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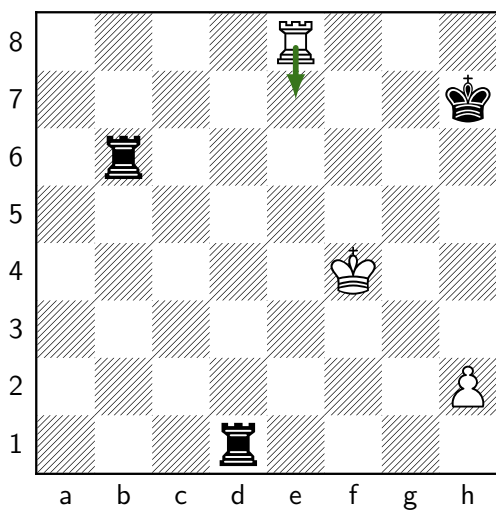


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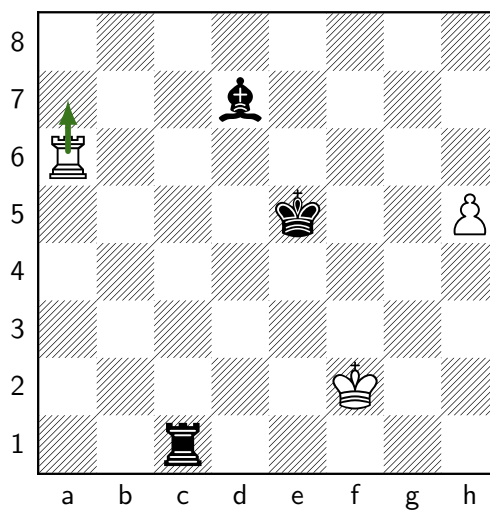


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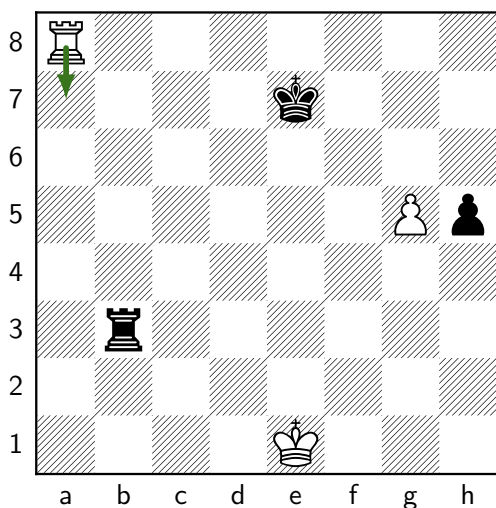
Figure 65: **Move 71, White: Re8+ (e7e8)**. (a) Our position; Melanie plays **Re8+ (e7e8)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1240` (ply 143) [structural distance $d=2$, QAP obj=1]: the master played **Ra3+**; under μ our *Re7* plays the role of that game's *Rb6*; cross-type roles: our *rd1* as their *pc6*; transferred move **e7e8 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#3466` (ply 142) [structural distance $d=2$, QAP obj=1]: the master played **Ra8+**; under μ our *Re7* plays the role of that game's *Ra7*; cross-type roles: our *rb6* as their *ph5*; transferred move **e7e8 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#2362` (ply 131) [structural distance $d=3$, QAP obj=1]: the master played **f3**; under μ our *Ph2* plays the role of that game's *Pf5*; cross-type roles: our *rd1* as their *ba3*; transferred move **h2h4**. Synthesis: 5 retrieved analogs transferred to **Re8+** (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



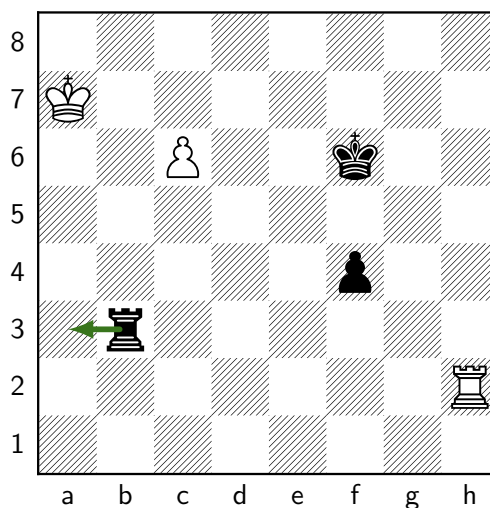
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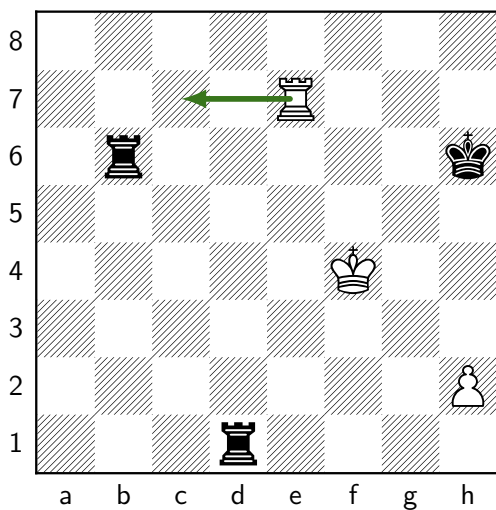


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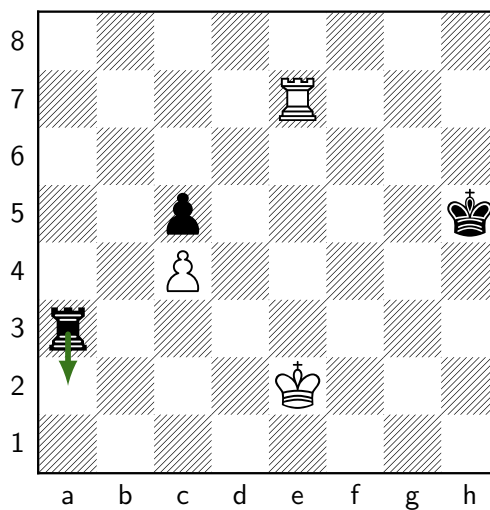


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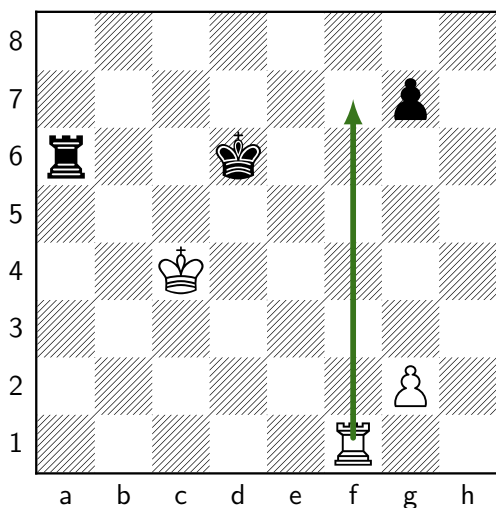
Figure 66: **Move 72, White: Re7+ (e8e7)**. (a) Our position; Melanie plays Re7+ (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3987` (ply 142) [structural distance $d=2$, QAP obj=1]: the master played Ra7; under μ our Re8 plays the role of that game’s Ra6; cross-type roles: our rb6 as their bd7; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3466` (ply 140) [structural distance $d=2$, QAP obj=1]: the master played Ra7+; under μ our Re8 plays the role of that game’s Ra8; cross-type roles: our rd1 as their Pg5; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1240` (ply 143) [structural distance $d=2$, QAP obj=1]: the master played Ra3+; under μ our Re8 plays the role of that game’s Rb6; cross-type roles: our rd1 as their pc6; transferred move e8e7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Re7+ (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



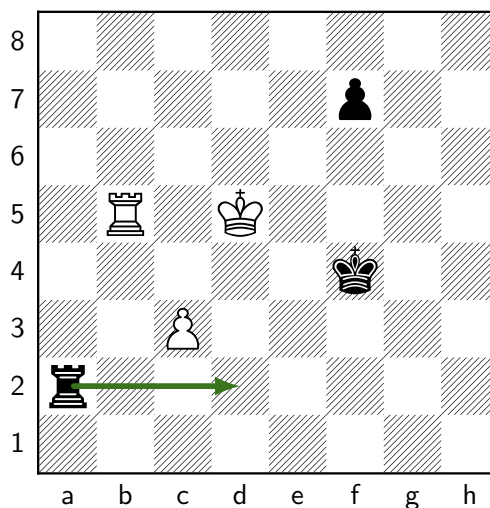
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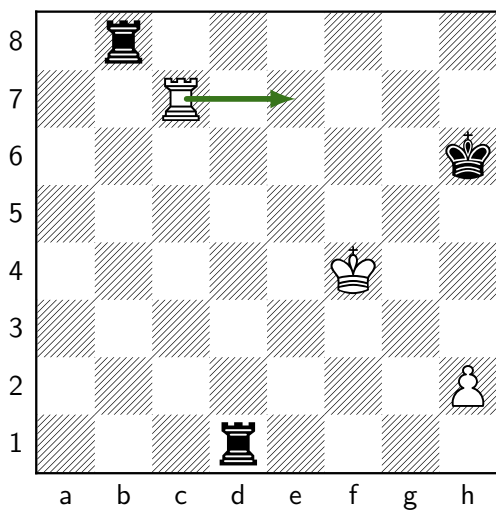


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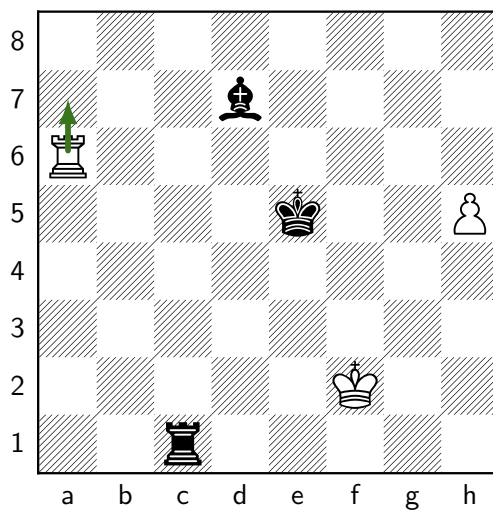


(d)

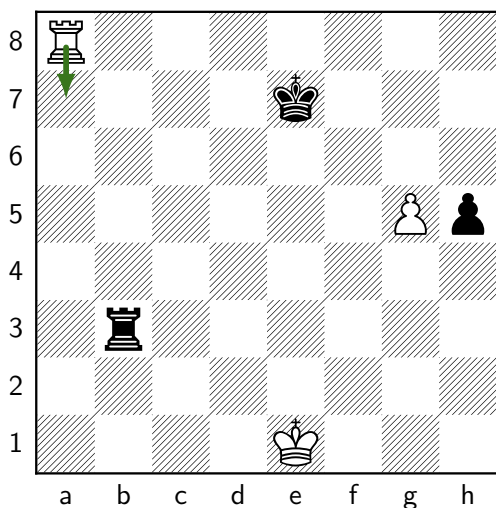
Figure 67: **Move 73, White: Rc7**. (a) Our position; Melanie plays Rc7 (e7c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#374` (ply 153) [structural distance $d=3$, QAP obj=3]: the master played Ra2+; under μ our Re7 plays the role of that game's Ra6; cross-type roles: our rd1 as their pc4; transferred move e7c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2903` (ply 130) [structural distance $d=2$, QAP obj=2]: the master played Rf7; under μ our Re7 plays the role of that game's Rf1; cross-type roles: our rd1 as their pg7; transferred move e7c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1609` (ply 147) [structural distance $d=3$, QAP obj=1]: the master played Rd2+; under μ our Re7 plays the role of that game's Ra7; cross-type roles: our rd1 as their Pf7; transferred move e7c7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rc7 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



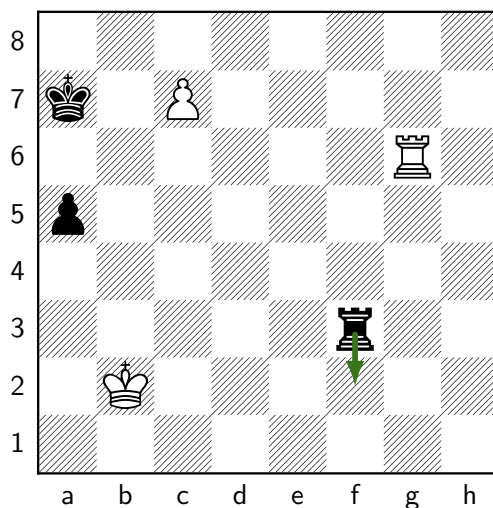
(a)



(b)

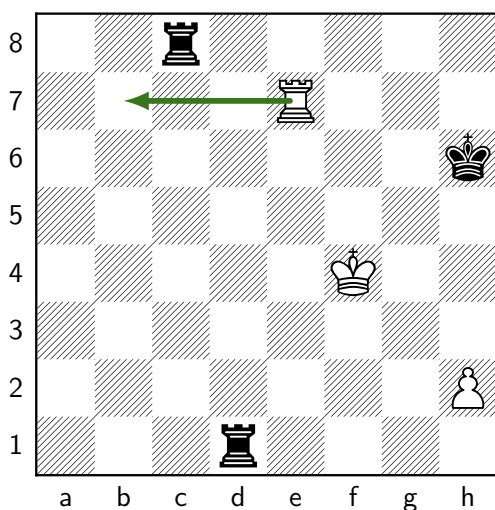


(c)

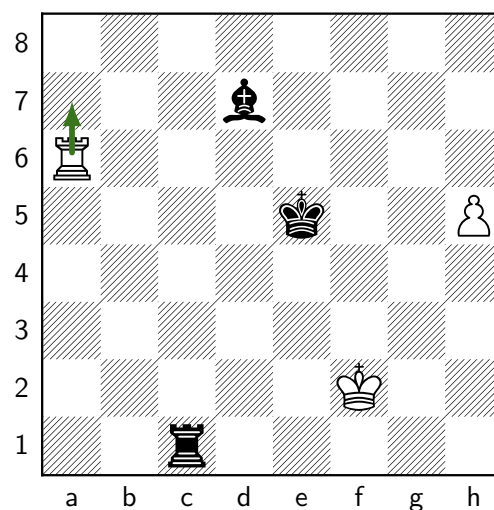


(d)

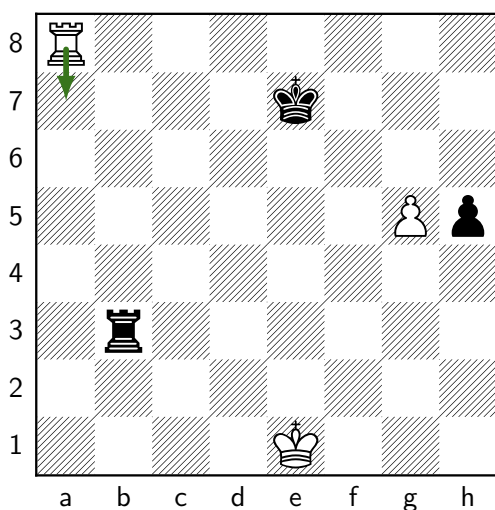
Figure 68: **Move 74, White: Re7**. (a) Our position; Melanie plays **Re7** (**c7e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3987` (ply 142) [structural distance $d=2$, QAP obj=1]: the master played **Ra7**; under μ our *Rc7* plays the role of that game's *Ra6*; cross-type roles: our *rb8* as their *bd7*; transferred move **c7e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3466` (ply 144) [structural distance $d=2$, QAP obj=1]: the master played **Ra7+**; under μ our *Rc7* plays the role of that game's *Ra8*; cross-type roles: our *rb8* as their *Pg5*; transferred move **c7e7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2511` (ply 159) [structural distance $d=3$, QAP obj=1]: the master played **Rf2+**; under μ our *Rc7* plays the role of that game's *Rf6*; cross-type roles: our *rd1* as their *pc7*; transferred move **c7e7** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **Re7** (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



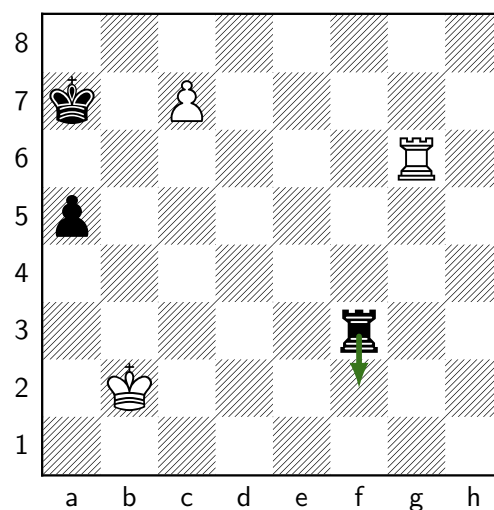
(a)



(b)



(c)



(d)

Figure 69: **Move 75, White: Rb7**. (a) Our position; Melanie plays Rb7 (e7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3987` (ply 142) [structural distance $d=2$, QAP obj=1]: the master played Ra7; under μ our *Re7* plays the role of that game’s *Ra6*; cross-type roles: our *rc8* as their *bd7*; transferred move e7b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3466` (ply 144) [structural distance $d=2$, QAP obj=1]: the master played Ra7+; under μ our *Re7* plays the role of that game’s *Ra8*; cross-type roles: our *rc8* as their *Pg5*; transferred move e7b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2511` (ply 159) [structural distance $d=3$, QAP obj=1]: the master played Rf2+; under μ our *Re7* plays the role of that game’s *Rf6*; cross-type roles: our *rd1* as their *pc7*; transferred move e7b7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rb7 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

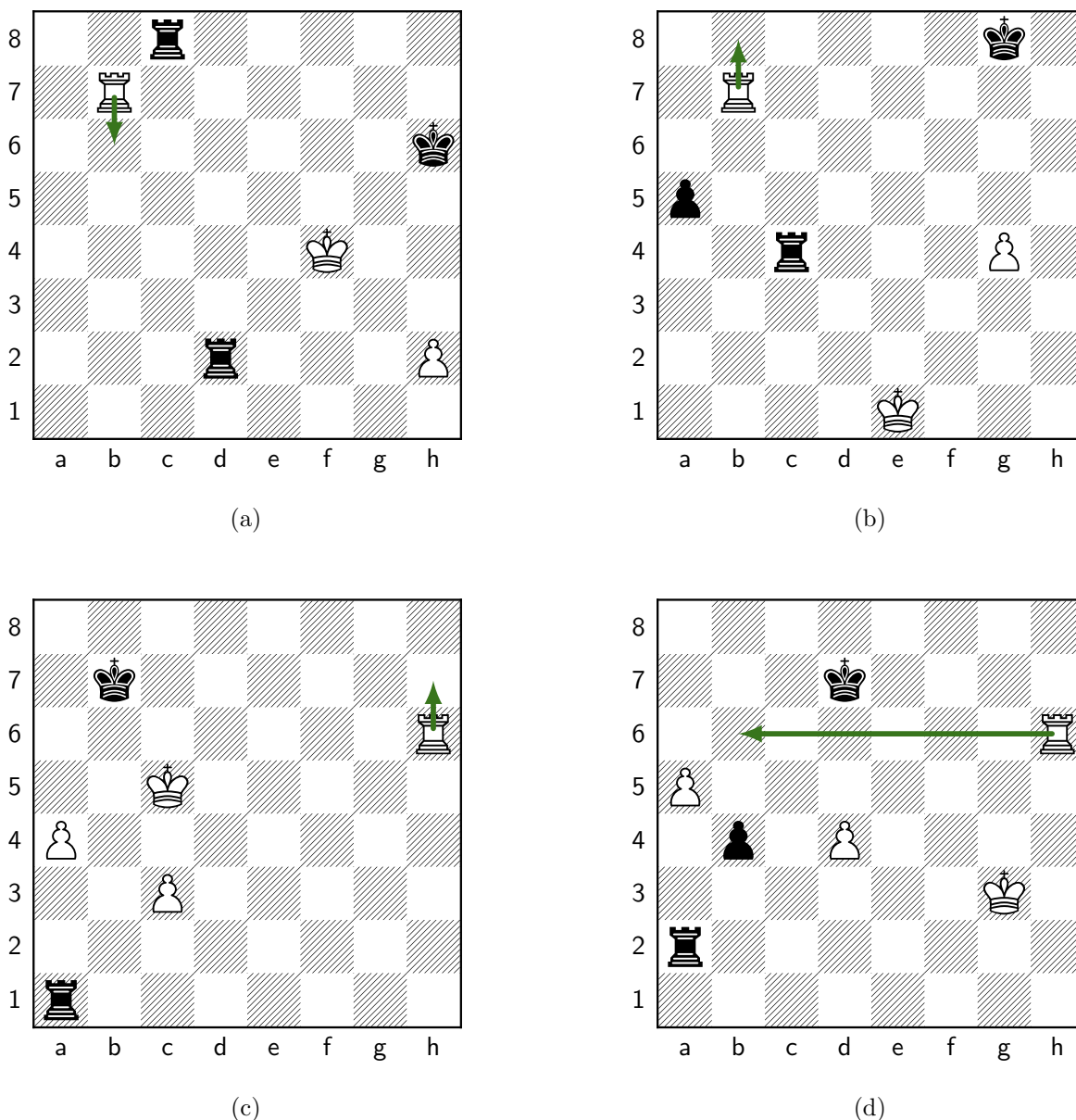
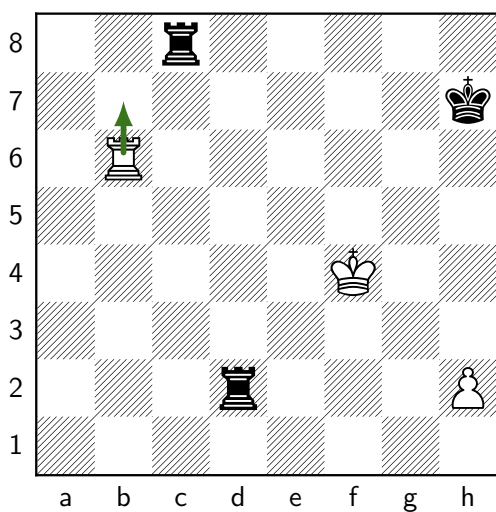
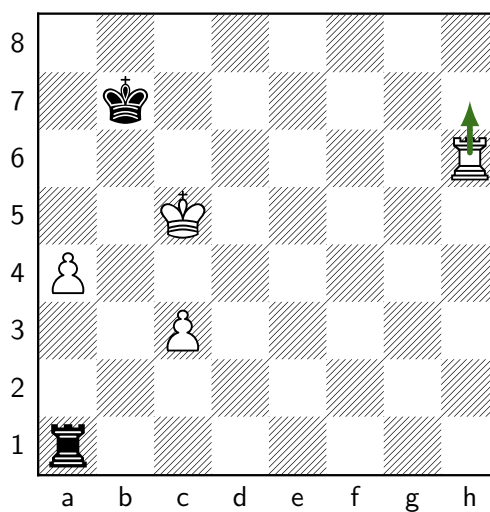


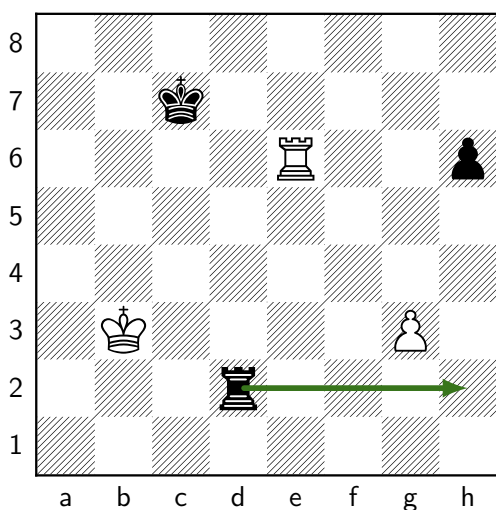
Figure 70: **Move 76, White: Rb6+.** (a) Our position; Melanie plays Rb6+ (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1203` (ply 134) [structural distance $d=3$, QAP obj=19]: the master played Rb8+; under μ our Rb7 plays the role of that game’s Rb7; cross-type roles: our rc8 as their pa5; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2011` (ply 136) [structural distance $d=2$, QAP obj=15]: the master played Rh7+; under μ our Rb7 plays the role of that game’s Rh6; cross-type roles: our rc8 as their Pc3; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3232` (ply 134) [structural distance $d=4$, QAP obj=14]: the master played Rb6; under μ our Rb7 plays the role of that game’s Rh6; cross-type roles: our rc8 as their Pd4; transferred move b7b6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rb6+ (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



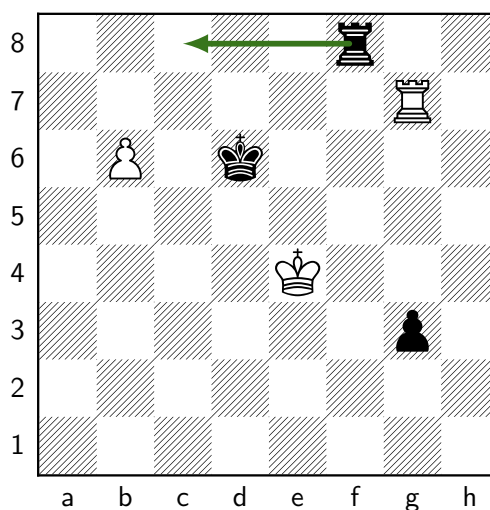
(a)



(b)

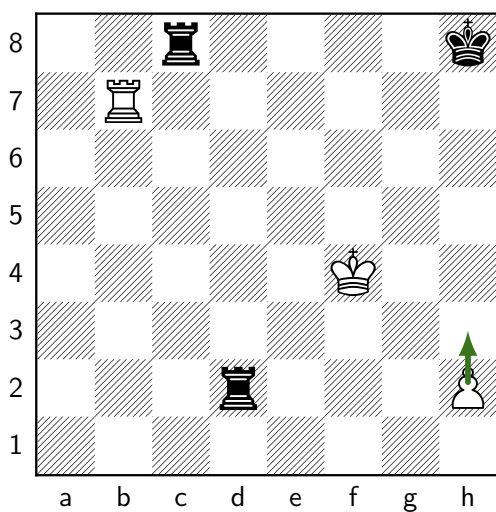


(c)

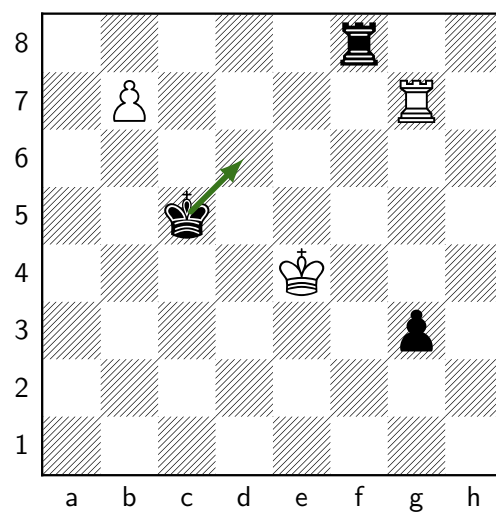


(d)

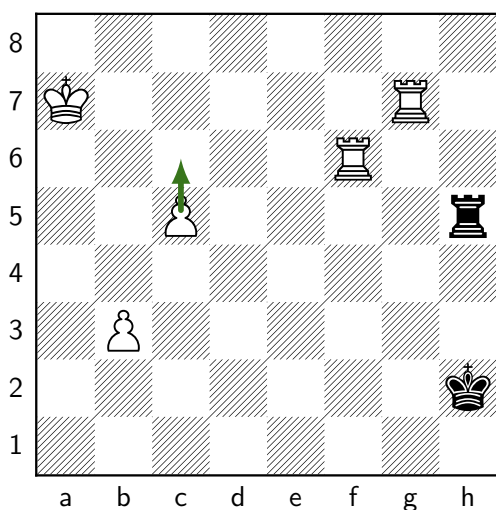
Figure 71: **Move 77, White: Rb7+.** (a) Our position; Melanie plays Rb7+ (b6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2011` (ply 136) [structural distance $d=2$, QAP obj=15]: the master played Rh7+; under μ our Rb6 plays the role of that game's Rh6; cross-type roles: our rc8 as their Pc3; transferred move b6b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2782` (ply 145) [structural distance $d=2$, QAP obj=15]: the master played Rh2; under μ our Rb6 plays the role of that game's Rd7; cross-type roles: our rc8 as their pg3; transferred move b6b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#438` (ply 155) [structural distance $d=4$, QAP obj=19]: the master played Rc8; under μ our Rb6 plays the role of that game's Rf1; cross-type roles: our rc8 as their pb6; transferred move b6b1. Synthesis: 2 retrieved analogs transferred to Rb7+ (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



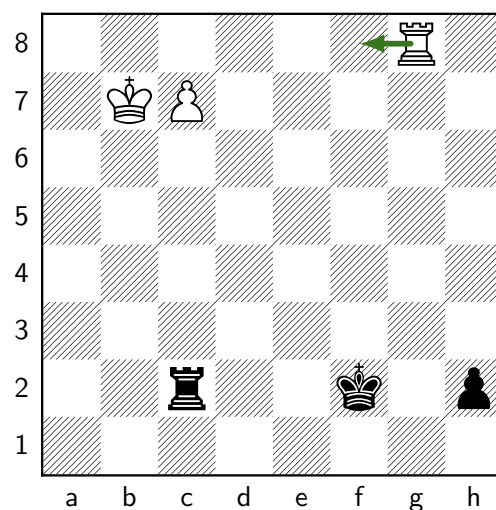
(a)



(b)

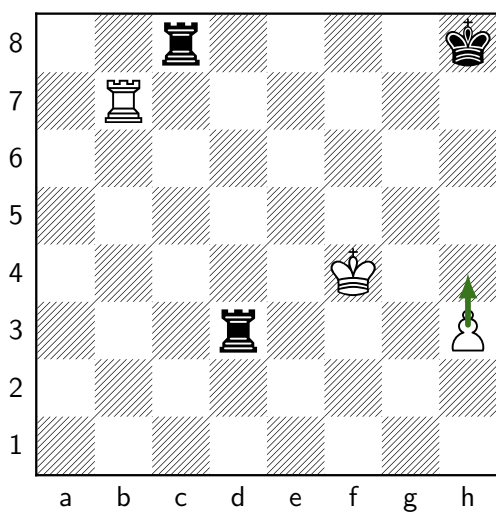


(c)

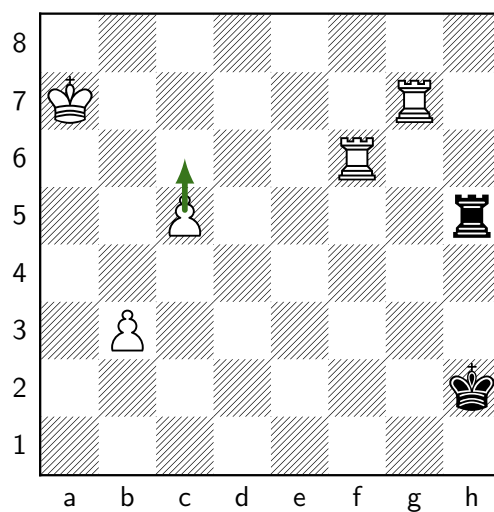


(d)

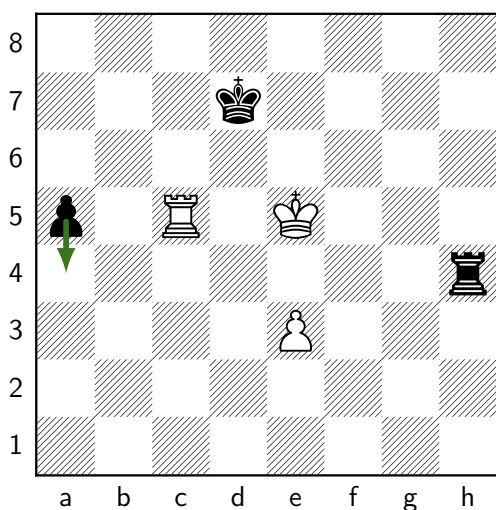
Figure 72: **Move 78, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#438` (ply 161) [structural distance $d=8$, QAP obj=19]: the master played Kd6; under μ our Kf4 plays the role of that game's Kc4; cross-type roles: our rc8 as their pb7; transferred move f4f3. (c) Retrieved from `ccr1_top_p0000#905` (ply 148) [structural distance $d=8$, QAP obj=15]: the master played c6; under μ our Ph2 plays the role of that game's Pc5; cross-type roles: our rc8 as their Pb3; transferred move h2h4. (d) Retrieved from `ccr1_top_p0002#1317` (ply 138) [structural distance $d=5$, QAP obj=15]: the master played Rf8+; under μ our Rb7 plays the role of that game's Rg8; cross-type roles: our rc8 as their ph2; transferred move b7e7. Synthesis: no retrieved analog transferred to h3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



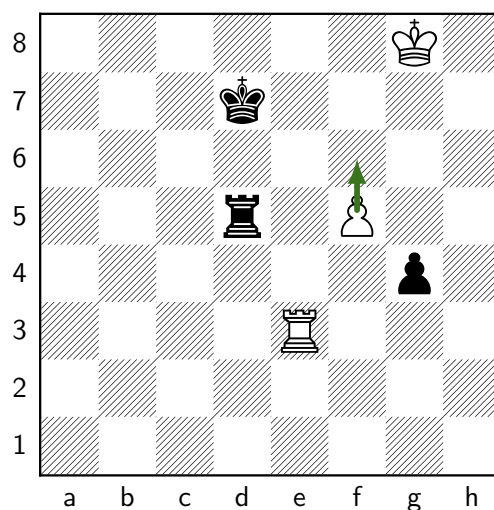
(a)



(b)

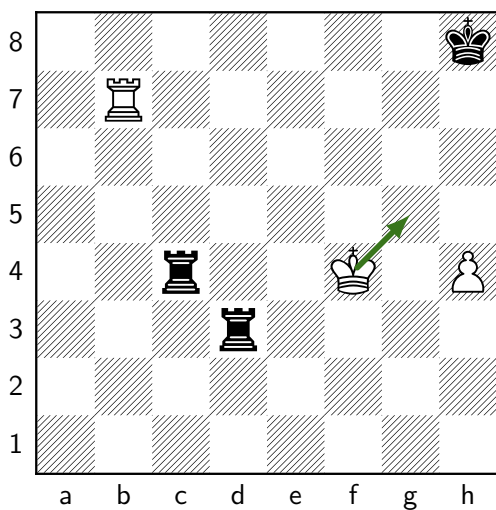


(c)

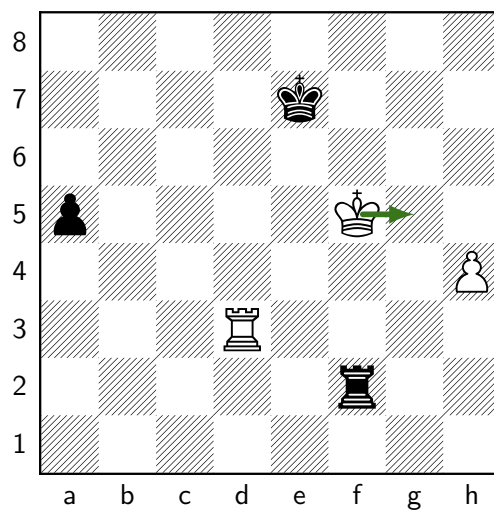


(d)

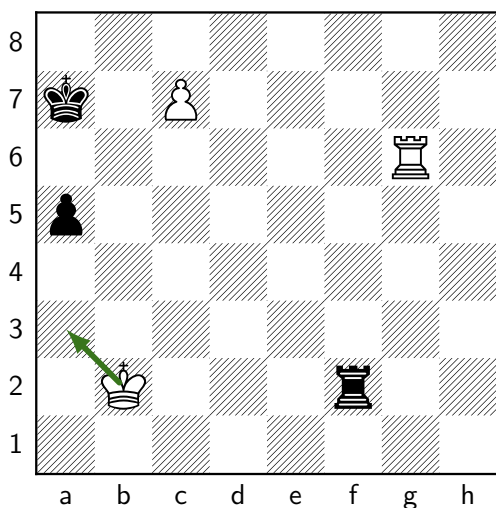
Figure 73: **Move 79, White: h4.** (a) Our position; Melanie plays **h4** (**h3h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#905` (ply 148) [structural distance $d=8$, QAP obj=15]: the master played **c6**; under μ our *Ph3* plays the role of that game’s *Pc5*; cross-type roles: our *rc8* as their *Pb3*; transferred move **h3h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1972` (ply 147) [structural distance $d=4$, QAP obj=10]: the master played **a4**; under μ our *Ph3* plays the role of that game’s *Pa4*; cross-type roles: our *rc8* as their *pe3*; transferred move **h3h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2966` (ply 156) [structural distance $d=4$, QAP obj=9]: the master played **f6**; under μ our *Ph3* plays the role of that game’s *Pf5*; cross-type roles: our *rc8* as their *pg4*; transferred move **h3h4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **h4** (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



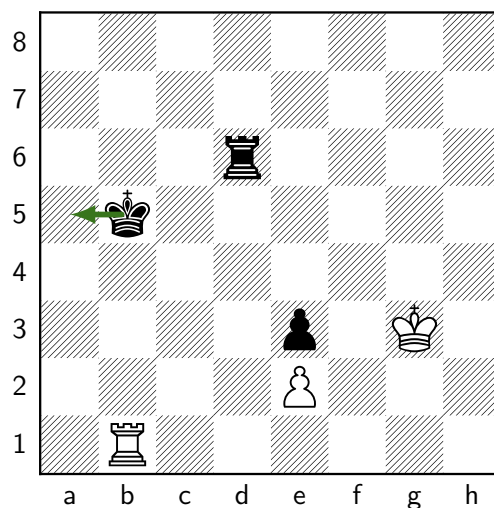
(a)



(b)

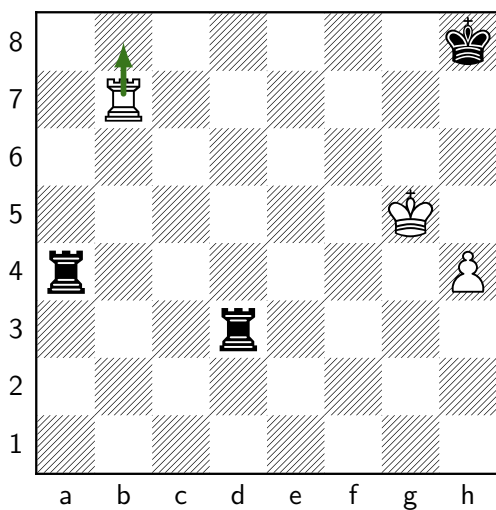


(c)

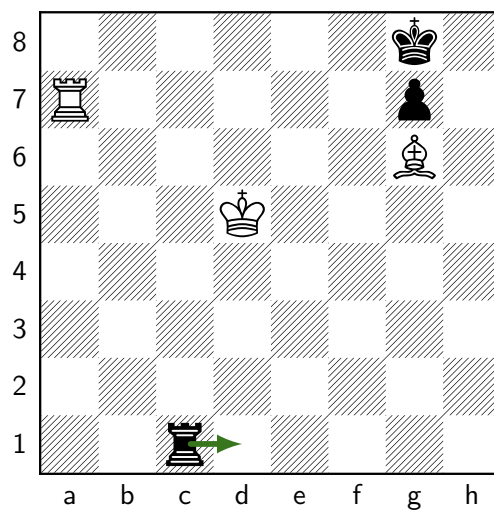


(d)

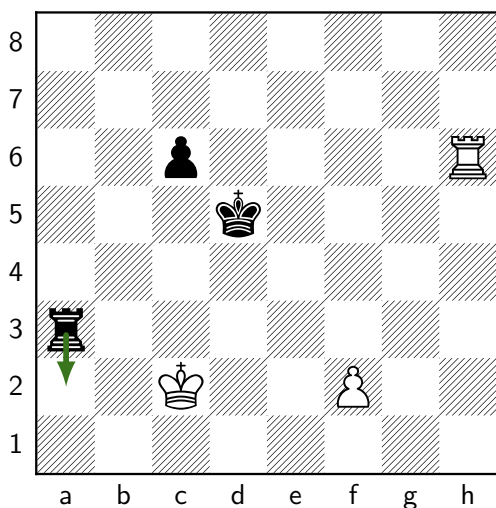
Figure 74: **Move 80, White: Kg5.** (a) Our position; Melanie plays Kg5 (f4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2643` (ply 160) [structural distance $d=3$, QAP obj=66]: the master played Kg5; under μ our Kf4 plays the role of that game’s Kf5; cross-type roles: our rd3 as their pa5; transferred move f4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2511` (ply 160) [structural distance $d=3$, QAP obj=50]: the master played Ka3; under μ our Kf4 plays the role of that game’s Kb2; cross-type roles: our rd3 as their pa5; transferred move f4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3287` (ply 175) [structural distance $d=5$, QAP obj=50]: the master played Ka5; under μ our Kf4 plays the role of that game’s Kb4; cross-type roles: our Ph4 as their Rd6, our Rb7 as their Pe3; transferred move f4g5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg5 (summed QAP objective 314, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



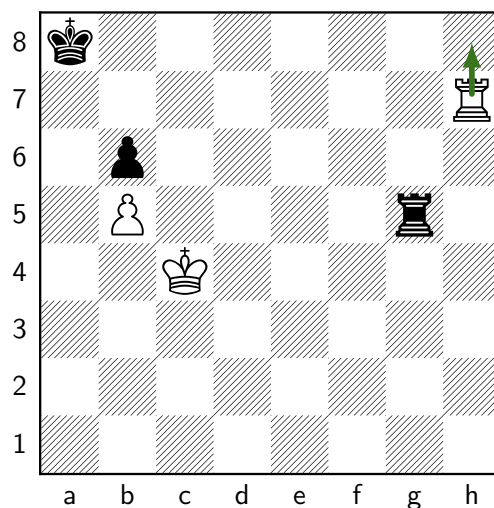
(a)



(b)

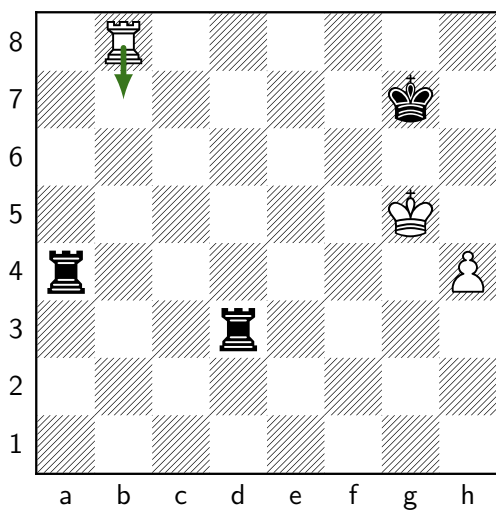


(c)

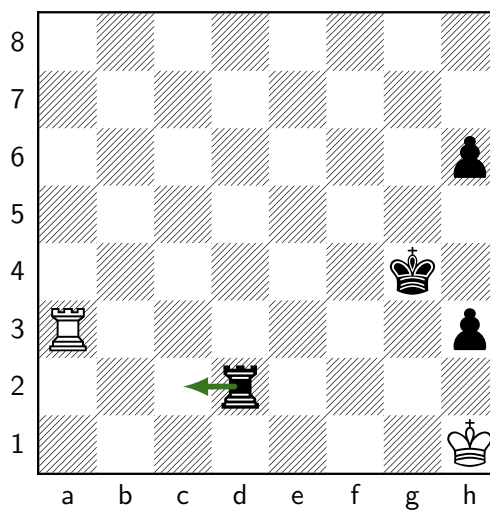


(d)

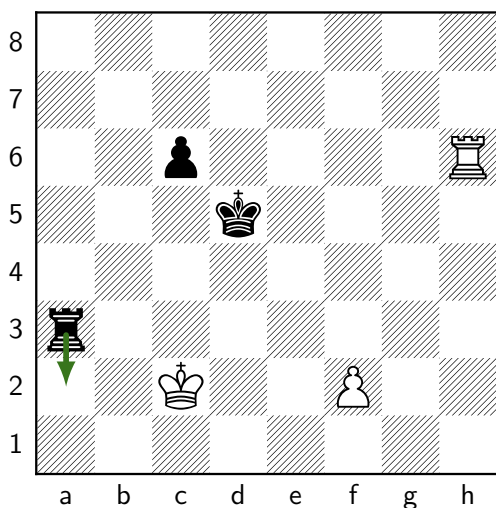
Figure 75: **Move 81, White: Rb8+ (b7b8).** (a) Our position; Melanie plays Rb8+ (b7b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1855` (ply 159) [structural distance $d=6$, QAP obj=18]: the master played Rd1+; under μ our Rb7 plays the role of that game’s Rc8; cross-type roles: our rd3 as their bg6; transferred move b7b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2428` (ply 153) [structural distance $d=4$, QAP obj=17]: the master played Ra2+; under μ our Rb7 plays the role of that game’s Ra6; cross-type roles: our rd3 as their pf2; transferred move b7b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3255` (ply 156) [structural distance $d=5$, QAP obj=16]: the master played Rh8+; under μ our Rb7 plays the role of that game’s Rh7; cross-type roles: our rd3 as their pb6; transferred move b7b8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rb8+ (summed QAP objective 97, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



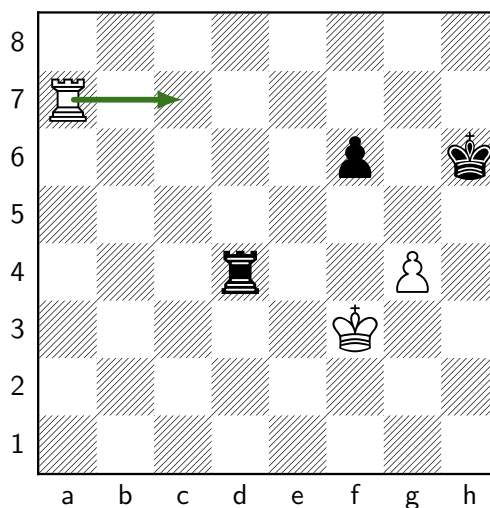
(a)



(b)

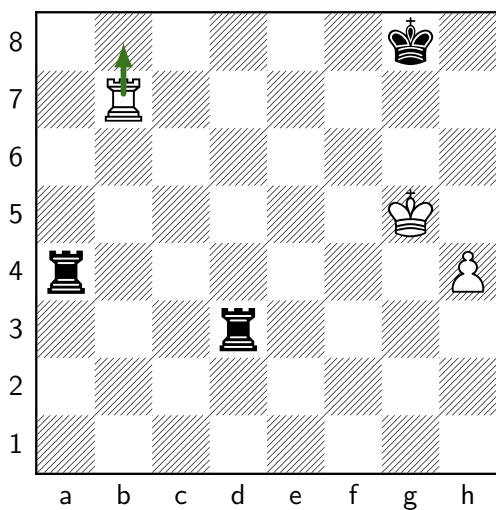


(c)

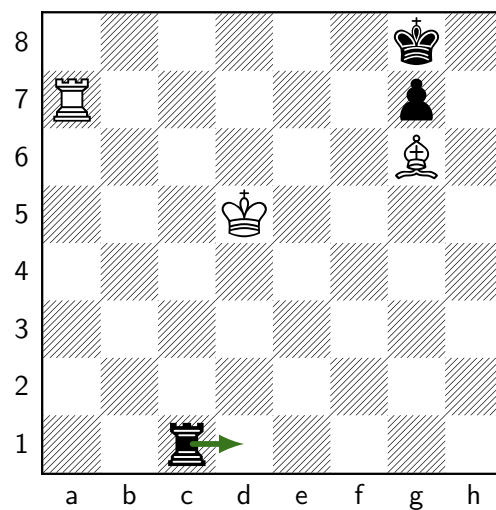


(d)

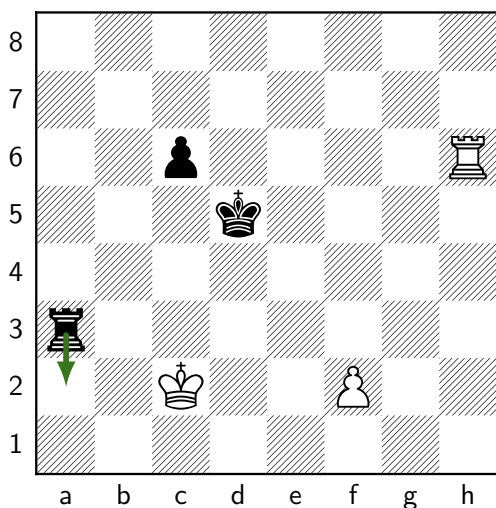
Figure 76: **Move 82, White: Rb7+.** (a) Our position; Melanie plays Rb7+ (b8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1095` (ply 161) [structural distance $d=6$, QAP obj=20]: the master played Rc2; under μ our Rb8 plays the role of that game’s Rd7; cross-type roles: our rd3 as their Ph6; transferred move b8b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2428` (ply 153) [structural distance $d=5$, QAP obj=17]: the master played Ra2+; under μ our Rb8 plays the role of that game’s Ra6; cross-type roles: our rd3 as their pf2; transferred move b8b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#63` (ply 166) [structural distance $d=6$, QAP obj=10]: the master played Rc7; under μ our Rb8 plays the role of that game’s Ra7; cross-type roles: our rd3 as their pf6; transferred move b8b7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb7+ (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



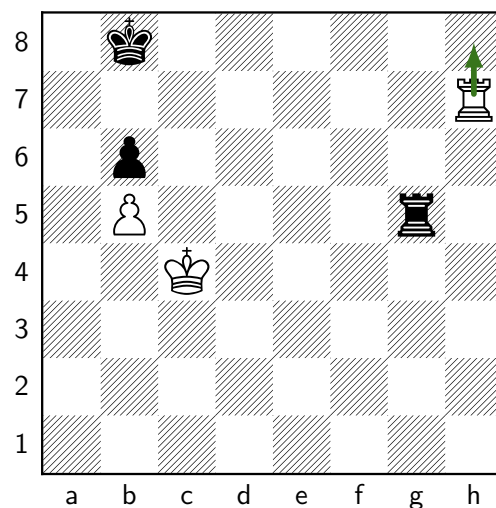
(a)



(b)



(c)



(d)

Figure 77: **Move 83, White: Rb8+ (b7b8).** (a) Our position; Melanie plays Rb8+ (b7b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1855` (ply 159) [structural distance $d=5$, QAP obj=18]: the master played Rd1+; under μ our Rb7 plays the role of that game’s Rc8; cross-type roles: our rd3 as their bg6; transferred move b7b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2428` (ply 153) [structural distance $d=4$, QAP obj=17]: the master played Ra2+; under μ our Rb7 plays the role of that game’s Ra6; cross-type roles: our rd3 as their pf2; transferred move b7b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3255` (ply 148) [structural distance $d=5$, QAP obj=16]: the master played Rh8+; under μ our Rb7 plays the role of that game’s Rh7; cross-type roles: our rd3 as their pb6; transferred move b7b8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rb8+ (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

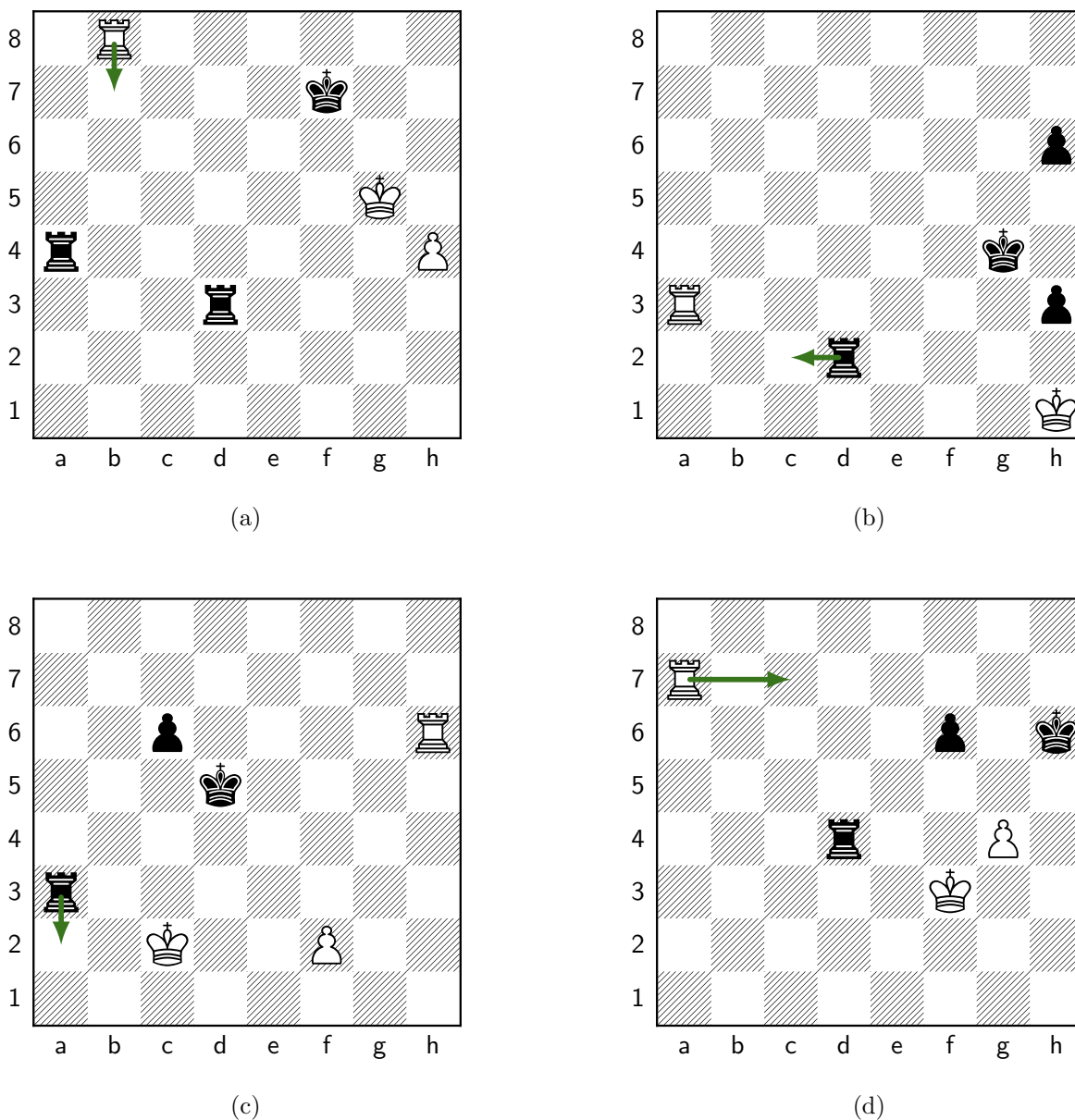
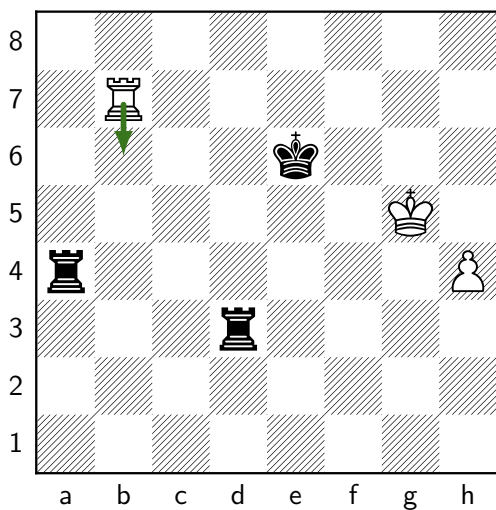
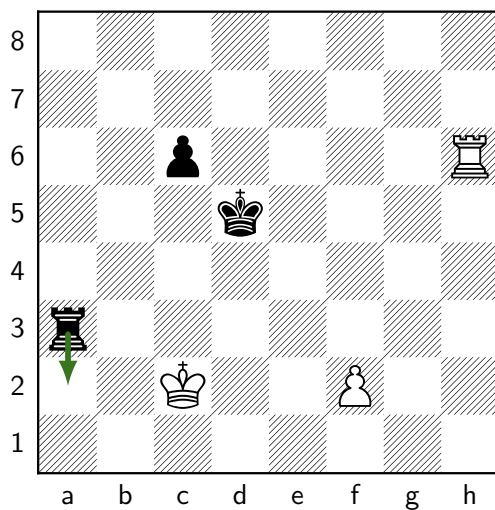


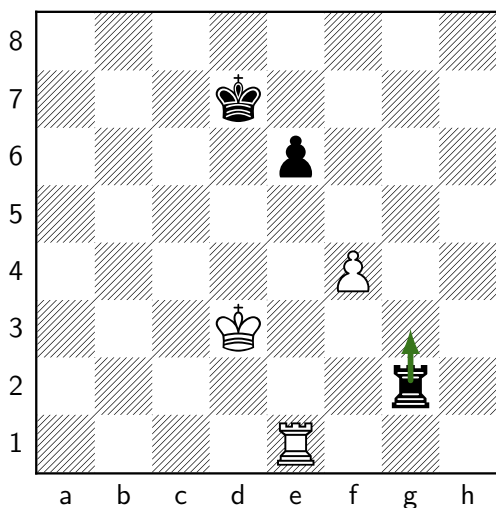
Figure 78: **Move 84, White: Rb7+.** (a) Our position; Melanie plays Rb7+ (b8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1095` (ply 161) [structural distance $d=6$, QAP obj=20]: the master played Rc2; under μ our Rb8 plays the role of that game’s Rd7; cross-type roles: our rd3 as their Ph6; transferred move b8b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2428` (ply 153) [structural distance $d=4$, QAP obj=17]: the master played Ra2+; under μ our Rb8 plays the role of that game’s Ra6; cross-type roles: our rd3 as their pf2; transferred move b8b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#63` (ply 166) [structural distance $d=6$, QAP obj=10]: the master played Rc7; under μ our Rb8 plays the role of that game’s Ra7; cross-type roles: our rd3 as their pf6; transferred move b8b7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb7+ (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



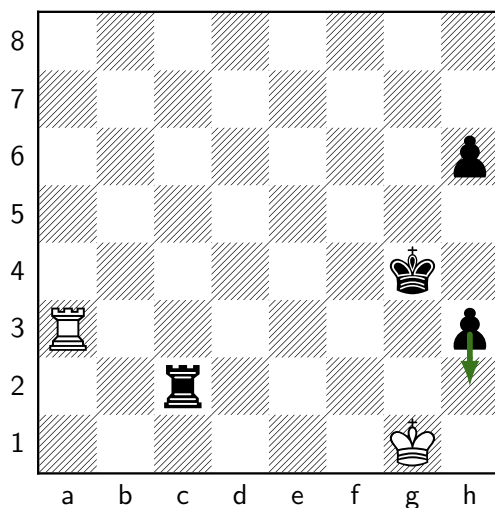
(a)



(b)

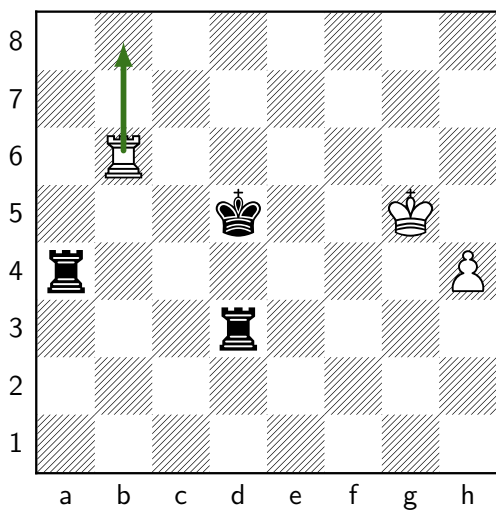


(c)

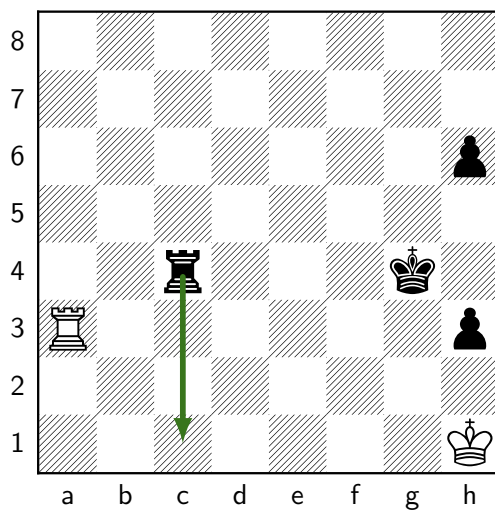


(d)

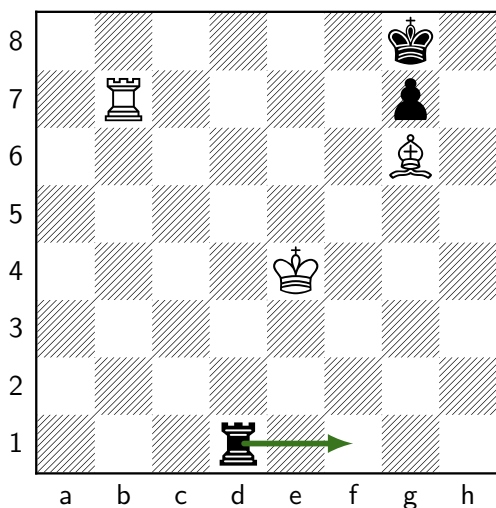
Figure 79: **Move 85, White: Rb6+.** (a) Our position; Melanie plays Rb6+ (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0002#2428` (ply 153) [structural distance $d=4$, QAP obj=17]: the master played Ra2+; under μ our Rb7 plays the role of that game’s Ra6; cross-type roles: our rd3 as their pf2; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top-p0001#3912` (ply 177) [structural distance $d=6$, QAP obj=16]: the master played Rg3+; under μ our Rb7 plays the role of that game’s Rg7; cross-type roles: our rd3 as their pf4; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccr1_top-p0001#1095` (ply 169) [structural distance $d=6$, QAP obj=20]: the master played h2+; under μ our Ph4 plays the role of that game’s Ph6; cross-type roles: our rd3 as their Ph6; transferred move h4h5. Synthesis: 2 retrieved analogs transferred to Rb6+ (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



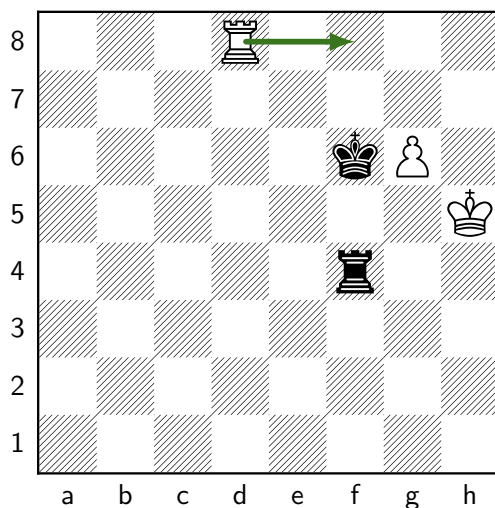
(a)



(b)



(c)



(d)

Figure 80: **Move 86, White: Rb8.** (a) Our position; Melanie plays Rb8 (b6b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1095` (ply 165) [structural distance $d=11$, QAP obj=20]: the master played Rc1+; under μ our Rb6 plays the role of that game’s Rc5; cross-type roles: our rd3 as their Ph6; transferred move b6b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1855` (ply 165) [structural distance $d=9$, QAP obj=18]: the master played Rf1; under μ our Rb6 plays the role of that game’s Rd8; cross-type roles: our rd3 as their bg6; transferred move b6b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1376` (ply 166) [structural distance $d=8$, QAP obj=2]: the master played Rf8+; under μ our Rb6 plays the role of that game’s Rd8; transferred move b6b8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb8 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

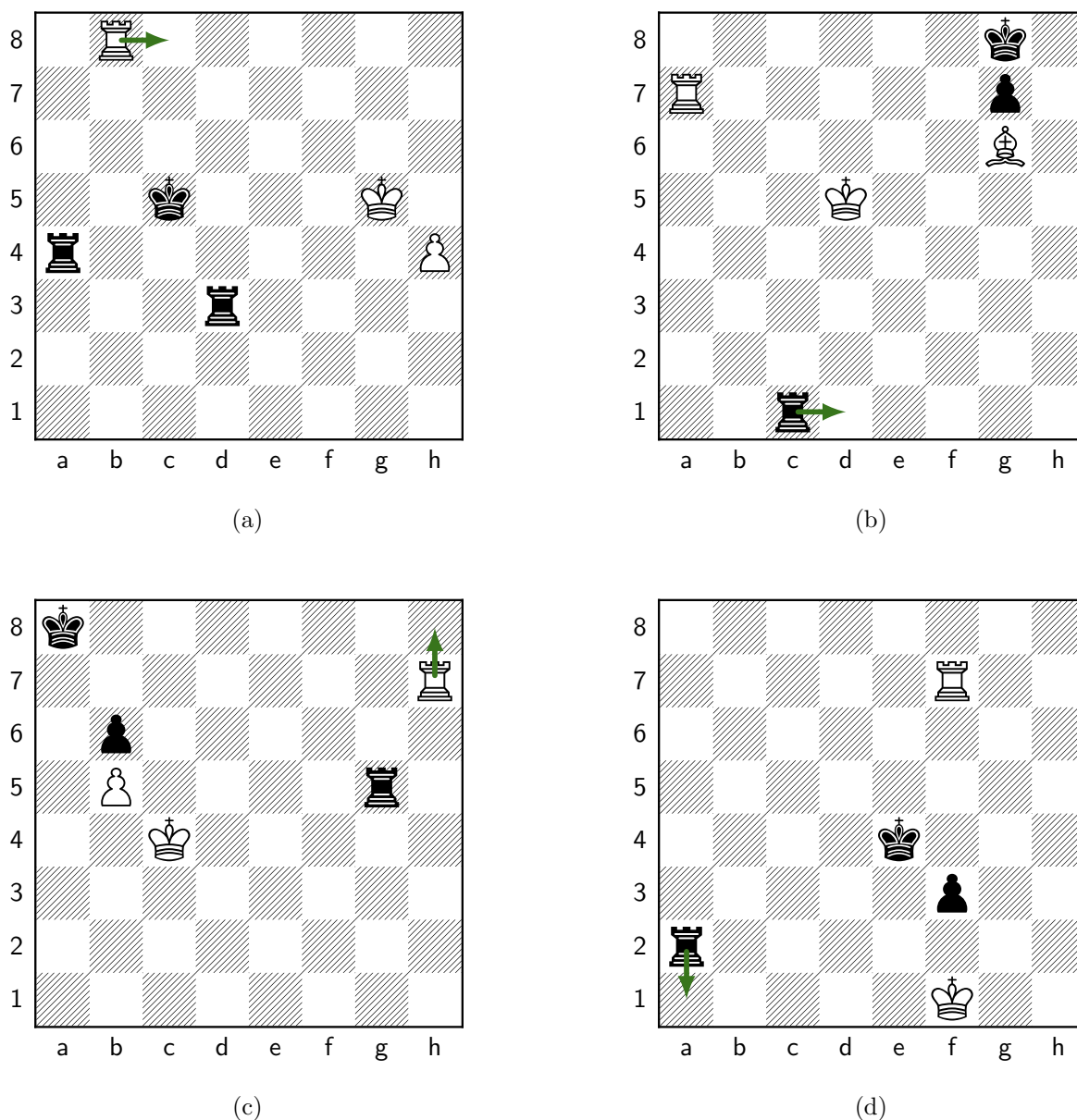
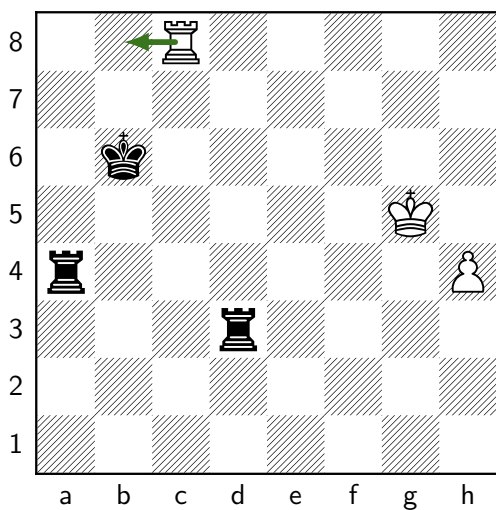
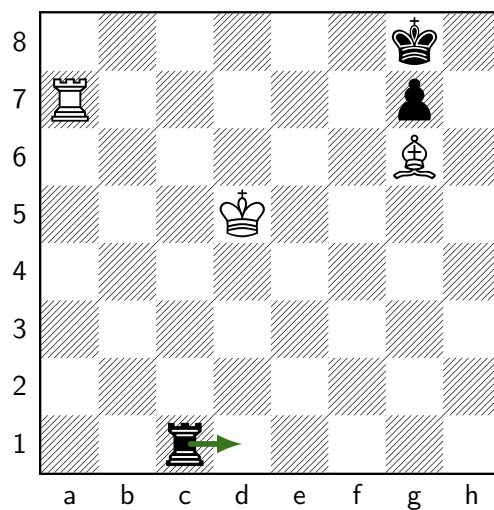


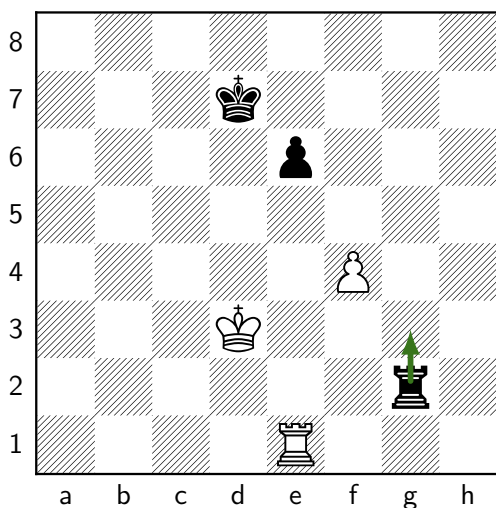
Figure 81: **Move 87, White: Rc8+.** (a) Our position; Melanie plays Rc8+ (b8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1855` (ply 159) [structural distance $d=5$, QAP obj=19]: the master played Rd1+; under μ our Rb8 plays the role of that game's Rc8; cross-type roles: our rd3 as their bg6; transferred move b8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3255` (ply 156) [structural distance $d=6$, QAP obj=15]: the master played Rh8+; under μ our Rb8 plays the role of that game's Rh7; cross-type roles: our rd3 as their pb6; transferred move b8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#139` (ply 171) [structural distance $d=6$, QAP obj=12]: the master played Ra1+; under μ our Rb8 plays the role of that game's Ra7; transferred move b8c8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rc8+ (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



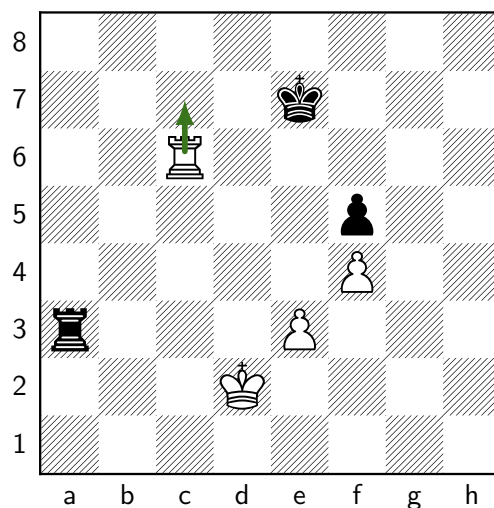
(a)



(b)

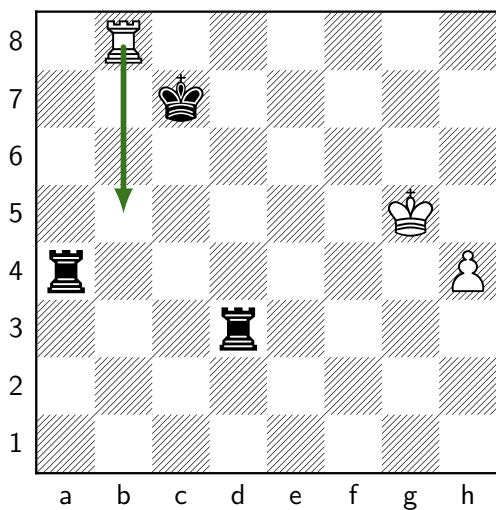


(c)

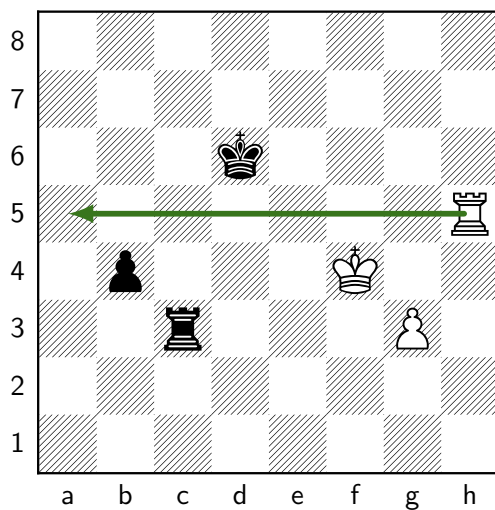


(d)

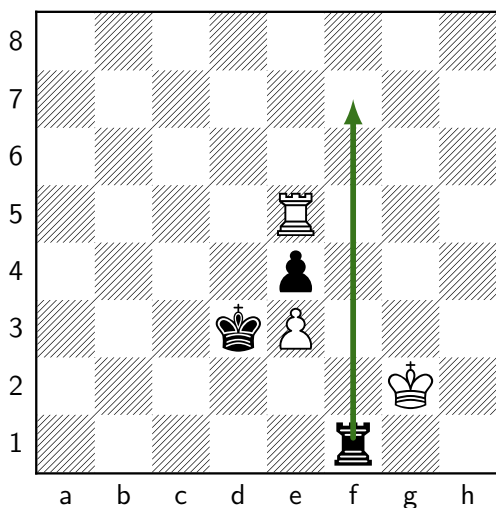
Figure 82: **Move 88, White: Rb8+.** (a) Our position; Melanie plays Rb8+ (c8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1855` (ply 159) [structural distance $d=5$, QAP obj=19]: the master played Rd1+; under μ our *Rc8* plays the role of that game’s *Rc8*; cross-type roles: our *rd3* as their *bg6*; transferred move c8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3912` (ply 177) [structural distance $d=5$, QAP obj=16]: the master played Rg3+; under μ our *Rc8* plays the role of that game’s *Rg7*; cross-type roles: our *rd3* as their *pf4*; transferred move c8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2003` (ply 176) [structural distance $d=8$, QAP obj=13]: the master played Rc7+; under μ our *Rc8* plays the role of that game’s *Rc6*; cross-type roles: our *rd3* as their *Pf4*; transferred move c8b8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rb8+ (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



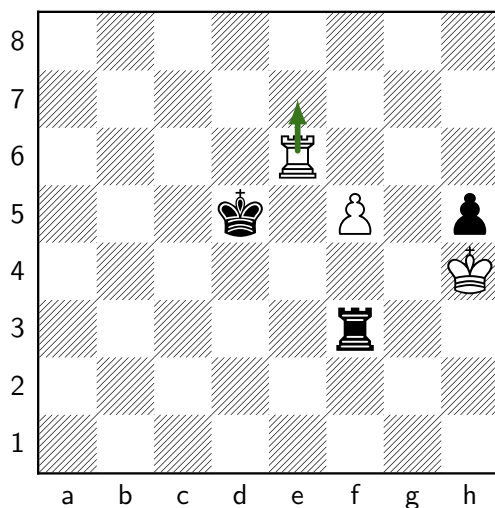
(a)



(b)

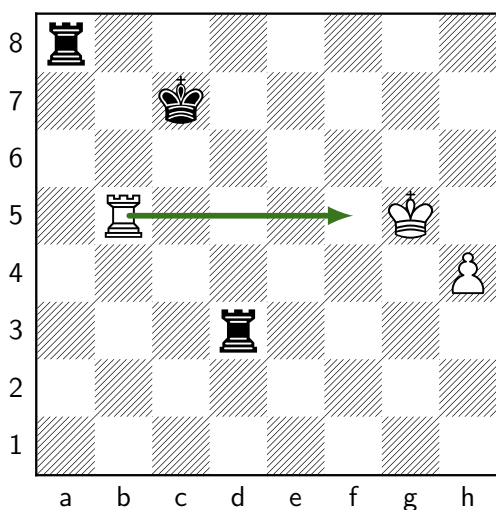


(c)

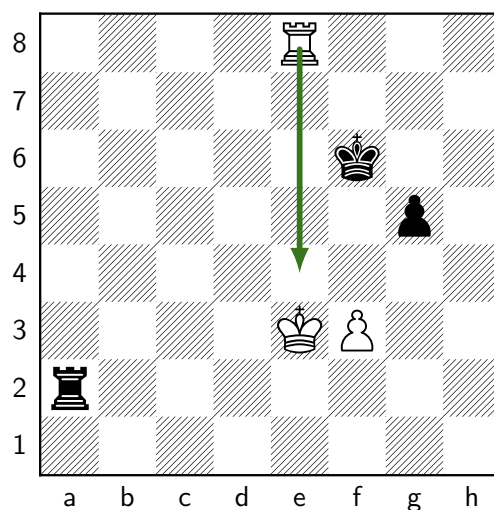


(d)

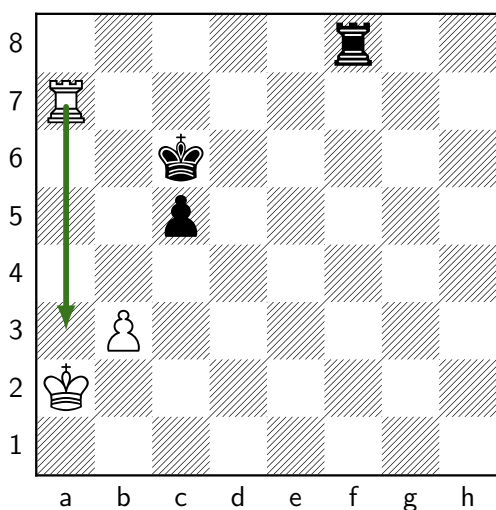
Figure 83: **Move 89, White: Rb5.** (a) Our position; Melanie plays Rb5 (b8b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1803` (ply 162) [structural distance $d=7$, QAP obj=3]: the master played Rb5; under μ our Rb8 plays the role of that game’s Rh5; cross-type roles: our rd3 as their pb4; transferred move b8b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2831` (ply 179) [structural distance $d=8$, QAP obj=21]: the master played Rf7; under μ our Rb8 plays the role of that game’s Rf8; cross-type roles: our rd3 as their pe3; transferred move b8b2. (d) Retrieved from `ccr1_top_p0001#2180` (ply 160) [structural distance $d=11$, QAP obj=19]: the master played Re7; under μ our Rb8 plays the role of that game’s Re6; cross-type roles: our rd3 as their Pf5; transferred move b8f8. Synthesis: 1 retrieved analog transferred to Rb5 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



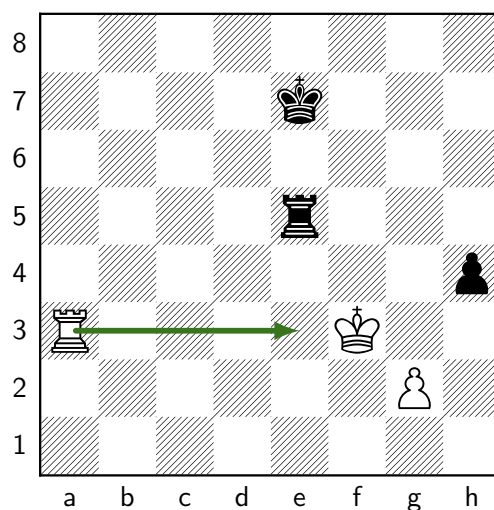
(a)



(b)

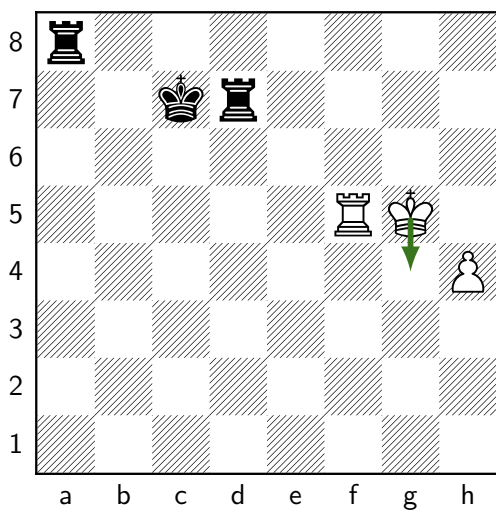


(c)

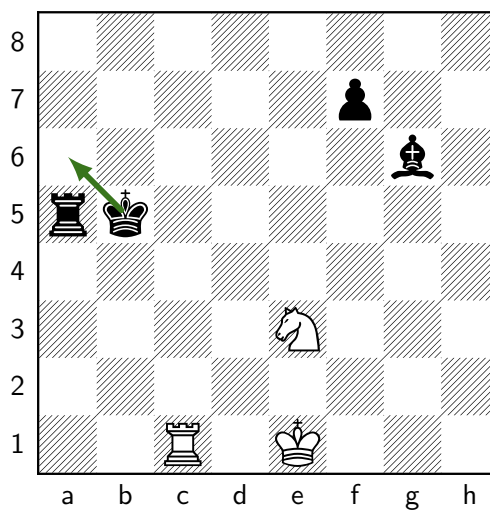


(d)

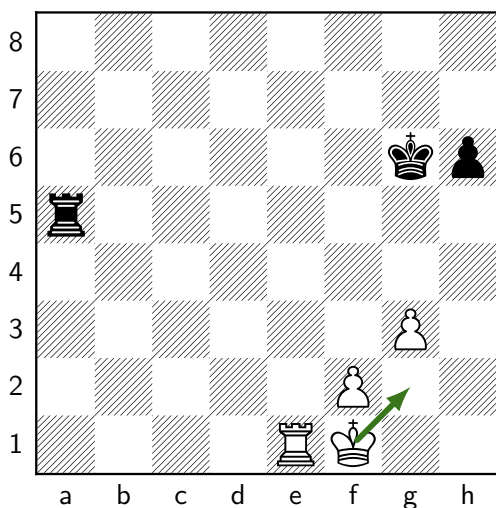
Figure 84: **Move 90, White: Rf5.** (a) Our position; Melanie plays Rf5 (b5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2474` (ply 170) [structural distance $d=9$, QAP obj=2]: the master played Re4; under μ our Rb5 plays the role of that game's Re8; cross-type roles: our ra8 as their Pf3; transferred move b5f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3960` (ply 178) [structural distance $d=9$, QAP obj=2]: the master played Ra3; under μ our Rb5 plays the role of that game's Ra7; cross-type roles: our rd3 as their pc5; transferred move b5f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1046` (ply 168) [structural distance $d=9$, QAP obj=2]: the master played Re3; under μ our Rb5 plays the role of that game's Ra3; cross-type roles: our ra8 as their ph4; transferred move b5f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rf5 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



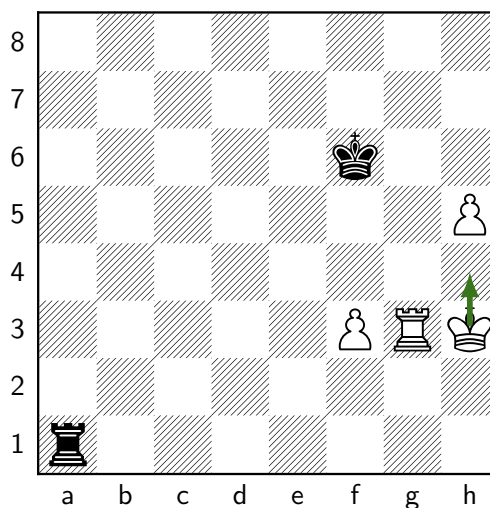
(a)



(b)

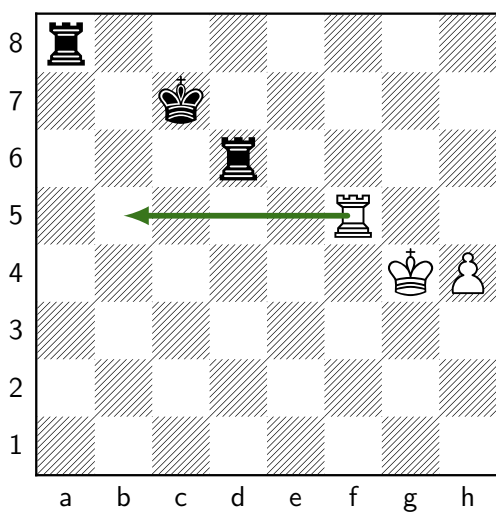


(c)

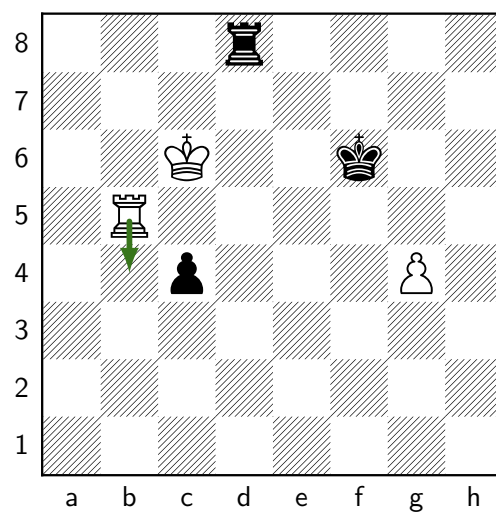


(d)

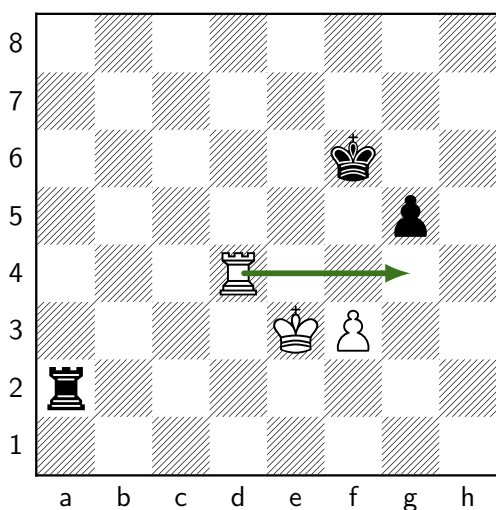
Figure 85: **Move 91, White: Kg4.** (a) Our position; Melanie plays Kg4 ($g5g4$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#7` (ply 177) [structural distance $d=16$, QAP obj=3]: the master played $Ka6$; under μ our $Kg5$ plays the role of that game’s $Kb4$; cross-type roles: our $ra8$ as their $Bg6$, our $Ph4$ as their $ne3$; transferred move $g5g4$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3386` (ply 176) [structural distance $d=15$, QAP obj=2]: the master played $Kg2$; under μ our $Kg5$ plays the role of that game’s $Kf1$; cross-type roles: our $rd7$ as their $ph6$; transferred move $g5g4$ (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3559` (ply 170) [structural distance $d=17$, QAP obj=2]: the master played $Kh4$; under μ our $Kg5$ plays the role of that game’s $Kh3$; cross-type roles: our $rd7$ as their $Ph5$; transferred move $g5g4$ (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg4 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



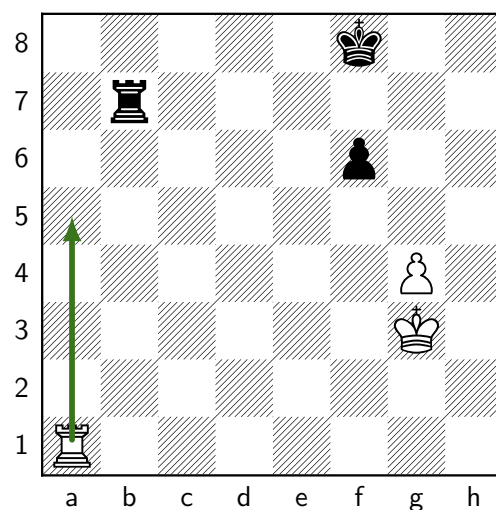
(a)



(b)

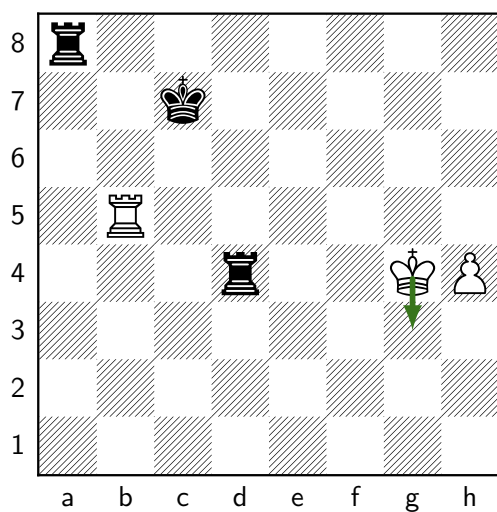


(c)

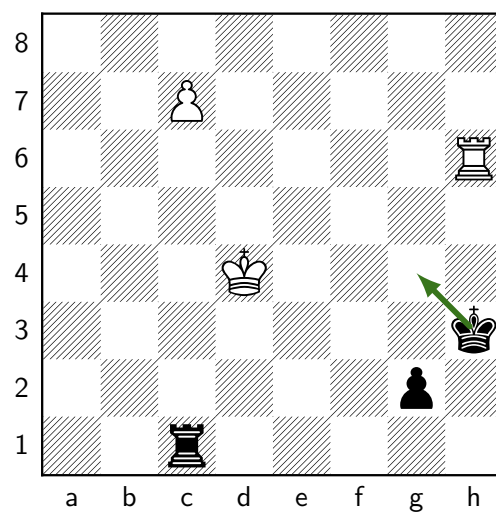


(d)

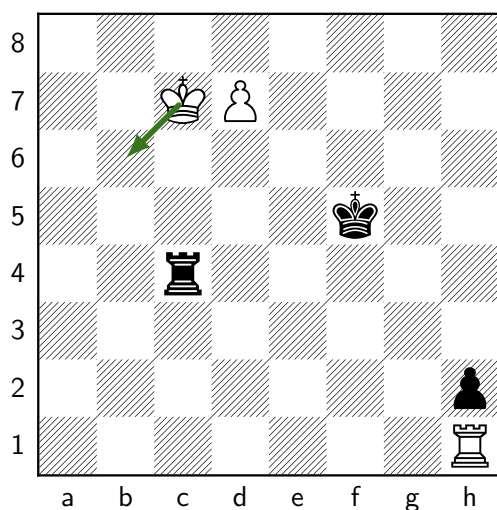
Figure 86: **Move 92, White: Rb5.** (a) Our position; Melanie plays Rb5 (f5b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1728` (ply 186) [structural distance $d=12$, QAP obj=2]: the master played Rb4; under μ our Rf5 plays the role of that game’s Rb5; cross-type roles: our rd6 as their pc4; transferred move f5b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2474` (ply 174) [structural distance $d=11$, QAP obj=2]: the master played Rg4; under μ our Rf5 plays the role of that game’s Rd4; cross-type roles: our ra8 as their pg5; transferred move f5b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#578` (ply 178) [structural distance $d=11$, QAP obj=1]: the master played Ra5; under μ our Rf5 plays the role of that game’s Ra1; cross-type roles: our rd6 as their pf6; transferred move f5b5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb5 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



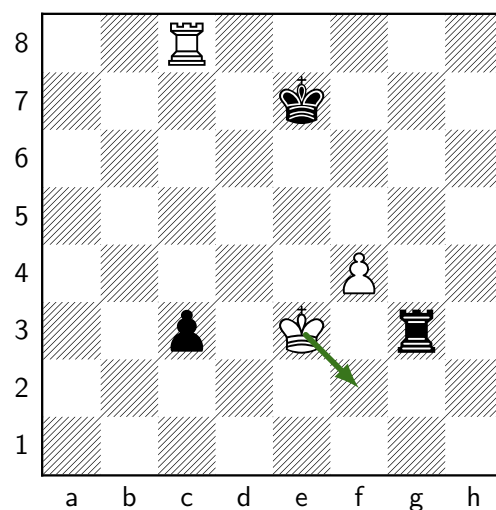
(a)



(b)

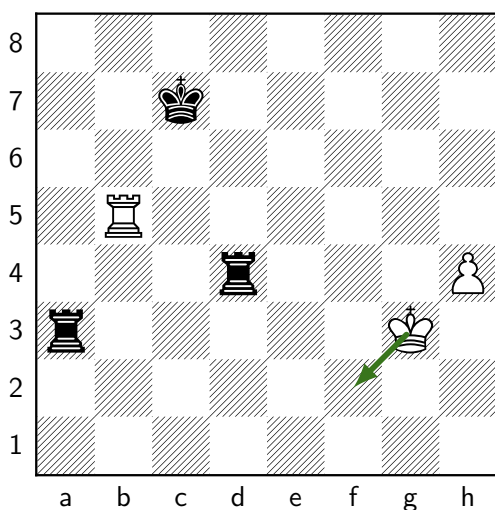


(c)

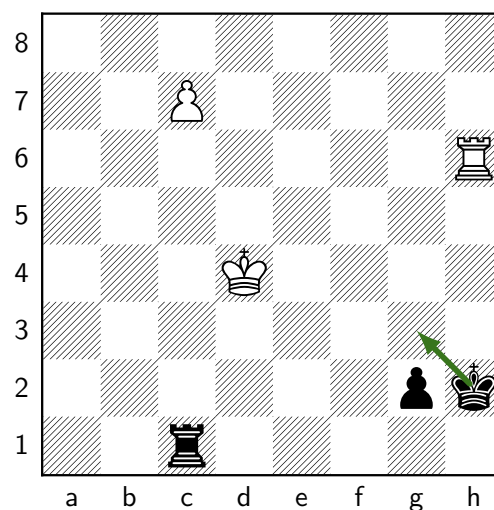


(d)

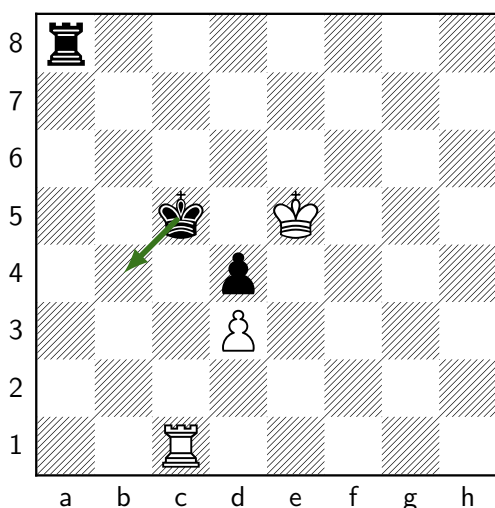
Figure 87: **Move 93, White: Kg3**. (a) Our position; Melanie plays Kg3 (g4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1553` (ply 175) [structural distance $d=5$, QAP obj=73]: the master played Kg4; under μ our Kg4 plays the role of that game's Kh6; cross-type roles: our ra8 as their pc7; transferred move g4g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1987` (ply 168) [structural distance $d=6$, QAP obj=70]: the master played Kb6; under μ our Kg4 plays the role of that game's Kc7; cross-type roles: our ra8 as their ph2; transferred move g4g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1355` (ply 170) [structural distance $d=5$, QAP obj=50]: the master played Kf2; under μ our Kg4 plays the role of that game's Ke3; cross-type roles: our ra8 as their pc3; transferred move g4g3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kg3 (summed QAP objective 324, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



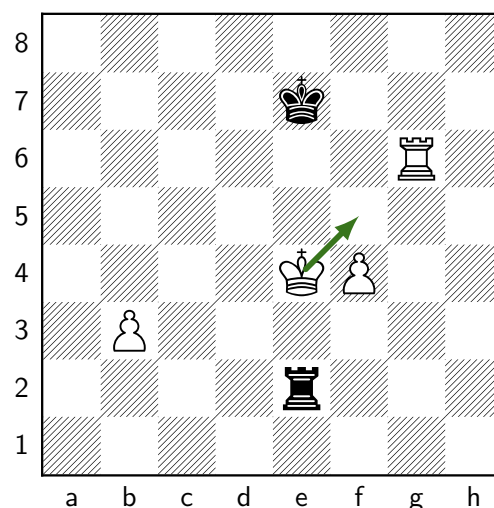
(a)



(b)

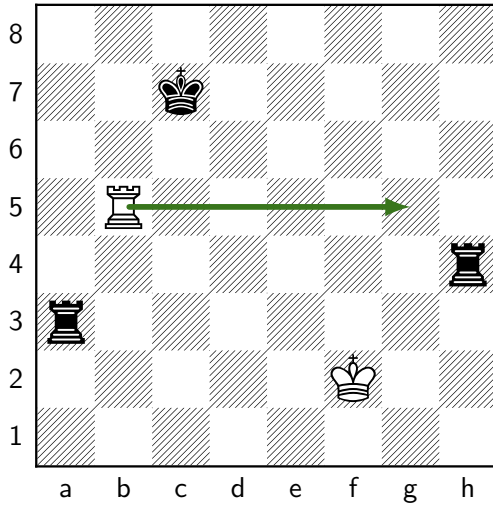


(c)

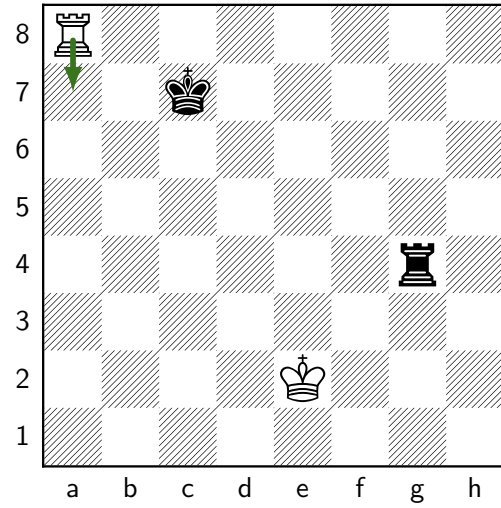


(d)

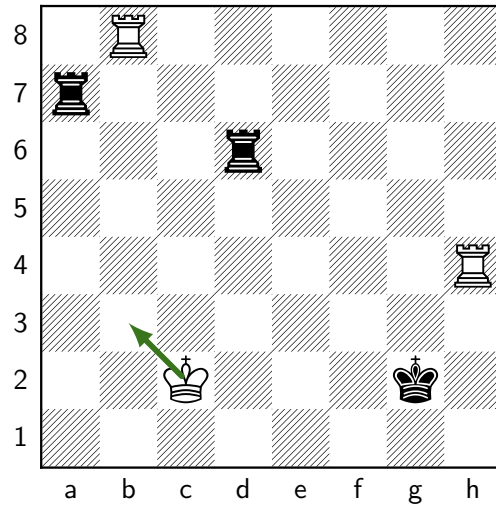
Figure 88: **Move 94, White: Kf2.** (a) Our position; Melanie plays Kf2 (g3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1553` (ply 171) [structural distance $d=9$, QAP obj=47]: the master played Kg3; under μ our Kg3 plays the role of that game’s Kh7; cross-type roles: our rd4 as their pc7; transferred move g3h2. (c) Retrieved from `ccrl_top_p0000#1150` (ply 179) [structural distance $d=7$, QAP obj=46]: the master played Kb4; under μ our Kg3 plays the role of that game’s Kc4; cross-type roles: our rd4 as their pd3; transferred move g3g2. (d) Retrieved from `ccrl_top_p0003#1490` (ply 190) [structural distance $d=8$, QAP obj=25]: the master played Kf5; under μ our Kg3 plays the role of that game’s Ke4; cross-type roles: our rd4 as their Pb3; transferred move g3g2. Synthesis: no retrieved analog transferred to Kf2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



(a)

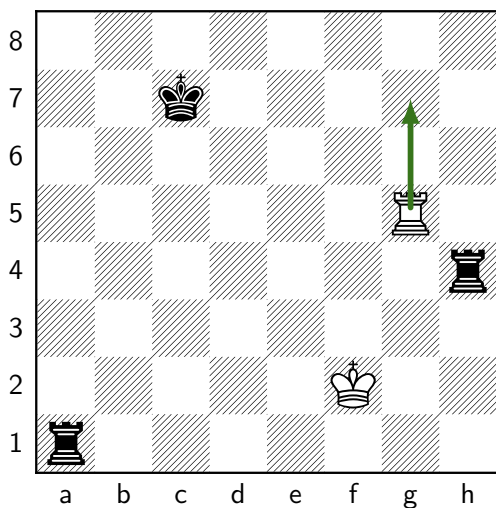


(b)

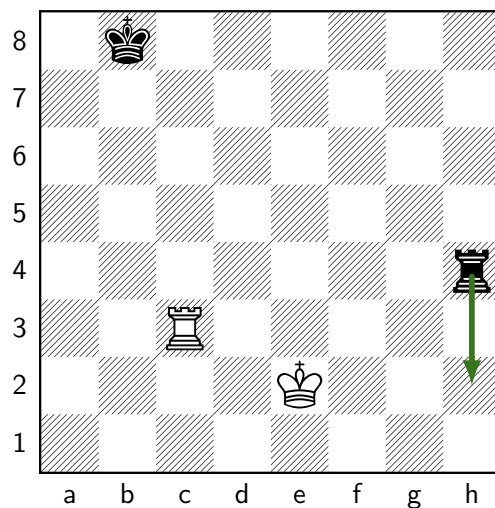


(c)

Figure 89: **Move 95, White: Rg5**. (a) Our position; Melanie plays Rg5 (b5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 172) [structural distance $d=3$, QAP obj=1]: the master played Ra7+; under μ our Rb5 plays the role of that game's Ra8; transferred move b5g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3387` (ply 182) [structural distance $d=2$, QAP obj=2]: the master played Kb3; under μ our Kf2 plays the role of that game's Kc2; transferred move f2e2. Synthesis: 6 retrieved analogs transferred to Rg5 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

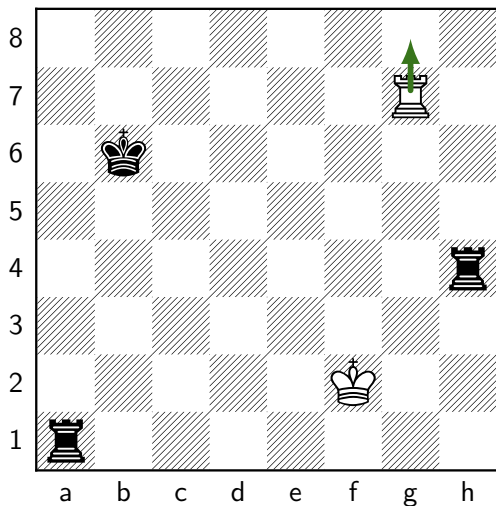


(a)

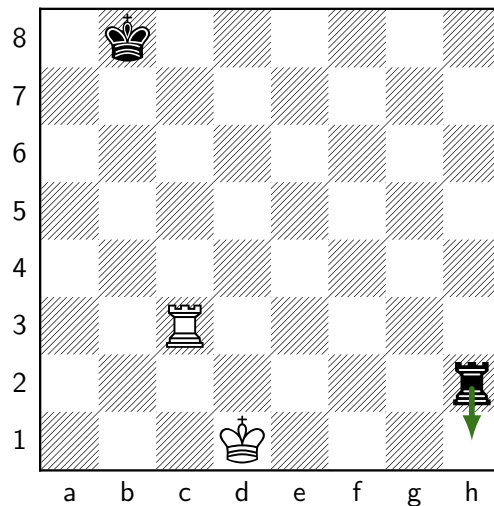


(b)

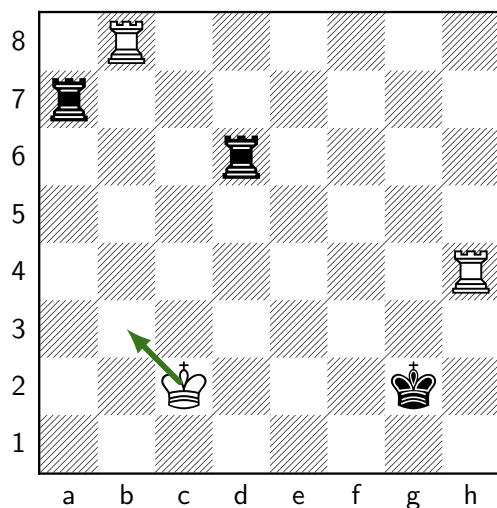
Figure 90: **Move 96, White: Rg7+.** (a) Our position; Melanie plays Rg7+ (g5g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#147` (ply 183) [structural distance $d=2$, QAP obj=1]: the master played Rh2+; under μ our Rg5 plays the role of that game’s Rh5; transferred move g5g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rg7+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

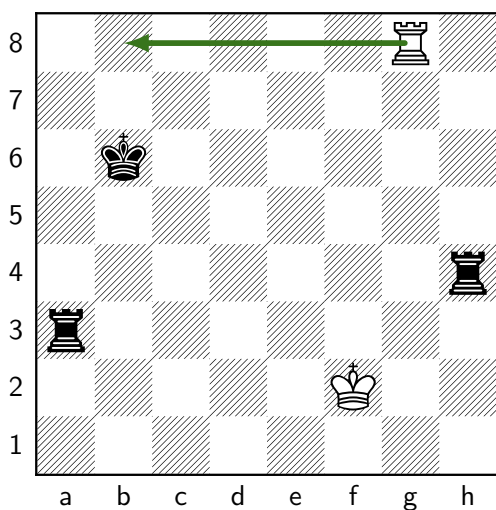


(b)

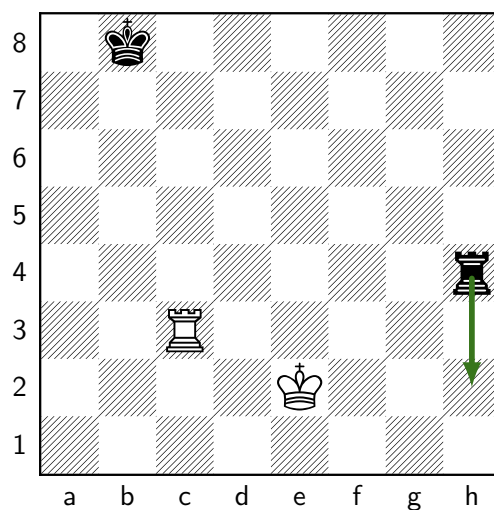


(c)

Figure 91: **Move 97, White: Rg8**. (a) Our position; Melanie plays Rg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 185) [structural distance $d=2$, QAP obj=1]: the master played Rh1+; under μ our Rg7 plays the role of that game's Rh7; transferred move g7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3387` (ply 182) [structural distance $d=2$, QAP obj=2]: the master played Kb3; under μ our Kf2 plays the role of that game's Kc2; transferred move f2g2. Synthesis: 3 retrieved analogs transferred to Rg8 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

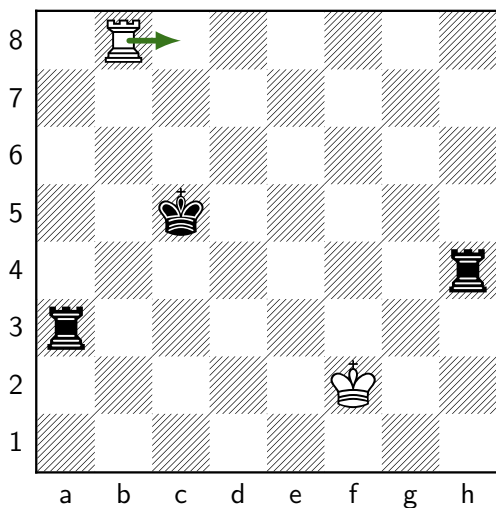


(a)

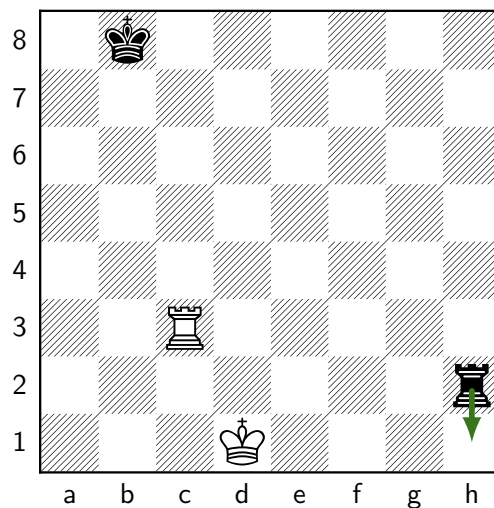


(b)

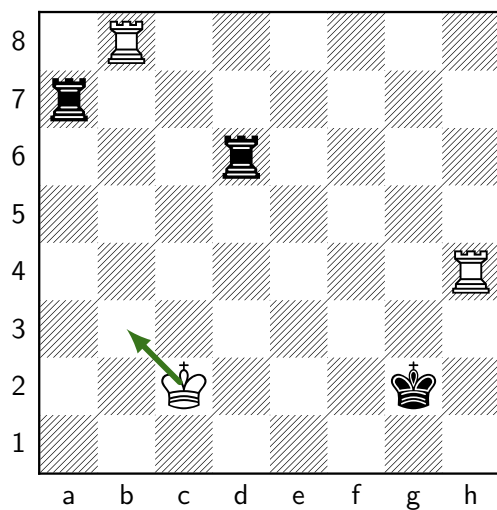
Figure 92: **Move 98, White: Rb8+.** (a) Our position; Melanie plays Rb8+ (g8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#147` (ply 183) [structural distance $d=2$, QAP obj=1]: the master played Rh2+; under μ our Rg8 plays the role of that game’s Rh5; transferred move g8b8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb8+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

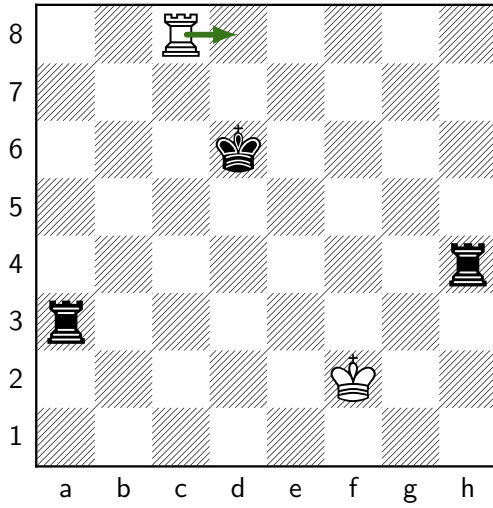


(b)

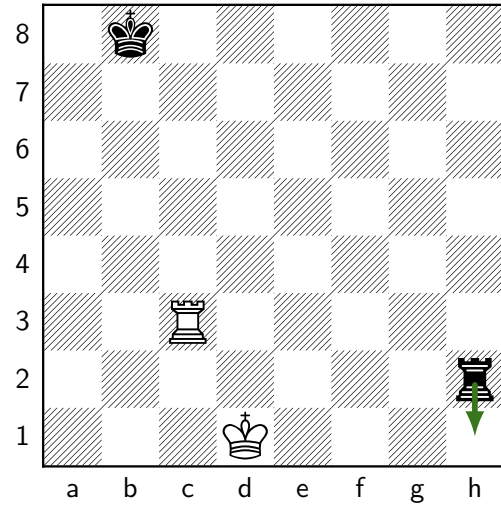


(c)

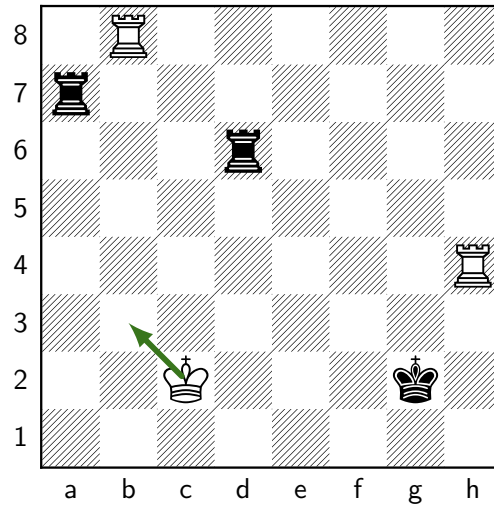
Figure 93: **Move 99, White: Rc8+.** (a) Our position; Melanie plays Rc8+ (b8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 185) [structural distance $d=2$, QAP obj=1]: the master played Rh1+; under μ our Rb8 plays the role of that game's Rh7; transferred move b8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3387` (ply 182) [structural distance $d=2$, QAP obj=2]: the master played Kb3; under μ our Kf2 plays the role of that game's Kc2; transferred move f2e2. Synthesis: 4 retrieved analogs transferred to Rc8+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

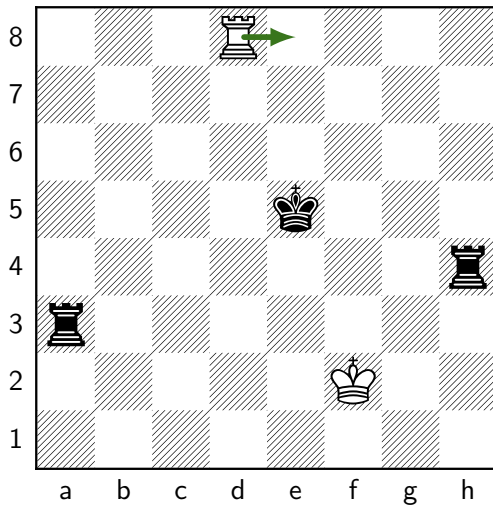


(b)

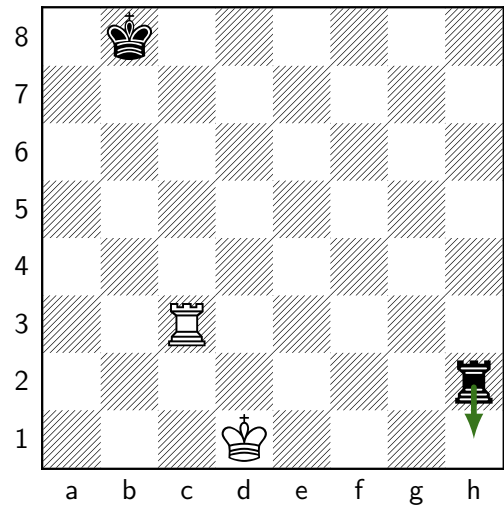


(c)

Figure 94: **Move 100, White: Rd8+.** (a) Our position; Melanie plays Rd8+ (c8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 185) [structural distance $d=2$, QAP obj=1]: the master played Rh1+; under μ our Rc8 plays the role of that game's Rh7; transferred move c8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3387` (ply 182) [structural distance $d=2$, QAP obj=2]: the master played Kb3; under μ our Kf2 plays the role of that game's Kc2; transferred move f2e2. Synthesis: 4 retrieved analogs transferred to Rd8+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

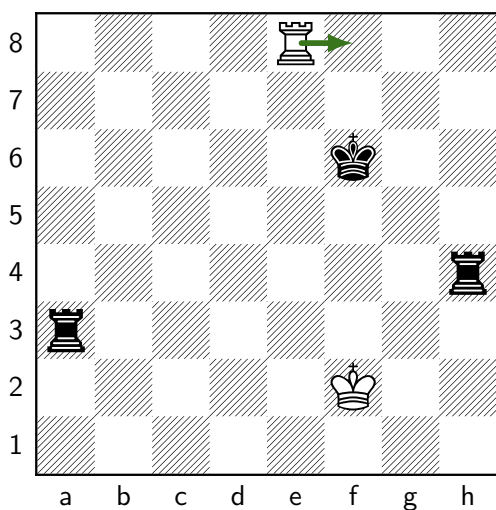


(a)

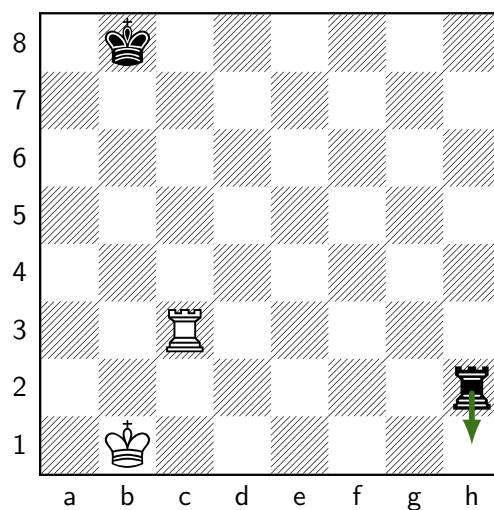


(b)

Figure 95: **Move 101, White: Re8+.** (a) Our position; Melanie plays **Re8+** (**d8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 185) [structural distance $d=2$, QAP obj=1]: the master played **Rh1+**; under μ our *Rd8* plays the role of that game’s *Rh7*; transferred move **d8e8** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Re8+** (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

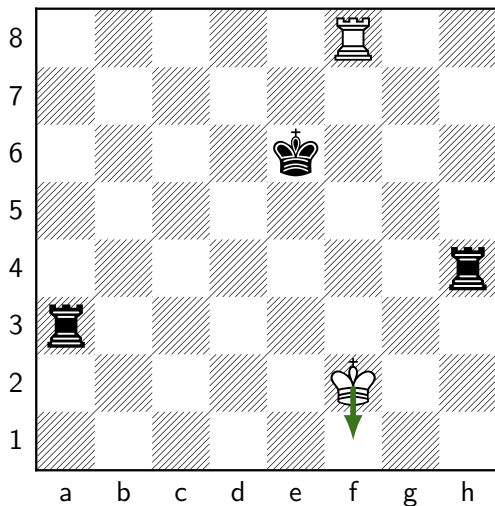


(a)

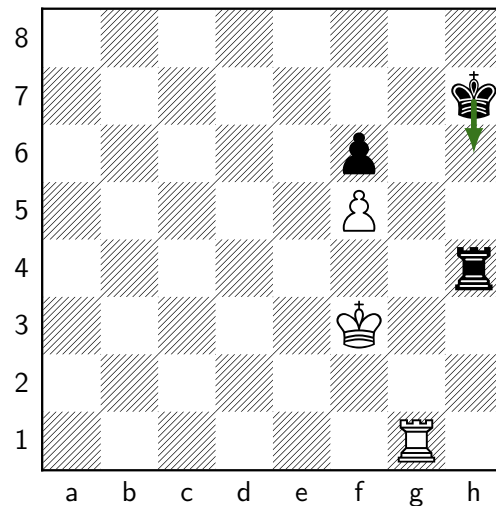


(b)

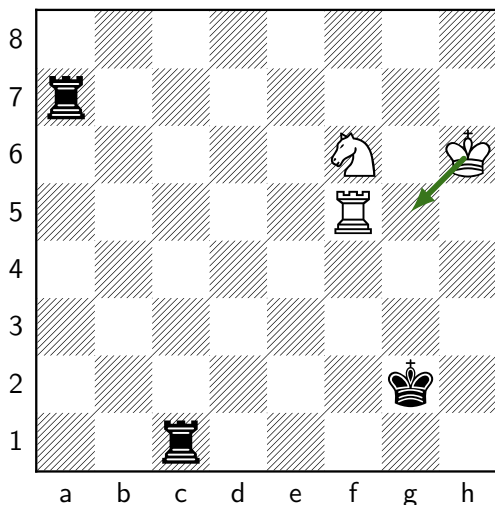
Figure 96: **Move 102, White: Rf8+.** (a) Our position; Melanie plays Rf8+ (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 189) [structural distance $d=2$, QAP obj=1]: the master played Rh1+; under μ our *Re8* plays the role of that game's *Rh7*; transferred move e8f8 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Rf8+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



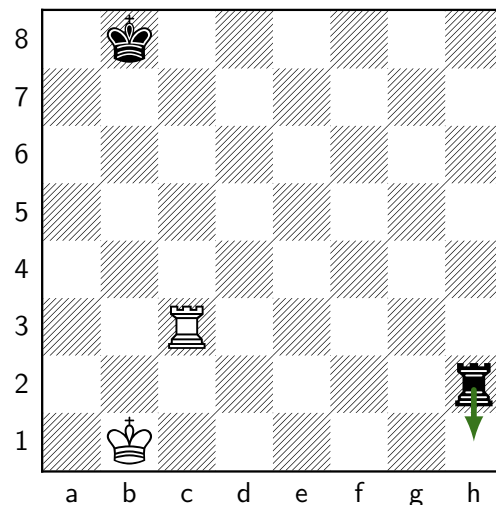
(a)



(b)

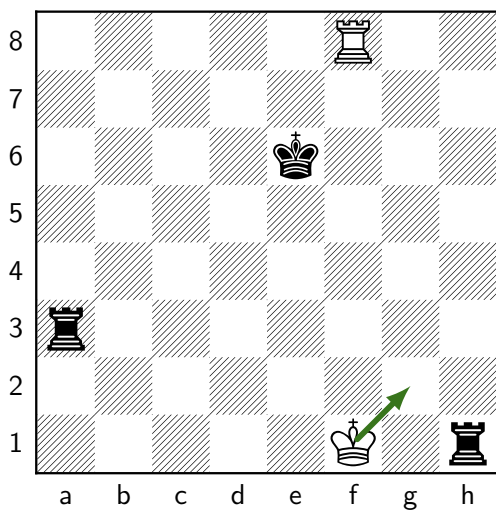


(c)

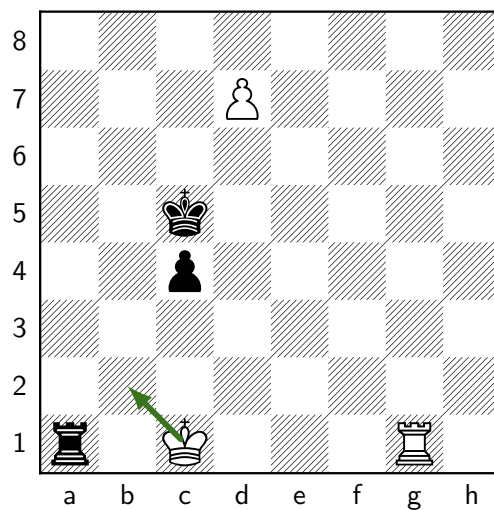


(d)

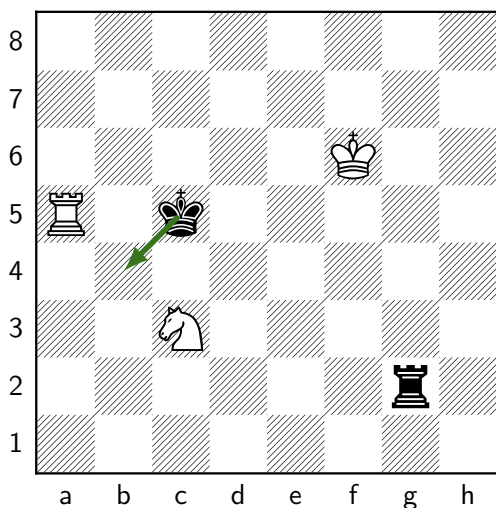
Figure 97: **Move 103, White: Kf1.** (a) Our position; Melanie plays Kf1 (f2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#1405` (ply 207) [structural distance $d=6$, QAP obj=2]: the master played Kh6; under μ our *Kf2* plays the role of that game’s *Kh2*; cross-type roles: our *rh4* as their *Pf6*; transferred move f2f1 (**endorses this move**). (c) Retrieved from `ccr1_top.p0002#1464` (ply 216) [structural distance $d=6$, QAP obj=2]: the master played Kg5; under μ our *Kf2* plays the role of that game’s *Kh6*; transferred move f2f1 (**endorses this move**). (d) Retrieved from `ccr1_top.p0000#147` (ply 189) [structural distance $d=6$, QAP obj=1]: the master played Rh1+; under μ our *Rf8* plays the role of that game’s *Rh7*; transferred move f8e8. Synthesis: 4 retrieved analogs transferred to Kf1 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



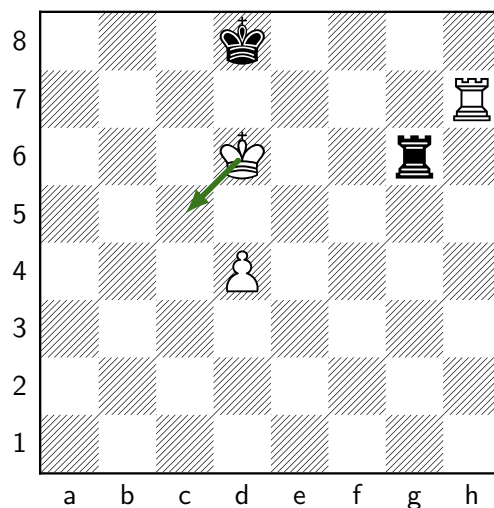
(a)



(b)

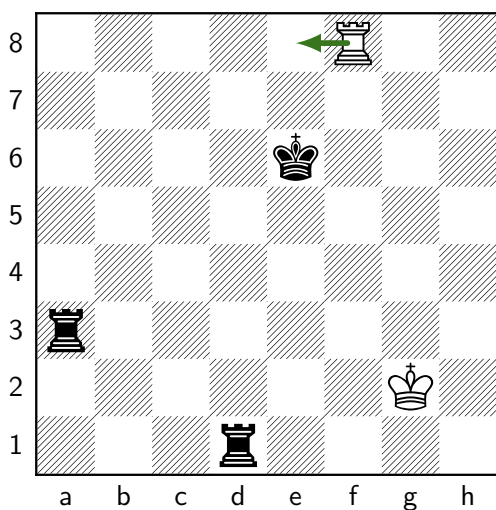


(c)

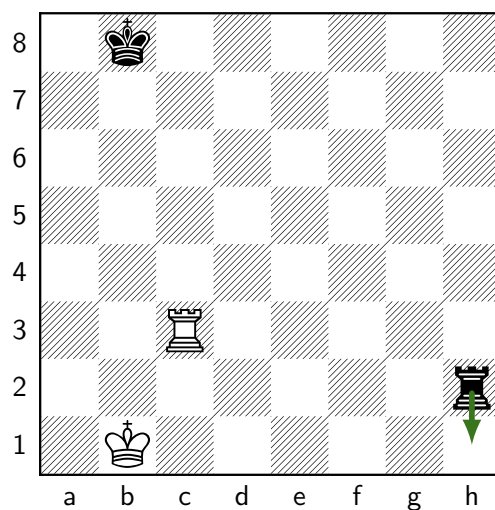


(d)

Figure 98: **Move 104, White: Kg2.** (a) Our position; Melanie plays Kg2 (f1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2549` (ply 194) [structural distance $d=8$, QAP obj=70]: the master played Kb2; under μ our *Kf1* plays the role of that game’s *Kc1*; cross-type roles: our *ra3* as their *pc4*; transferred move f1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1627` (ply 199) [structural distance $d=7$, QAP obj=67]: the master played Kb4; under μ our *Kf1* plays the role of that game’s *Kc4*; cross-type roles: our *ra3* as their *nc3*; transferred move f1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1611` (ply 190) [structural distance $d=7$, QAP obj=45]: the master played Kc5; under μ our *Kf1* plays the role of that game’s *Kd6*; cross-type roles: our *ra3* as their *Pd4*; transferred move f1f2. Synthesis: 2 retrieved analogs transferred to Kg2 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

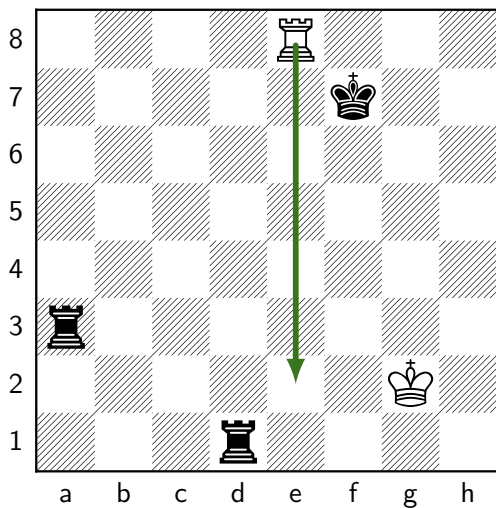


(a)

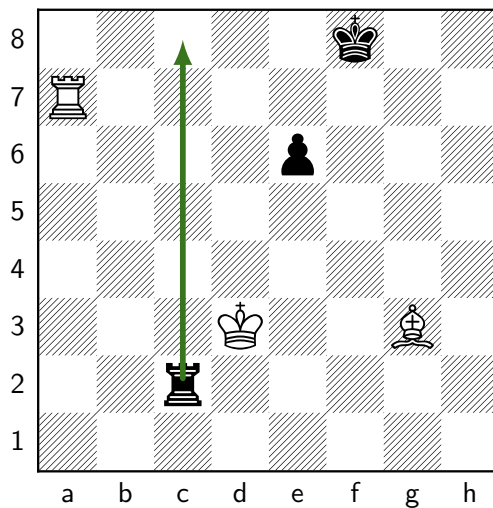


(b)

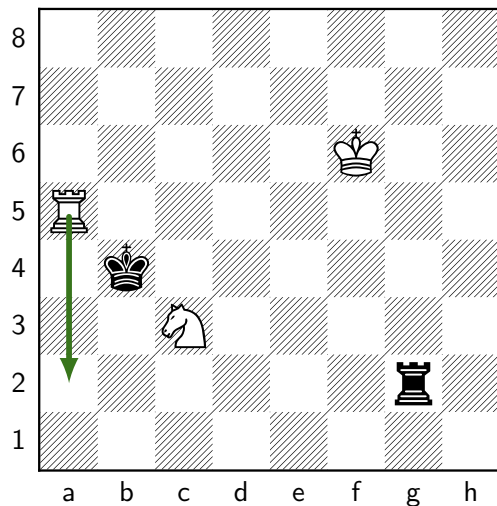
Figure 99: **Move 105, White: Re8+.** (a) Our position; Melanie plays **Re8+** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#147` (ply 193) [structural distance $d=3$, QAP obj=1]: the master played **Rh1+**; under μ our *Rf8* plays the role of that game’s *Rh7*; transferred move **f8e8** (**endorses this move**). Synthesis: 1 retrieved analog transferred to **Re8+** (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

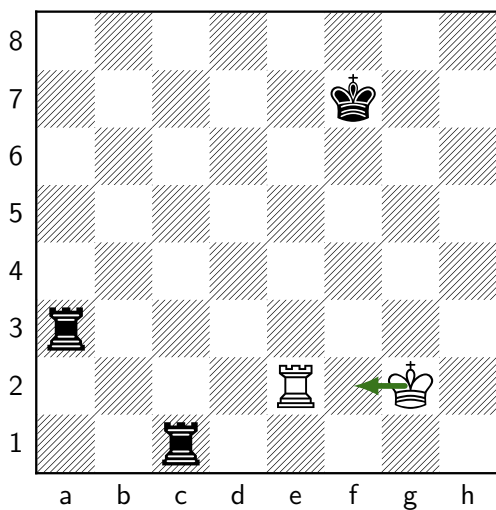


(b)

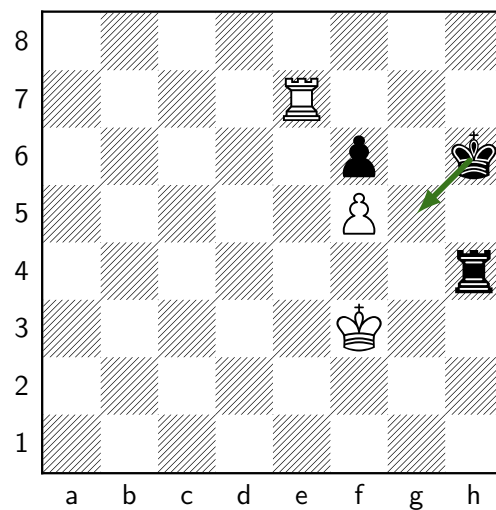


(c)

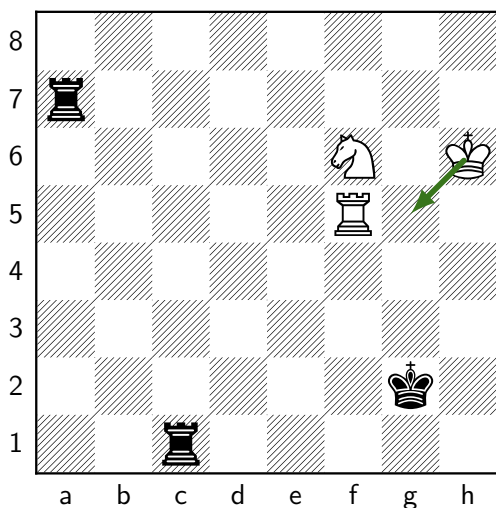
Figure 100: **Move 106, White: Re2.** (a) Our position; Melanie plays **Re2** (**e8e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2823` (ply 221) [structural distance $d=5$, QAP obj=20]: the master played **Rc8**; under μ our *Re8* plays the role of that game’s *Rc7*; cross-type roles: our *ra3* as their *bg3*; transferred move **e8e2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1627` (ply 200) [structural distance $d=6$, QAP obj=26]: the master played **Ra2**; under μ our *Re8* plays the role of that game’s *Ra5*; cross-type roles: our *Re8* as their *Nc3*; transferred move **e8e1**. Synthesis: 1 retrieved analog transferred to **Re2** (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



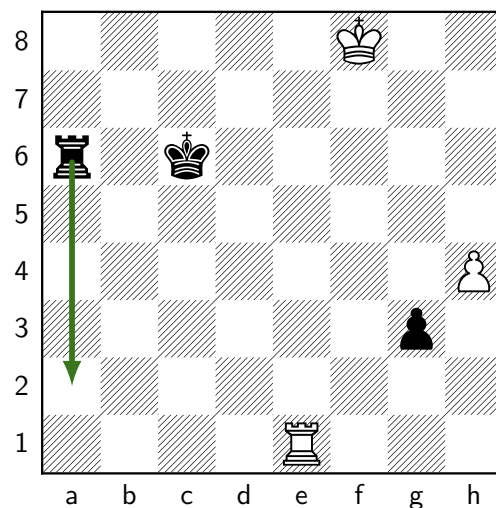
(a)



(b)

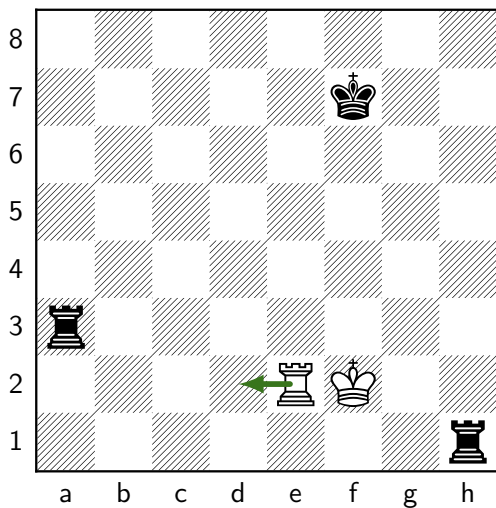


(c)

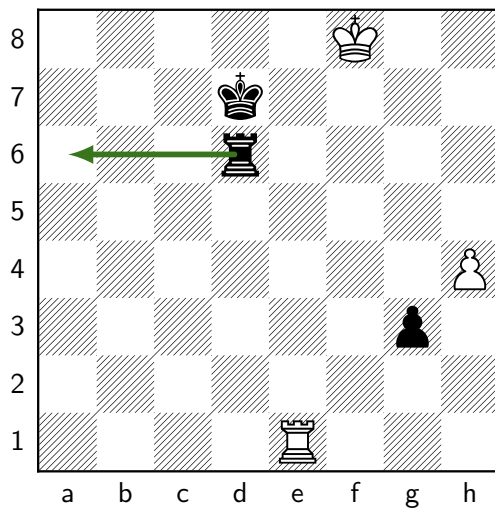


(d)

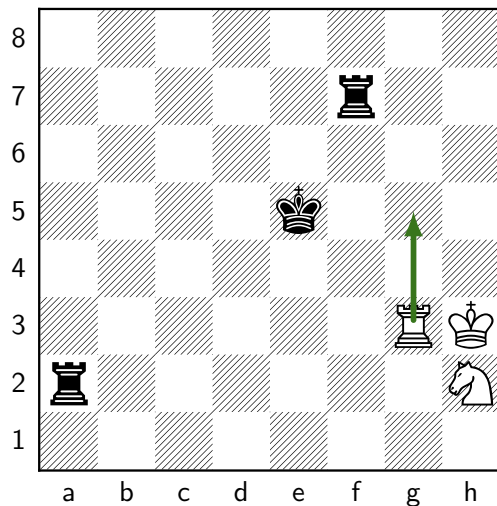
Figure 101: **Move 107, White: Kf2**. (a) Our position; Melanie plays Kf2 (g2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1405` (ply 201) [structural distance $d=6$, QAP obj=4]: the master played Kg5; under μ our Kg2 plays the role of that game's Kh3; cross-type roles: our ra3 as their pf5; transferred move g2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1464` (ply 216) [structural distance $d=6$, QAP obj=2]: the master played Kg5; under μ our Kg2 plays the role of that game's Kh6; transferred move g2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1481` (ply 211) [structural distance $d=7$, QAP obj=2]: the master played Ra2; under μ our Re2 plays the role of that game's Ra3; cross-type roles: our ra3 as their ph4; transferred move e2e5. Synthesis: 4 retrieved analogs transferred to Kf2 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

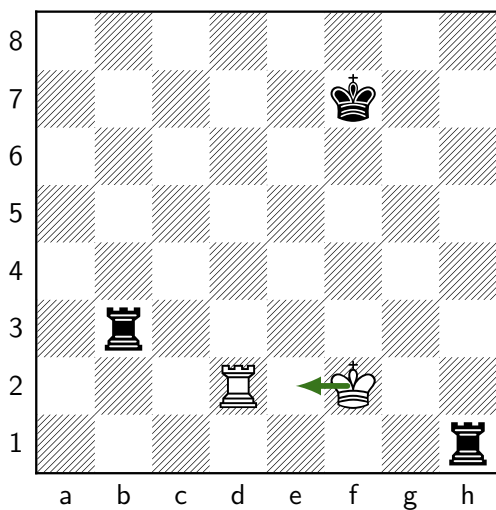


(b)

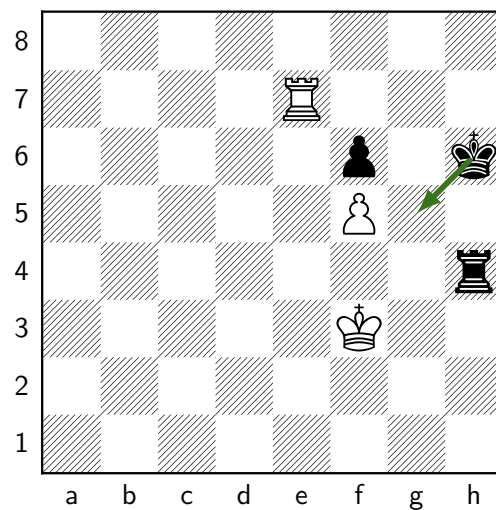


(c)

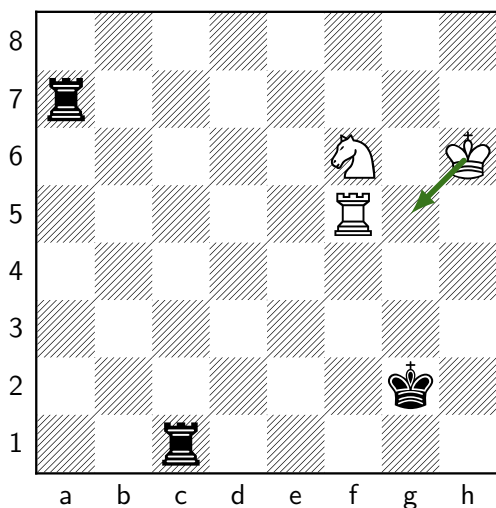
Figure 102: **Move 108, White: Rd2.** (a) Our position; Melanie plays Rd2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1481` (ply 207) [structural distance $d=5$, QAP obj=3]: the master played Ra6; under μ our *Re2* plays the role of that game’s *Rd3*; cross-type roles: our *rh1* as their *ph4*; transferred move e2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2385` (ply 212) [structural distance $d=8$, QAP obj=3]: the master played Rg5+; under μ our *Re2* plays the role of that game’s *Rg3*; cross-type roles: our *rh1* as their *Nh2*; transferred move e2d2 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Rd2 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



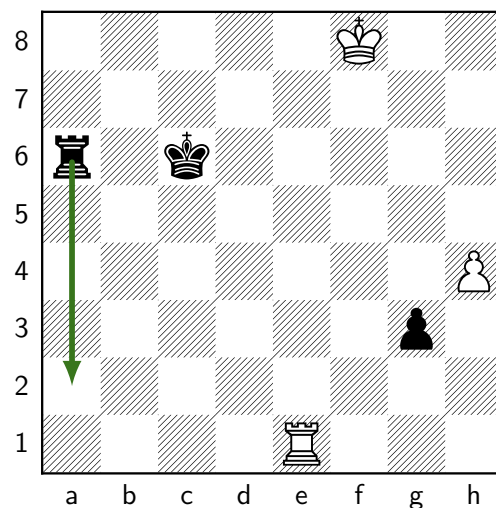
(a)



(b)

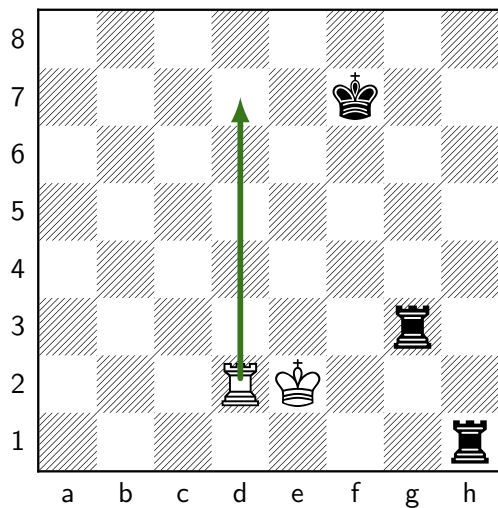


(c)

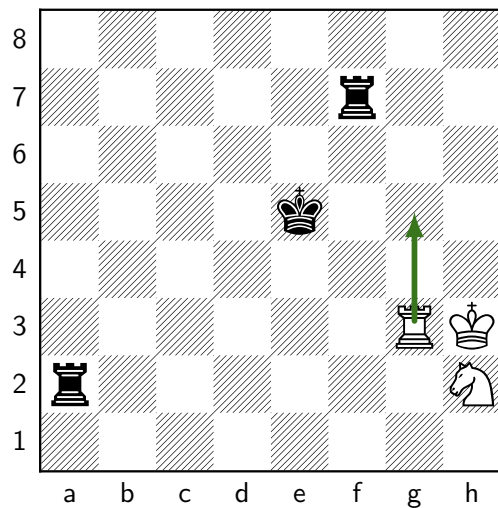


(d)

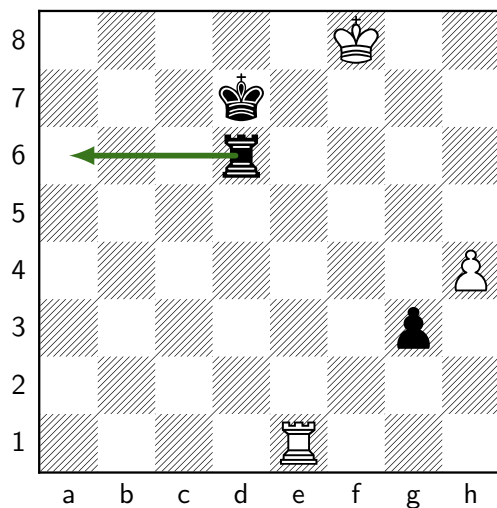
Figure 103: **Move 109, White: Ke2.** (a) Our position; Melanie plays Ke2 (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1405` (ply 201) [structural distance $d=6$, QAP obj=3]: the master played Kg5; under μ our Kf2 plays the role of that game's Kh3; cross-type roles: our rb3 as their pf5; transferred move f2e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1464` (ply 216) [structural distance $d=7$, QAP obj=1]: the master played Kg5; under μ our Kf2 plays the role of that game's Kh6; transferred move f2e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1481` (ply 211) [structural distance $d=6$, QAP obj=2]: the master played Ra2; under μ our Rd2 plays the role of that game's Ra3; cross-type roles: our rb3 as their ph4; transferred move d2d7. Synthesis: 4 retrieved analogs transferred to Ke2 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)



(b)



(c)

Figure 104: **Move 110, White: Rd7+.** (a) Our position; Melanie plays Rd7+ (d2d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2385` (ply 212) [structural distance $d=8$, QAP obj=2]: the master played Rg5+; under μ our Rd2 plays the role of that game's Rg3; cross-type roles: our rg3 as their Nh2; transferred move d2d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1481` (ply 207) [structural distance $d=5$, QAP obj=3]: the master played Ra6; under μ our Rd2 plays the role of that game's Rd3; cross-type roles: our rg3 as their ph4; transferred move d2c2. Synthesis: 1 retrieved analog transferred to Rd7+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.